

Probing Viterbo's Conjecture

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Abstract

Haim–Kislev and Ostrover disproved Viterbo's conjecture with a Lagrangian product of two regular pentagons. We investigate the geometry near their example and use finite computation to look for further structure in the systolic ratio of four-dimensional convex polytopes.

Our main theorem shows that the HKO body is a local maximizer among polytopes with exactly ten facets. Nearby equality occurs precisely under translations, positive dilations and linear symplectic transformations.

We implement, optimize and extend two capacity algorithms, based on the quadratic program of Haim–Kislev and the flow graph of Chaidez and Hutchings. We also study the first-order variation of capacity. As a side theorem, we prove an explicit capacity formula for the relative-rotation family of two regular pentagons. A short analytic proof shows that the HKO body is a global maximizer of the systolic ratio in this family. We then extend the capacity formula and sharp bound to products of two independently linearly deformed regular pentagons.

An empirical investigation looks for statistical patterns in geometric measurements of sampled polytopes, with the aim of turning those patterns into mathematical conjectures. We find patterns among bodies with systolic ratio below one, but do not establish that these patterns extend to the regime above one. This leaves open whether further progress requires different statistical methods or new geometrically motivated measurements.

Disclosure of AI Use

Generative artificial intelligence was used materially throughout the research and production of this thesis, principally through Codex and Claude Code agents. Agents were not limited to language polishing or executing fully specified tasks. They participated in choosing questions and methods; developing mathematical arguments, software, experiments, and prose; interpreting and criticizing results; revising work; and planning, delegating, and reviewing work by other agents. This breadth of AI use was approved by my advisor.

There was no stable division of labor by category. I am not aware of any kind of work I performed on this thesis that I did not also delegate to agents at some point. Agents and I both planned, produced, interpreted, corrected, and reviewed work, often through repeated handoffs. This overlap does not imply equal labor shares or identical contributions to every result. The research and production process, and many retained components, therefore have mixed human–AI provenance. I do not claim sole production or sole intellectual origin of every part of the thesis. Because work often passed between me and agents, I cannot reliably say who first originated every idea, passage, program, or decision.

The analytic endpoint-comparison proof of the rotated-pentagon profile in Section 9 was supplied during an AI-assisted review by ChatGPT Pro. I subsequently checked and accepted that mathematical argument. The same review supplied the restricted-family ridge-capacity identity in equation 35. These contributions are distinct from the earlier computational exploration of the rotation family.

The independent-linear-deformation argument in Section 10 originated in a further ChatGPT Pro return on 18 September 2026. Codex agents checked the argument and its capacity-formula conventions and prepared the repository and thesis versions. This records agent review, not a completed personal verification of the new argument.

As of 13 July 2026, I had proofread the then-current thesis and personally worked to verify its mathematical and factual claims and the overall message it presented. The manuscript was revised substantially afterwards. This statement therefore does not assert that I have completed a final review of the current candidate.

Contents

Disclosure of AI Use	2
1 Introduction	5
1.1 Capacities, dynamics and convex geometry	5
1.2 A local theorem around the counterexample	6
1.3 From boundary dynamics to finite calculations	6
1.4 Looking for mathematical structure in data	7
1.5 How to read the thesis	7
2 Preliminaries	7
2.1 Convex and symplectic notation	7
2.2 Convex Hamiltonians	8
2.3 EHZ capacity	9
2.4 Polytope input language	11
2.5 Lagrangian products	11
2.6 Clarke’s dual action principle	12
3 Generalized Reeb orbits on polytopes	14
3.1 Facet velocities	14
3.2 Words, dwell times, and closure	15
3.3 Existence of simple minimizers	17
4 The Haim–Kislev Quadratic Program	20
4.1 The finite quadratic program	21
4.2 Solving one word and comparing the results	23
4.3 Finite enumeration for Lagrangian products	24
4.4 A six-facet reduction for the product capacity	26
4.5 Use in later certificates	28
5 Computing closed trajectories with a flow graph	28
5.1 Affine tubes	28
5.2 Exhaustive search and correctness	32
5.3 Why the regular class is generic	33
5.4 Relation to CH2021	34
5.5 Scope and rational implementation	34
6 First-Order Variation in Facet Coordinates	35
6.1 Row coordinates and the volume variation	35
6.2 Feasible sections and smooth maxima	36
6.3 From branches to the systolic ratio	38
6.4 What the first-order model contributes to search	38
6.5 The non-generic boundary	39
6.6 Application at HKO	39
6.7 Implementation and validation status	39
7 HKO Local Maximality	41
7.1 The proof strategy	41
7.2 Local coordinates and volume	41
7.3 The touching upper bounds	43
7.4 Computing the twenty-six derivatives	45
7.5 The transverse slope estimate	46
7.6 From slopes to a neighborhood	46

8	Finding mathematical patterns with data science	49
8.1	Ridge area and systolic ratio	50
8.2	Looking for patterns in the measurements	50
8.3	The sampled bodies and the measured ratio	50
8.4	Predictive information in ridge area	53
8.5	Returning from a pattern to geometry	53
8.6	Testing whether a pattern can guide fresh selection	55
8.7	Transfer and the limit of a stronger filter	56
8.8	Changing support heights and symplectic alignment	56
8.9	Adapting the sampling distribution	57
8.10	Local search from matched starting bodies	58
8.11	What a failed local probe tells us	59
8.12	What remains unexplained	59
9	The rotated-pentagon profile	60
9.1	Symmetries and the Haim–Kislev–Ostrover position	61
9.2	Feasible weights under rotation	61
9.3	Bounding all choices by their endpoint values	62
10	Independent linear deformations of the pentagon factors	63
10.1	Reducing the possible facet orders	64
10.2	Two identities cover all six-group competitors	65
10.3	Capacity and the sharp ratio	65
11	Rotating a product out of symplectic position	66
12	Three-Dimensional Visualization	68
13	Numerics	70
13.1	Input and output contracts	70
13.2	Certified fixed-word decisions	70
13.3	Complete general and product routes	71
13.4	Empirical verification and limitations	71
14	Availability of Code and Data	72
15	AI in the research process	73
15.1	Delegating an incomplete proof strategy	73
15.2	Matching the check to the claim	74
15.3	What still requires a reader	74
15.4	A connection with training on verifiable outcomes	74
16	Conclusion	76
A	Data-science methods and additional results	76
A.1	Sampling and retained data	77
A.2	Numerical evaluation and retrospective checks	77
A.3	Selection and comparison rules	77
A.4	Finite-budget local optimizer comparison	78
A.5	Adaptive product sampling	79
A.6	Evidence and reproduction index	82
B	HKO computation: an annotated exact program	84
B.1	Executable exact verification of the HKO bounds	84
	References	86

1 Introduction

How large can a convex body’s symplectic capacity be if its volume is fixed? Viterbo’s conjecture proposed that a ball gives the largest value [Vit00]. For the Ekeland–Hofer–Zehnder capacity in dimension four, the proposed inequality was

$$\text{sys}(K) := \frac{c_{\text{EHZ}}(K)^2}{2 \text{vol}_4(K)} \leq 1.$$

The quantity on the left is called the systolic ratio. Capacity scales by the square of a dilation and four-dimensional volume by its fourth power, so this ratio compares shape independently of overall size. Translations and linear symplectic maps also preserve it.

The inequality is false. Haim–Kislev and Ostrover found a counterexample formed from two regular pentagons in complementary Lagrangian planes [HO26]. Writing the coordinates as $(q, p) \in \mathbb{R}^2 \oplus \mathbb{R}^2$, one representative is

$$K_{\text{HKO}} = P_5 \times R(-\pi/2)P_5, \quad \text{sys}(K_{\text{HKO}}) = \frac{3 + \sqrt{5}}{5} > 1.$$

Here P_5 is a centred regular pentagon with a vertex on the positive horizontal axis, and $R(\theta)$ is planar rotation. A product placed in the q - and p -planes is called a Lagrangian product. The relative position of its factors affects capacity even though it does not affect volume.

The counterexample leaves several geometric questions. Is the HKO body a local maximum of the systolic ratio among convex bodies? Are there other families with large ratios? Can the computations reveal geometric relationships involving capacity? This thesis investigates these questions through local analysis, finite algorithms and data science. This is the sense in which we continue to probe Viterbo’s conjecture: although its proposed bound is false, the geometric relationship between symplectic capacity and volume remains to be understood.

The contributions grew out of a broad computational investigation. We developed methods for evaluating capacity and used them to explore examples, following up on questions that the computations suggested. This led to several separate results, from a local maximality theorem to algorithms and statistical observations. The following overview explains what each part contributes to that investigation.

1.1 Capacities, dynamics and convex geometry

Symplectic capacities measure constraints on embeddings that volume alone cannot detect. Two basic examples are the Gromov width c_G , defined by the largest symplectic ball that embeds into a domain, and the cylindrical capacity c_Z , defined by the smallest symplectic cylinder into which the domain embeds. With the usual normalization, every symplectic capacity lies between these two [Edt24]. The EHZ capacity has a different description: on convex bodies it is the least action of a closed characteristic. Thus comparing capacities connects an embedding question to a question about Hamiltonian trajectories.

The conjecture that all normalized capacities agree on convex domains was stronger than Viterbo’s volume bound. Indeed, the Gromov width satisfies that bound directly by volume preservation. The HKO example therefore also separates the EHZ capacity from the Gromov width [HO26]. Abbondandolo, Edtmair and Kang prove that EHZ and cylindrical capacity coincide on every convex body in four dimensions [AEK24, Corollary 1]. Thus the capacity computed in this thesis also determines cylindrical capacity. Their proof builds on Edtmair’s earlier work, which characterizes cylindrical capacity on smooth star-shaped domains in $\mathbb{R}^4$ with dynamically convex boundary by the least area of a disk-like global surface of section, and proves coincidence of all normalized capacities for convex domains sufficiently C^3 -close to the ball [Edt24]. A global surface of section is a surface whose interior is transverse to the flow and

is met in both forward and backward time by every trajectory outside its boundary. These results illustrate how information about the flow can determine an embedding obstruction.

There is also a direct connection to a central problem in convex geometry. For a centrally symmetric convex body $L \subset \mathbb{R}^n$ and its polar L° , Artstein–Avidan, Karasev and Ostrover prove $c_{\text{EHZ}}(L \times L^\circ) = 4$ [AKO14]. Viterbo’s proposed inequality on these particular products is consequently equivalent to the lower bound $\text{vol}(L) \text{vol}(L^\circ) \geq 4^n/n!$ in the symmetric Mahler conjecture. The general counterexample does not invalidate this equivalence for the restricted family. Capacity–volume questions therefore retain significance beyond the original universal bound.

Local maximality has a dynamical precedent as well. Abbondandolo and Benedetti prove that Zoll contact forms, whose Reeb orbits are all closed with the same minimal period, locally maximize the contact systolic ratio [AB23]. Their results include the volume–capacity inequality near the ball. The HKO body presents a different setting: it has corners and a ratio larger than the ball’s. Our local theorem studies how such a nonsmooth example behaves under perturbations of its supporting facets.

1.2 A local theorem around the counterexample

Our main result partially answers the first question: every sufficiently Hausdorff-close ten-facet polytope has ratio at most that of K_{HKO} . In a smaller neighborhood, equality occurs precisely for translations, positive dilations and linear symplectic images of K_{HKO} ; see Theorem 7.1. Thus a small change of shape that is not one of these symmetries lowers the ratio. Allowing facets to appear or disappear is a different local problem, which the theorem leaves open.

The proof uses upper bounds that agree with the systolic ratio at HKO. For each nearby direction, at least one of these bounds decreases. It follows that the ratio itself decreases. The finite task is to construct enough such bounds and check their derivatives. Feasible weights in Haim–Kislev’s capacity formula provide the bounds. An algebraic computation establishes the derivative conditions; the proof explains why those conditions imply the geometric local maximum.

1.3 From boundary dynamics to finite calculations

For a smooth convex body, the EHZ capacity is the least action of a closed characteristic on its boundary. Polytopes have corners, so the relevant curves are generalized characteristics. Their velocities are governed by the normals of the facets they meet. Haim–Kislev’s theorem makes the capacity computable by a finite optimization over facet orders and nonnegative weights [Hai19].

We expand the proof in Haim–Kislev’s original paper into a self-contained presentation for a master’s-level reader. We explain the dual action principle and the existence of a minimizing curve with a finite list of distinct facet directions.

For Lagrangian products, we derive a further reduction from the bilinear structure of the objective: some maximizing word uses at most three facets from each factor. This gives a smaller finite search for the capacity, without classifying every minimizing orbit.

We pursued the flow-graph method of Chaidez and Hutchings [CH21] as an alternative way to compute capacity. Its performance in our tests made it less useful for large searches, but it provides a reference implementation for cross-checking the quadratic-program method. Agreement between the two gives additional evidence that the implementations compute the intended quantity.

The same finite methods yield a side result for the family $P_5 \times R(\theta)P_5$: its systolic ratio is

$$\frac{5 + 2\sqrt{5}}{10} \sec^2 d(\theta),$$

where $d(\theta)$ is the distance to the nearest multiple of $\pi/5$. A short analytic comparison proves the formula, whose maximum is the HKO value. A separate matrix-containment argument extends the capacity formula and sharp systolic bound to independent linear deformations of both regular pentagons; see Section 10. This family is larger than the rotation family but does not include arbitrary pentagons.

We also vary the placement of the factor planes themselves. For any two full-dimensional planar convex bodies, a small rigid rotation away from the symplectic-product arrangement increases capacity. The Lagrangian endpoint need not be better: an explicit triangle product has smaller capacity there; see Section 11.

1.4 Looking for mathematical structure in data

We also use data science to seek geometric relationships involving capacity. The aim is to discover statistical patterns, interpret them geometrically, and turn them into conjectures that can be proved or refuted. The experiments find patterns below systolic ratio one, but do not establish their extension above one. One geometric interpretation yields an exact edge-width formula for a measured quantity on Lagrangian products. On rotated regular pentagon products, this descriptor has an exact inverse-square relation to the systolic ratio; see equation 35. The relation does not extend to arbitrary products.

1.5 How to read the thesis

The foundations in Sections 2–4 explain how nonsmooth boundary dynamics become a finite quadratic program. Section 5 presents the alternative flow-graph method, and Section 6 develops first-order variation.

A reader primarily interested in the HKO theorem can start with its statement and proof strategy in Section 7, returning to the foundations for the capacity formula and to Section 6 for variation. The flow-graph and data-science chapters are not prerequisites for that proof. Section 8 describes the empirical investigation. Sections 9 and 10 give the rotation profile and its extension to independently deformed pentagons. Section 11 then rotates the full product out of symplectic position, rather than rotating one factor within its plane. The later sections document visualizations, numerical methods, code and data, and reflect on the use of AI. The appendices contain computational detail for checking or reproducing the results.

2 Preliminaries

A polytope has edges and vertices where the smooth definition of a characteristic no longer applies. We first extend that definition to convex bodies with nonsmooth boundaries. We then introduce a variational description of capacity in terms of closed curves that need not lie on the boundary. This will allow us to modify curves when deriving the finite capacity formula.

All convex bodies below are compact convex subsets of $\mathbb{R}^4$ with non-empty interior. Unless stated otherwise, the origin lies in the interior.

2.1 Convex and symplectic notation

We write $\mathbb{R}^4$ with coordinates (q_1, q_2, p_1, p_2) . The standard complex structure is

$$J_0 = \begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix},$$

and the standard symplectic form is

$$\omega_0(u, v) = \langle J_0 u, v \rangle = \sum_{i=1}^2 (u_{q_i} v_{p_i} - u_{p_i} v_{q_i}).$$

We use the standard primitive

$$\lambda_0 = \frac{1}{2} \sum_{i=1}^2 (q_i dp_i - p_i dq_i).$$

Thus $d\lambda_0 = \omega_0$ and $\lambda_0|_x(v) = \frac{1}{2} \langle -J_0 v, x \rangle$. For an absolutely continuous closed curve $\gamma: [0, T] \rightarrow \mathbb{R}^4$, its action is

$$A(\gamma) := \int_{\gamma} \lambda_0 = \frac{1}{2} \int_0^T \langle -J_0 \dot{\gamma}(t), \gamma(t) \rangle dt. \quad (1)$$

This action is invariant under orientation-preserving reparametrization. If γ bounds a smooth map $u: D^2 \rightarrow \mathbb{R}^4$, then Stokes' theorem gives

$$A(\gamma) = \int_{D^2} u^* \omega_0.$$

Thus action is the signed symplectic area enclosed by the curve. For Reeb orbits this is the quantity that the EHZ capacity minimizes.

Let $K \subset \mathbb{R}^4$ be a smooth convex body with $0 \in \text{int}(K)$. At $x \in \partial K$, the restriction of ω_0 to the three-dimensional space $T_x \partial K$ has a one-dimensional kernel,

$$\ell_x = \{v \in T_x \partial K : \omega_0(v, w) = 0 \text{ for all } w \in T_x \partial K\}.$$

This is the characteristic line. A *closed characteristic* is a closed curve on ∂K whose velocity lies in the positive characteristic direction. We choose the positive direction by requiring $\lambda_0(\dot{\gamma}) > 0$.

The restriction $\alpha = \lambda_0|_{\partial K}$ is a contact form. Its Reeb vector field R is the contact-normalized characteristic direction,

$$\alpha(R_x) = 1, \quad \omega_0(R_x, v) = 0 \quad \text{for all } v \in T_x \partial K.$$

A Reeb orbit is a periodic integral curve of R . It is the same geometric object as a closed characteristic, with the parametrization fixed by $\lambda_0(\dot{\gamma}) = 1$. Consequently, if γ is a Reeb orbit on $[0, T]$, then

$$A(\gamma) = \int_0^T \lambda_0(\dot{\gamma}(t)) dt = T.$$

This is the sense in which the action of a contact-normalized Reeb orbit is also its period.

2.2 Convex Hamiltonians

The smooth characteristic line field has a convex-analytic formulation that continues to make sense when ∂K is not smooth. For a convex body K containing the origin, its support and gauge functions are

$$h_K(y) := \max_{x \in K} \langle x, y \rangle, \quad g_K(x) := \inf\{r > 0 : x \in rK\}.$$

The boundary is the level set $g_K = 1$. We use the convex, 2-homogeneous Hamiltonian

$$H_K(x) := g_K(x)^2.$$

For a convex function H , its subdifferential at x is

$$\partial H(x) = \{y : H(z) \geq H(x) + \langle y, z - x \rangle \text{ for every } z\}.$$

When H is differentiable, this set consists only of $\nabla H(x)$.

Definition 2.1 (Contact-normalized generalized characteristic). *A contact-normalized generalized characteristic on ∂K is a curve $\gamma \in W^{1,2}([0, T], \mathbb{R}^4)$, with $T > 0$ and $\gamma(0) = \gamma(T)$, whose image lies in ∂K and which satisfies*

$$-J_0 \dot{\gamma}(t) \in \partial H_K(\gamma(t)) \quad \text{for almost every } t \in [0, T]. \quad (2)$$

For smooth K , the subdifferential in (2) is the singleton $\{\nabla H_K\}$. The resulting Hamiltonian vector is tangent to the characteristic line on ∂K . Thus the definition recovers the Reeb orbits described above, including their parametrization.

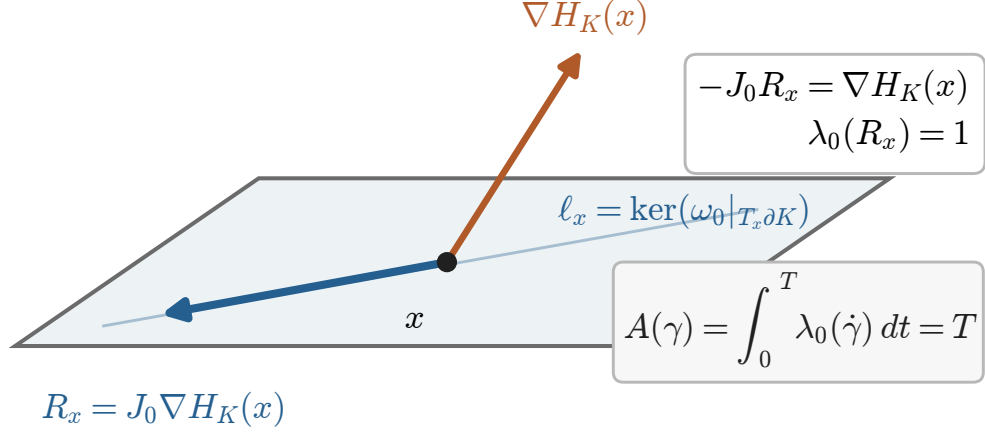


Figure 1: The characteristic and contact normalizations at a smooth boundary point. This is a dimension-reduced tangent-space schematic: it records that $R_x = J_0 \nabla H_K(x)$ spans the characteristic line inside $T_x \partial K$, but projected angles and lengths carry no meaning. The contact condition $\lambda_0(R_x) = 1$ turns elapsed time into action.

The normalization also persists without smoothness. Indeed, if $y \in \partial H_K(x)$, then the 2-homogeneity of H_K gives

$$\langle y, x \rangle = 2H_K(x).$$

Along a generalized characteristic, $H_K(\gamma) = 1$, and hence

$$\lambda_0|_{\gamma(t)}(\dot{\gamma}(t)) = \frac{1}{2} \langle -J_0 \dot{\gamma}(t), \gamma(t) \rangle = 1 \quad \text{for almost every } t.$$

Consequently every contact-normalized generalized characteristic satisfies

$$A(\gamma) = T. \quad (3)$$

This formulation separates the general nonsmooth dynamical object from its finite description on a polytope. The latter will follow by calculating the subdifferential of H_K from the active facets.

2.3 EHZ capacity

For smooth convex bodies, the Ekeland–Hofer and Hofer–Zehnder capacities coincide. Their common value is the minimum action among closed characteristics on the boundary. We use the resulting EHZ capacity in the form

$$c_{\text{EHZ}}(K) = \min\{A(\gamma) : \gamma \text{ is a closed characteristic on } \partial K\}. \quad (4)$$

This is the smooth least-action characterization of [AO14, Theorem 2.2]. Equivalently, the minimum may be taken over Reeb orbits. The contact normalization only chooses the parametrization

of each characteristic, while the action is a line integral and is invariant under orientation-preserving reparametrization. For a contact-normalized representative, $A(\gamma)$ is also the period. The same least-action theorem gives a positive attained minimum.

For a nonsmooth convex body, we define capacity as the limit of the capacities of smooth convex approximations in the Hausdorff metric. This limit is independent of the approximating sequence. The least-action description remains valid when closed characteristics are replaced by the generalized characteristics of Definition 2.1 [AO14, Proposition 2.7]. Their finite facet description on a polytope is derived in Section 3.

As a symplectic capacity on convex bodies, c_{EHZ} is monotone under inclusion, invariant under symplectomorphisms, and conformal:

$$c_{\text{EHZ}}(\lambda K) = \lambda^2 c_{\text{EHZ}}(K) \quad (\lambda > 0).$$

It is normalized by

$$c_{\text{EHZ}}(B^4(r)) = c_{\text{EHZ}}(Z^4(r)) = \pi r^2,$$

where $B^4(r)$ is the radius- r ball and $Z^4(r) = B^2(r) \times \mathbb{R}^2$ is the symplectic cylinder.

Throughout the thesis, vol_4 denotes four-dimensional Lebesgue measure in the coordinate order (q_1, q_2, p_1, p_2) . In dimension four we define the systolic ratio by

$$\text{sys}(K) := \frac{c_{\text{EHZ}}(K)^2}{2 \text{vol}_4(K)}. \quad (5)$$

With this normalization, Viterbo's proposed inequality for four-dimensional convex bodies was

$$\text{sys}(K) \leq 1, \quad \text{equivalently} \quad c_{\text{EHZ}}(K)^2 \leq 2 \text{vol}_4(K)$$

[Vit00].

Definition 2.2 (Systolic-ratio symmetries). *Let*

$$G_{\text{sys}} = \{x \mapsto b + \rho Mx : b \in \mathbb{R}^4, \rho > 0, M \in \text{Sp}(4, \mathbb{R})\}.$$

This is the group of affine transformations generated by translations, positive scalings, and linear symplectic transformations.

Lemma 2.3 (Systolic-ratio invariance). *For every convex body $K \subset \mathbb{R}^4$ and every $g \in G_{\text{sys}}$,*

$$\text{sys}(gK) = \text{sys}(K).$$

Proof. Translations and linear symplectic transformations are symplectomorphisms, so they preserve c_{EHZ} . They also preserve four-dimensional volume, because translations do not change volume and every symplectic matrix has determinant 1. A positive scaling by ρ gives

$$c_{\text{EHZ}}(\rho K) = \rho^2 c_{\text{EHZ}}(K), \quad \text{vol}_4(\rho K) = \rho^4 \text{vol}_4(K).$$

Substituting these identities in (5) shows that the ratio is unchanged. $\square$

Proposition 2.4 (Continuity). *Let K_t be a family of four-dimensional convex bodies which is continuous in the Hausdorff topology. Then $c_{\text{EHZ}}(K_t)$ is continuous in t . If, moreover, $t \mapsto \text{vol}_4(K_t)$ is positive, finite, and continuous, then the systolic ratio of the family is continuous.*

Proof. The EHZ capacity is C^0 -continuous on convex bodies with respect to the Hausdorff metric [CH21]. The assertion for sys is then ordinary continuity of the quotient in (5). In the applications below the volume hypothesis is usually automatic, since four-volume is continuous under Hausdorff convergence of convex bodies; for the rotated-polygon family it is even constant. $\square$

2.4 Polytope input language

The finite methods represent polytopes by normalized halfspaces. Let $K \subset \mathbb{R}^4$ be full-dimensional with $0 \in \text{int}(K)$. We write

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1, i = 1, \dots, F\}. \quad (6)$$

If n_i is the outward unit normal to the corresponding supporting hyperplane and $h_i = h_K(n_i) > 0$ is its height, then

$$a_i = \frac{n_i}{h_i}.$$

For an irredundant description, the vectors a_i are precisely the vertices of the polar body

$$K^\circ = \text{conv}\{a_1, \dots, a_F\}.$$

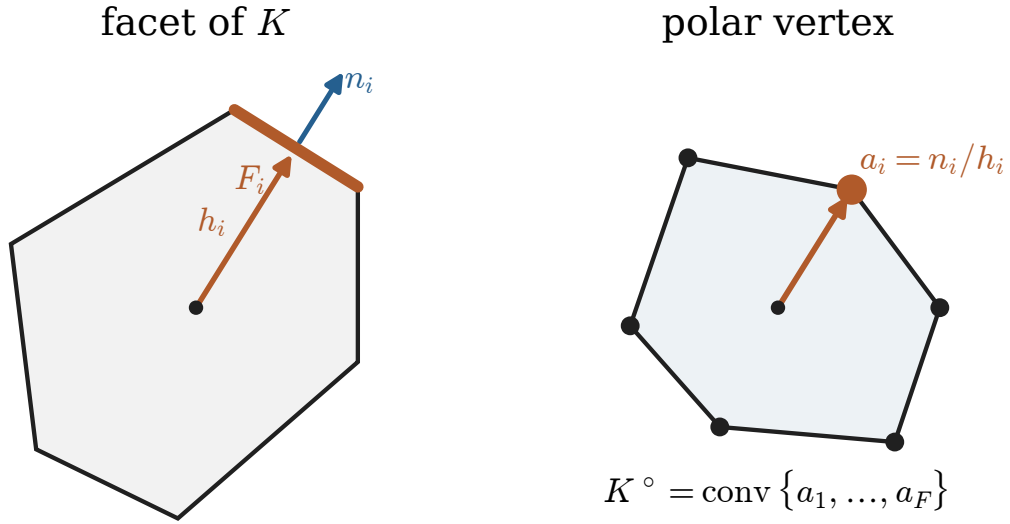


Figure 2: Facet normalization and polarity in a planar model. The supporting line $\langle n_i, x \rangle = h_i$ becomes $\langle a_i, x \rangle = 1$ after setting $a_i = n_i/h_i$, and that row is a vertex of the polar polygon. The same facet–polar-vertex correspondence holds in four dimensions; the planar metric proportions are specific to this example and are not part of the duality statement.

Conversely, the normalized halfspaces define a bounded full-dimensional polytope containing the origin in its interior exactly when $0 \in \text{int conv}\{a_i\}$. The description is irredundant exactly when the listed vectors are pairwise distinct and every a_i is an extreme point of this convex hull.

The ordered list $(a_1, \dots, a_F)$ is a labelled presentation. Relabelling the same inequalities does not change the unlabelled convex body, but labels matter when an algorithm records facet words or when a local calculation uses the a_i as coordinates.

2.5 Lagrangian products

Write

$$\mathbb{R}^4 = \mathbb{R}_q^2 \oplus \mathbb{R}_p^2 = \{(q_1, q_2, p_1, p_2)\}.$$

Definition 2.5 (Lagrangian product). *For planar convex bodies $K_q, K_p \subset \mathbb{R}^2$, their Lagrangian product is*

$$K_q \times_L K_p = \{(q_1, q_2, p_1, p_2) : (q_1, q_2) \in K_q, (p_1, p_2) \in K_p\}. \quad (7)$$

Thus K_q occupies the Lagrangian q -plane and K_p occupies the Lagrangian p -plane. This is different from taking a product of planar domains in the symplectic coordinate planes (q_1, p_1) and (q_2, p_2) .

The two decompositions pair the same four coordinates in different ways:

$$\underbrace{(q_1, q_2)}_{K_q} \mid \underbrace{(p_1, p_2)}_{K_p} \quad \text{versus} \quad \underbrace{(q_1, p_1)}_{\text{first symplectic plane}} \mid \underbrace{(q_2, p_2)}_{\text{second symplectic plane}}.$$

In block notation $J_0(u, v) = (-v, u)$. Consequently, the Reeb direction associated later with a supporting row in one Lagrangian factor moves in the other factor. This exchange, rather than a Cartesian-product decoupling of the dynamics, is what makes Lagrangian products useful here.

The central examples of the thesis are Lagrangian products of polygons; their lifted facet rows and the resulting restricted word enumeration are introduced with the Haim–Kislev formula, where they are first used.

2.6 Clarke’s dual action principle

A curve constrained to lie on ∂K is difficult to modify: changing one velocity can move the rest of it off the boundary. We will instead work with closed curves in $\mathbb{R}^4$ and a functional that depends only on velocity. The dual action principle says that minimizing this functional recovers the capacity, and that a minimizer can be placed back on the boundary. This is the freedom used in the next section to rearrange facet velocities.

Recall $H_K = g_K^2$, and define its convex conjugate by $H_K^*(y) = \sup_x (\langle x, y \rangle - H_K(x))$. For a closed curve $z \in W^{1,2}([0, T], \mathbb{R}^4)$ put

$$I_K(z) = \int_0^T H_K^*(-J_0 \dot{z}(t)) dt = \frac{1}{4} \int_0^T h_K(-J_0 \dot{z}(t))^2 dt. \quad (8)$$

The last identity will be proved below. Our normalization is

$$d(K) = \inf \{ I_K(z) : T > 0, z(0) = z(T), A(z) = T \}. \quad (9)$$

The infimum is over Sobolev curves as above. Their positions are unrestricted; the action constraint fixes their size relative to their period.

Theorem 2.6 (Dual action principle). *For a convex body $K \subset \mathbb{R}^4$ with $0 \in \text{int } K$, the infimum in (9) is attained and*

$$d(K) = c_{\text{EHZ}}(K) = \min_{\gamma \subset \partial K} A(\gamma),$$

where the last minimum is over contact-normalized generalized characteristics. Every such characteristic satisfies $I_K(\gamma) = A(\gamma)$. Every dual minimizer can be rescaled and translated into a minimizing generalized characteristic, without changing its value of I_K . If a dual minimizer already has $I_K(z) = T$, only a translation is needed.

We use the nonsmooth variational results underlying Clarke’s principle [AO14, Proposition 2.5 and Lemmas 5.1–5.2]; for polytopes the correspondence is also stated in [Hai19, Lemma 2.1]. The following argument explains the conjugacy, normalization and reconstruction used here.

The conjugate and its equality condition.

Lemma 2.7. *One has $H_K^*(y) = h_K(y)^2/4$, and*

$$H_K(x) + H_K^*(y) \geq \langle x, y \rangle.$$

Equality holds exactly when $y \in \partial H_K(x)$, equivalently $x \in \partial H_K^(y)$.*

Proof. Write $x = ru$ with $r \geq 0$ and $g_K(u) = 1$. For a fixed r , the largest value of $\langle x, y \rangle$ is $rh_K(y)$. Thus

$$H_K^*(y) = \sup_{r \geq 0} (rh_K(y) - r^2) = \frac{1}{4}h_K(y)^2.$$

The inequality follows from the definition of the supremum. Equality says that x maximizes $\langle x, y \rangle - H_K(x)$, which is precisely the subgradient inequality defining $y \in \partial H_K(x)$. Applying conjugacy in the other direction gives the equivalent condition. $\square$

The variational equation. A minimizer satisfies a weak Euler–Lagrange equation. Its useful form is

$$\nu z(t) + b \in \partial H_K^*(-J_0 \dot{z}(t)) \quad \text{for almost every } t, \quad (10)$$

for a constant vector b and a real multiplier ν .

Definition 2.8 (Weak dual critical curve). *A feasible curve satisfying this equation is called weak dual critical.*

The constant b reflects the freedom to translate the curve. Conjugacy rewrites the same equation as $-J_0 \dot{z} \in \partial H_K(\nu z + b)$, bringing it close to the characteristic equation.

To see the form of the variational equation, choose $p(t) \in \partial H_K^*(-J_0 \dot{z}(t))$. The derivative of action in a periodic variation v is $DA(z)[v] = \int \langle -J_0 \dot{z}, v \rangle$. The cited nonsmooth multiplier theorem provides a measurable L^2 choice of p such that, for every smooth periodic variation v ,

$$\int_0^T \langle p, -J_0 \dot{v} \rangle dt = \nu \int_0^T \langle -J_0 \dot{z}, v \rangle dt.$$

The left side pairs the chosen subgradient with the change in velocity; the right side is the variation of the action constraint. Integration by parts gives $-J_0 \dot{p} = -\nu J_0 \dot{z}$ in distributions, hence $\dot{p} = \nu \dot{z}$ and $p = \nu z + b$. The action constraint is regular because $DA(z)[z] = 2T \neq 0$. These are the variational assertions invoked from the cited nonsmooth theorem; the calculation fixes their signs and constants in our convention.

For any weak critical curve, the multiplier is determined by its value. The two-homogeneity of H_K^* gives $\langle p, y \rangle = 2H_K^*(y)$ when $p \in \partial H_K^*(y)$. Pairing (10) with $-J_0 \dot{z}$ and integrating, the constant vector contributes zero because the curve is closed. We obtain

$$2I_K(z) = 2\nu A(z) = 2\nu T, \quad \nu = \frac{I_K(z)}{T} > 0. \quad (11)$$

Positivity follows since a feasible curve has positive action and is nonconstant, while H_K^* is positive away from zero.

Reconstructing a boundary curve. Let z be weak critical and set $x = \nu z + b$ and $y = -J_0 \dot{z}$. Then $y \in \partial H_K(x)$. The convex chain rule gives

$$\frac{d}{dt} H_K(x(t)) = \langle y(t), \dot{x}(t) \rangle = \nu \langle -J_0 \dot{z}(t), \dot{z}(t) \rangle = 0.$$

Thus $H_K(x)$ is constant. Since $y \in \partial H_K(x)$, Fenchel equality gives $H_K(x) + H_K^*(y) = \langle x, y \rangle$. Two-homogeneity gives $\langle x, y \rangle = 2H_K(x)$, so $H_K^*(y) = H_K(x)$ almost everywhere. Integrating the constant value yields

$$I_K(z) = \int_0^T H_K^*(y) dt = TH_K(x), \quad H_K(x) = I_K(z)/T = \nu.$$

Put $c = \sqrt{\nu}$ and define

$$\gamma(t) = cz(t/c^2) + b/c, \quad 0 \leq t \leq c^2 T.$$

Since $\gamma = x(t/c^2)/c$, it satisfies $H_K(\gamma) = 1$. Subgradient homogeneity also gives

$$-J_0\dot{\gamma}(t) = y(t/c^2)/c \in \partial H_K(\gamma(t)).$$

This is the contact-normalized characteristic equation. Its period and action are $c^2T = I_K(z)$. Direct substitution in (8) gives $I_K(\gamma) = I_K(z)$. In particular, when $I_K(z) = T$ we have $c = 1$, so reconstruction only translates the curve.

Conversely, a contact-normalized characteristic satisfies $H_K(\gamma) = 1$ and $-J_0\dot{\gamma} \in \partial H_K(\gamma)$. Fenchel equality therefore gives $H_K^*(-J_0\dot{\gamma}) = 1$ almost everywhere. Consequently $I_K(\gamma) = T = A(\gamma)$, and it is a feasible dual curve. An attained dual minimum reconstructs to a characteristic with the same value; every characteristic supplies a feasible dual value. This proves equality of the minima and, by the least-action characterization, their equality with capacity.

Period convention and attainment. The cited fixed-period formulation uses $w : [0, 1] \rightarrow \mathbb{R}^4$ with $A(w) = 1/2$. If z is feasible in our convention, then

$$w(s) = \frac{z(Ts)}{\sqrt{2T}}, \quad I_K(w) = \frac{1}{2}I_K(z).$$

Conversely, $z(t) = \sqrt{2T}w(t/T)$ is feasible for any chosen $T > 0$ and has $I_K(z) = 2I_K(w)$. If the fixed-period formulation imposes mean zero, subtract the mean of w ; translation changes neither its action nor I_K . Thus the fixed-period attainment theorem applies to our infimum, with its factor of two accounted for. Choosing $T = 2I_K(w)$ gives an attained representative with $I_K(z) = T$.

The distinction needed below is now concrete. Feasibility allows us to compare functional values, but does not place a curve on the boundary. Once a modified curve is shown to minimize I_K , the variational equation and reconstruction justify placing it there. This is why rearranging velocities will be done in the dual problem first.

3 Generalized Reeb orbits on polytopes

On the interior of a polytope facet, the Reeb velocity is constant. At an intersection of facets, the generalized characteristic equation permits convex combinations of their velocities. Haim–Kislev’s theorem nevertheless guarantees a capacity minimizer that uses only the individual facet velocities, each at most once. Its action is therefore determined by a finite ordered list of facets and the time spent in each direction.

We first derive this finite description and distinguish closure of the curve from its placement on the boundary. We then prove the existence theorem by rearranging curves in the dual variational problem.

3.1 Facet velocities

Let

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1, i = 1, \dots, F\}$$

be a full-dimensional polytope with $0 \in \text{int}(K)$. We assume that the halfspace description is irredundant, so the facet corresponding to a_i is

$$F_i = \{x \in K : \langle a_i, x \rangle = 1\}.$$

The normalized halfspace description gives

$$g_K(x) = \max_{1 \leq i \leq F} \langle a_i, x \rangle.$$

If $x \in \partial K$, let $I(x) = \{i : x \in F_i\}$ be its active facets. The subdifferential rules for a finite maximum and for the square on the level set $g_K = 1$ give

$$\partial H_K(x) = 2 \operatorname{conv}\{a_i : i \in I(x)\}. \quad (12)$$

Thus Definition 2.1 becomes a finite differential inclusion. In this normalization the pure contact-normalized Reeb direction on F_i is

$$R_i = 2J_0 a_i. \quad (13)$$

Indeed R_i is tangent to F_i , because $\langle a_i, J_0 a_i \rangle = 0$, and for $x \in F_i$,

$$\lambda_0|_x(R_i) = \frac{1}{2} \langle -J_0 R_i, x \rangle = \langle a_i, x \rangle = 1.$$

For a Lagrangian product, this formula makes the exchange between the two factors explicit. A row $a_i = (u, 0)$ from a facet of the q -factor and a row $a_j = (0, v)$ from a facet of the p -factor give

$$R_i = 2J_0(u, 0) = (0, 2u), \quad R_j = 2J_0(0, v) = (-2v, 0).$$

Thus a segment on a q -facet moves only in the p -coordinates, while a segment on a p -facet moves only in the q -coordinates.

A *generalized Reeb orbit* on K is a contact-normalized generalized characteristic in the sense of Definition 2.1. By (12), this means precisely that $\gamma \in W^{1,2}([0, T], \mathbb{R}^4)$ is closed, has image in ∂K , and satisfies

$$\dot{\gamma}(t) \in \operatorname{conv}\{R_i : \gamma(t) \in F_i\} \quad \text{for a.e. } t \in [0, T]. \quad (14)$$

The general normalization already shows that such an orbit has

$$A(\gamma) = T.$$

We call such an orbit *simple* if, after choosing its starting point at a velocity change, it consists of finitely many segments with velocities R_i , and each R_i occurs in at most one segment. Here simplicity refers to the use of facet directions. In particular, it includes the requirement that the velocities be individual R_i , rather than nontrivial convex combinations of them.

3.2 Words, dwell times, and closure

A simple orbit is described by the ordered facet labels it uses, the time spent in each direction, and a starting point. The *active word* contains only directions used for positive time. A cyclic word

$$\sigma = (\sigma_1, \dots, \sigma_m)$$

with distinct entries, together with dwell times $\tau_k > 0$ and a base point $b = \gamma(0)$, describes the velocity data

$$\dot{\gamma}(t) = R_{\sigma_k} \quad \text{for a time interval of length } \tau_k.$$

Such data can represent a closed orbit only if the closing equation holds:

$$\sum_{k=1}^m \tau_k R_{\sigma_k} = 0. \quad (15)$$

This equation closes the broken path. A further condition is needed to place each segment on its prescribed facet: there must be a suitable starting point b . Lemma 3.2 expresses that condition by linear equalities and inequalities.

For example, closure alone fails to give an orbit in the planar square $[-1, 1]^2$, the right and left facet rows give opposite directions $R_r = (0, 2)$ and $R_\ell = (0, -2)$. Equal dwell times satisfy

$\tau R_r + \tau R_\ell = 0$, but the breakpoint between the two segments would have to lie in the disjoint facets F_r and F_ℓ . No translation can realize this closed two-segment path on the prescribed facets.

The period and action of a closed pure-word orbit are

$$T = \sum_{k=1}^m \tau_k, \quad A(\gamma) = T.$$

Cyclically shifting the facet–duration pairs and moving the starting point to the corresponding breakpoint describes the same orbit. We therefore regard the word as cyclic. A zero duration, when allowed in a limiting argument, is simply deleted from the active word.

The action of a finite word can be calculated directly from its velocities and dwell times.

Lemma 3.1 (Symplectic shoelace formula). *Let $z: [0, T] \rightarrow \mathbb{R}^4$ be a closed piecewise affine curve. Suppose that z has constant velocity w_k on consecutive intervals of length ℓ_k , for $k = 1, \dots, m$. Then*

$$2A(z) = \sum_{1 \leq j < k \leq m} \ell_j \ell_k \omega_0(w_j, w_k). \quad (16)$$

In particular, the action of a closed pure-word Reeb orbit depends only on the cyclic velocity word and the dwell times, not on the base point.

Proof. Write $t_k = \sum_{j < k} \ell_j$. On the k -th interval,

$$z(t) = z(0) + \sum_{j < k} \ell_j w_j + (t - t_k) w_k.$$

The contribution of $z(0)$ to $\int_0^T \langle -J_0 \dot{z}, z \rangle dt$ vanishes because z is closed, so $\sum_k \ell_k w_k = 0$. The self-term $\langle -J_0 w_k, w_k \rangle$ vanishes by skew-symmetry. Hence

$$2A(z) = \sum_k \ell_k \left\langle -J_0 w_k, \sum_{j < k} \ell_j w_j \right\rangle = \sum_{j < k} \ell_j \ell_k \omega_0(w_j, w_k),$$

using $\omega_0(u, v) = \langle J_0 u, v \rangle$. □

Breakpoints of $\dot{\gamma}$ are velocity changes, not necessarily changes in the minimal face containing $\gamma(t)$: an orbit may move inside a nonsmooth face while switching from one active facet direction to another.

Lemma 3.2 (Base-point recovery). *Fix an active word $\sigma = (\sigma_1, \dots, \sigma_m)$ and dwell times $\tau_k > 0$ satisfying the closure equation (15). Put*

$$v_1 = 0, \quad v_k = \sum_{j=1}^{k-1} \tau_j R_{\sigma_j} \quad (k = 2, \dots, m+1).$$

Then $v_{m+1} = 0$. A point $b \in \mathbb{R}^4$ recovers a simple generalized Reeb orbit with these velocity data, by

$$\gamma(t) = b + v_k + s R_{\sigma_k}, \quad 0 \leq s \leq \tau_k$$

on the k -th segment, if and only if

$$\langle a_{\sigma_k}, b + v_k \rangle = 1 \quad \text{for } k = 1, \dots, m, \quad (17)$$

and

$$\langle a_i, b + v_k \rangle \leq 1 \quad \text{for } i = 1, \dots, F \text{ and } k = 1, \dots, m. \quad (18)$$

Equivalently, the equalities have the form

$$N_\sigma b = r_\sigma, \quad (N_\sigma)_{k\bullet} = a_{\sigma_k}^\top, \quad (r_\sigma)_k = 1 - \langle a_{\sigma_k}, v_k \rangle.$$

When this system is nonempty, its solution space is an affine space of dimension $4 - \text{rank } N_\sigma$. The inequalities cut out the valid base points as a convex polytope inside this affine space. In particular, the set of valid base points is bounded, because $v_1 = 0$ and (18) imply $b \in K$.

Proof. On the k -th segment, the prescribed facet equation is

$$\langle a_{\sigma_k}, b + v_k + sR_{\sigma_k} \rangle = 1.$$

The s -term vanishes because $\langle a_{\sigma_k}, R_{\sigma_k} \rangle = 2\langle a_{\sigma_k}, J_0 a_{\sigma_k} \rangle = 0$. Hence the whole segment lies on F_{σ_k} exactly when (17) holds.

The segment lies in K exactly when

$$\langle a_i, b + v_k + sR_{\sigma_k} \rangle \leq 1 \quad \text{for all } i \text{ and all } s \in [0, \tau_k].$$

For fixed i, k this expression is affine in s , so it is enough to check the endpoints. The endpoint at $s = 0$ is $b + v_k$, and the endpoint at $s = \tau_k$ is $b + v_{k+1}$. As k runs cyclically and $v_{m+1} = v_1 = 0$, these endpoint checks are exactly (18). With these equalities and inequalities, the resulting closed broken path lies on ∂K and has velocity R_{σ_k} on the k -th segment, hence is a simple Reeb orbit with the prescribed data. $\square$

Remark 3.3 (Non-uniqueness and computation). *The base point is not part of the finite action data. If $\text{rank } N_\sigma = 4$, the facet equations determine it uniquely when they are consistent. If $\text{rank } N_\sigma < 4$, the inequalities may leave a positive-dimensional set of base points. This non-uniqueness does not affect the action, which is determined by the closed velocity polygon via Lemma 3.1. Computationally, base-point recovery is therefore a finite linear feasibility check after the word and dwell times have been chosen.*

3.3 Existence of simple minimizers

The next theorem is the reason a finite search can compute capacity. It is Haim–Kislev’s simple-minimizer theorem [Hai19, Theorem 1.5], written with the contact-normalized velocities $R_i = 2J_0 a_i$.

Theorem 3.4 (Simple minimizer). *Let $K \subset \mathbb{R}^4$ be a full-dimensional convex polytope with $0 \in \text{int}(K)$. Among all generalized Reeb orbits on ∂K , there exists a minimum-action orbit that is simple in the contact-normalized convention above. Hence*

$$c_{\text{EHZ}}(K) = \min\{A(\gamma) : \gamma \text{ is a simple Reeb orbit on } \partial K\}.$$

The difficulty is that splitting or rearranging velocities need not leave a curve on ∂K . We make those changes to closed curves in the dual problem instead. The changes preserve the total time and can be chosen to increase action. Shortening all durations then restores the constraint $A(z) = T$ and lowers the dual value. Starting near the minimum forces this improvement to tend to zero, and the resulting curves have a convergent subsequence with the desired finite form.

During the proof, a *simple pure-velocity dual loop* means a closed piecewise affine curve using each R_i on at most one cyclic interval. It need not lie on the boundary. The next five lemmas give the approximation, rearrangement and compactness steps.

Lemma 3.5 (Piecewise affine approximation). *Let $z \in W^{1,2}([0, T], \mathbb{R}^4)$ be closed and satisfy*

$$\dot{z}(t) \in \text{conv}\{R_1, \dots, R_F\} \quad \text{for a.e. } t.$$

For every $\varepsilon > 0$, there is a closed piecewise affine curve z' with

$$\dot{z}'(t) \in \text{conv}\{R_1, \dots, R_F\}$$

on each affine piece and $\|z' - z\|_{W^{1,2}} < \varepsilon$. Consequently $A(z') \rightarrow A(z)$ and $I_K(z') \rightarrow I_K(z)$ as $\varepsilon \rightarrow 0$.

Proof. Let $C = \text{conv}\{R_1, \dots, R_F\}$. Averages over intervals of refining uniform partitions converge to $\dot{z}$ in L^2 . Choose a partition $0 = t_0 < \dots < t_m = T$ so that the conditional averages of $\dot{z}$ on the partition intervals are L^2 -close to $\dot{z}$. On $[t_k, t_{k+1}]$, put

$$w_k = \frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} \dot{z}(s) ds.$$

Because C is convex and closed, each w_k lies in C . Define z' by $z'(0) = z(0)$ and $\dot{z}' = w_k$ on $[t_k, t_{k+1}]$. Then

$$z'(T) - z'(0) = \int_0^T \dot{z}'(t) dt = \int_0^T \dot{z}(t) dt = 0,$$

so z' is closed. Since $\dot{z}' \rightarrow \dot{z}$ in L^2 , integration gives $z' \rightarrow z$ uniformly, hence in $W^{1,2}$. The convergence of A follows from bilinearity in $(z, \dot{z})$. Both $-J_0 \dot{z}'$ and $-J_0 \dot{z}$ take values in the compact set $-J_0 C$, on which h_K^2 is Lipschitz; hence the L^2 -convergence also gives $I_K(z') \rightarrow I_K(z)$. $\square$

Lemma 3.6 (Splitting mixed velocities). *Let z be a closed piecewise affine curve of period T whose velocities lie in $\text{conv}\{R_1, \dots, R_F\}$. Then there is a closed piecewise affine curve z' , with the same period, whose velocity on each segment is one pure Reeb vector R_i , such that $A(z') \geq A(z)$ and*

$$I_K(z') = T.$$

Proof. On a segment with velocity $v = \sum_i \theta_i R_i$, $\theta_i \geq 0$ and $\sum_i \theta_i = 1$, replace the segment by subsegments with velocities R_i and lengths θ_i times the original length. The total displacement and period are unchanged. The only change in the action is the sum of the new internal cross-terms from Lemma 3.1. Reversing the order of the inserted subsegments changes the sign of this internal contribution, so one of the two orders does not decrease the action.

For a pure Reeb velocity $R_i = 2J_0 a_i$,

$$-J_0 R_i = 2a_i, \quad h_K(-J_0 R_i) = 2h_K(a_i) = 2,$$

because $F_i \neq \emptyset$ and $\max_{x \in K} \langle a_i, x \rangle = 1$. Thus $h_K^2(-J_0 R_i) = 4$ on every pure segment, and

$$I_K(z') = \frac{1}{4} \int_0^T 4 dt = T.$$

$\square$

Lemma 3.7 (Merging repeated velocities). *Let z be a closed piecewise affine curve whose velocity on each segment is one of the R_i . If some R_i occurs in two non-adjacent segments, then there is a rearrangement z' that merges those two occurrences into one contiguous segment and satisfies*

$$A(z') \geq A(z), \quad I_K(z') = I_K(z).$$

Repeating this operation, and merging adjacent equal-velocity intervals, gives a closed piecewise affine curve in which each pure Reeb vector occurs in at most one interval. Here adjacency is understood cyclically, since the curve is closed.

Proof. Suppose a velocity v occurs in two segments I_r and I_s , with middle segments $M = \{r+1, \dots, s-1\}$. Merging I_s forward into I_r changes only the action terms involving v and the middle velocities, and the change is

$$\Delta A_{\text{fwd}} = |I_s| \sum_{i \in M} |I_i| \omega_0(v, w_i).$$

Merging I_r backward into I_s gives

$$\Delta A_{\text{bwd}} = -|I_r| \sum_{i \in M} |I_i| \omega_0(v, w_i).$$

The two changes have opposite signs, or both vanish, so one merge has nonnegative action change. Closure is preserved because the total duration assigned to each velocity is unchanged, and I_K is unchanged because it depends only on the velocities and their durations. $\square$

Lemma 3.8 (Rescaling). *Let z be a closed piecewise affine curve with pure Reeb velocities and period T . If all durations are multiplied by $\beta > 0$, the resulting curve z' satisfies*

$$A(z') = \beta^2 A(z), \quad I_K(z') = \beta I_K(z).$$

Proof. In Lemma 3.1, every action term contains a product of two durations, so A scales by β^2 . The integrand in I_K is unchanged and the time intervals are all multiplied by β , so I_K scales by β . $\square$

Lemma 3.9 (Compactness of simple pure-velocity dual loops). *Let z_n be simple pure-velocity dual loops whose periods satisfy $T_n \rightarrow T > 0$. After translating the curves and passing to a subsequence, z_n converges to a simple pure-velocity dual loop z_* with period T , and*

$$A(z_*) = \lim_n A(z_n), \quad I_K(z_*) = T.$$

Proof. Choose the initial point of each periodic curve at a breakpoint, and insert zero dwell times for unused facet directions. Each curve is then determined up to translation by a permutation of the finitely many facet directions and a nonnegative dwell-time vector of total mass T_n . Pass first to a subsequence with a fixed permutation. Since $T_n \rightarrow T$, the dwell-time vectors lie in a common compact set and have a convergent subsequence whose entries sum to T . Translating the starting points gives uniform convergence after identifying each domain with $[0, 1]$ by linear time rescaling. Closure passes to the limit because $\sum_i \tau_i R_{\sigma(i)} = 0$ for every curve. Some limiting dwell times may be zero; deleting those zero-duration entries recovers the active word of the limiting simple curve. The action is continuous in the dwell-time data by Lemma 3.1. Since the remaining positive-duration segments still have pure Reeb velocities, the same computation as in Lemma 3.6 gives $I_K(z_*) = T$. $\square$

Proof of Theorem 3.4. Choose a capacity-realizing generalized Reeb orbit γ . By Theorem 2.6, γ is also a dual minimizer in the thesis convention, with

$$I_K(\gamma) = A(\gamma) = T = c_{\text{EHZ}}(K).$$

Moreover $\dot{\gamma}(t) \in \text{conv}\{R_1, \dots, R_F\}$ for almost every t .

Choose, by Lemma 3.5, a sequence of closed piecewise affine curves $z_{1,n} \rightarrow \gamma$ in $W^{1,2}$, with velocities in the convex hull of the R_i . Apply Lemma 3.6 to obtain closed pure-velocity curves $z_{2,n}$, and then Lemma 3.7 to obtain closed simple pure-velocity curves $z_{3,n}$, such that

$$A(z_{3,n}) \geq A(z_{2,n}) \geq A(z_{1,n}), \quad I_K(z_{3,n}) = T.$$

Since $A(z_{1,n}) \rightarrow A(\gamma) = T$, discard finitely many terms so that $A(z_{3,n}) > 0$. Rescale all durations of $z_{3,n}$ by

$$\beta_n = \frac{T}{A(z_{3,n})},$$

and denote the resulting curve by $z_{4,n}$. It satisfies

$$A(z_{4,n}) = \frac{T^2}{A(z_{3,n})}, \quad I_K(z_{4,n}) = \frac{T^2}{A(z_{3,n})}.$$

Thus $z_{4,n}$ is feasible for the dual problem, with period $T^2/A(z_{3,n})$. Since γ is a dual minimizer, $I_K(z_{4,n}) \geq T$, and therefore $A(z_{3,n}) \leq T$. Together with $A(z_{3,n}) \geq A(z_{1,n}) \rightarrow T$, this gives

$$A(z_{3,n}) \rightarrow T, \quad \beta_n \rightarrow 1, \quad \text{period}(z_{4,n}) \rightarrow T.$$

By Lemma 3.9, a subsequence of the $z_{4,n}$ converges up to translation to a simple pure-velocity dual loop z_* with $A(z_*) = T$ and $I_K(z_*) = T$. This curve is a dual minimizer. Applying Theorem 2.6 again gives a contact-normalized generalized Reeb orbit on ∂K in the same translate/rescale family. In the reconstruction part of that theorem, the corresponding characteristic has the form

$$t \mapsto c z_*(t/c^2) + b/c$$

for some $c > 0$, with

$$c^2 = \nu = \frac{I_K(z_*)}{T} = 1$$

by (11). Thus the reconstruction only translates z_* . Its velocities remain the pure facet directions R_i , each used in one connected interval, so the resulting orbit is simple and has minimum action. The translation also recovers the prescribed facet labels. Indeed, on an interval with $\dot{\gamma} = R_i$, the reconstructed inclusion gives

$$2a_i = -J_0 R_i \in 2 \operatorname{conv}\{a_j : \gamma(t) \in F_j\}.$$

Because the irredundant row a_i is an extreme point of K° , this forces F_i to be active along that interval. The reconstructed curve lies in K , so its starting point $\gamma(0)$ is a valid base point in Lemma 3.2; closure supplies the remaining displacement equation. $\square$

The theorem supplies one simple capacity minimizer, which is enough to compute the capacity by a finite optimization. It does not require every minimizer to be simple. The next section expresses the action in terms of the word and normalized dwell times, obtaining Haim–Kislev’s quadratic program.

4 The Haim–Kislev Quadratic Program

The preceding section showed that some capacity-minimizing generalized Reeb orbit uses a finite list of distinct facet directions. We now express its action in terms of that list and its dwell times. Haim–Kislev’s formula turns the result into a finite quadratic program [Hai19].

The chapter has two tasks. First, it explains the variables and how to compute the global optimum by solving linear stationarity systems and comparing their values. Second, it proves reductions for Lagrangian products of planar polygons. Those reductions make the later family calculations practical.

4.1 The finite quadratic program

Let

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1, i = 1, \dots, F\}$$

be a full-dimensional polytope with $0 \in \text{int}(K)$, written as in Section 3. Thus $a_i = n_i/h_i$ if n_i is the outward unit normal to the i -th facet and $h_i = h_K(n_i) > 0$ is its height. Haim–Kislev state their formula in the (n_i, h_i) -variables. We use the equivalent dual-vertex variables

$$\beta_i^{\text{here}} = \beta_i^{\text{HK}} h_i.$$

For a contact-normalized orbit represented by a pure facet word, β_i is the fraction of its period spent with velocity $2J_0 a_i$. The weights therefore sum to one. Closure of the curve says that the weighted sum of its velocities vanishes; since J_0 is invertible, this is equivalent to $\sum_i \beta_i a_i = 0$. Thus the constraints are

$$\beta_i \geq 0, \quad \sum_i \beta_i = 1, \quad \sum_i \beta_i a_i = 0.$$

For an ordered facet word $\sigma = (\sigma_1, \dots, \sigma_m)$ with distinct entries, define

$$M_\sigma = \left\{ \beta \in \mathbb{R}_{\geq 0}^m : \sum_{k=1}^m \beta_k = 1, \sum_{k=1}^m \beta_k a_{\sigma_k} = 0 \right\}.$$

The quadratic objective for this active word is

$$Q_\sigma(\beta) = \sum_{1 \leq j < k \leq m} \beta_j \beta_k \omega_0(a_{\sigma_j}, a_{\sigma_k}). \quad (19)$$

Here σ is written in the Reeb traversal order used in Section 3. Haim–Kislev’s displayed formula uses the reversed ordered word after translating their symplectic convention to ω_0 . Since word reversal is a bijection on the ordered candidate family, the maximum is unchanged; the convention matters only when a single fixed word is interpreted.

Let

$$Q_{\max} = \max_{\sigma} \max_{\beta \in M_\sigma} Q_\sigma(\beta),$$

where σ ranges over ordered words with distinct facets and $M_\sigma \neq \emptyset$. Equivalently, one may range over all permutations of all facets and allow zero weights; the zero-weight entries are then deleted to recover the active word. This is only an enumeration convention: HK’s theorem is the full-permutation statement, and the active-word form is obtained by deleting unused zero-weight facets and reindexing the remaining entries. This formulation intentionally permits duplicates of the same geometric orbit, for example from changing the starting segment.

Theorem 4.1 (Haim–Kislev quadratic program). *For a full-dimensional convex polytope $K \subset \mathbb{R}^4$ with $0 \in \text{int}(K)$, written in the facet normalization above, $Q_{\max} > 0$ and*

$$c_{\text{EHZ}}(K) = \frac{1}{2Q_{\max}}.$$

Proof. Write $c = c_{\text{EHZ}}(K)$. We derive the formula from the dual principle and the simple-minimizer theorem proved above.

First let $\beta \in M_\sigma$ have $q = Q_\sigma(\beta) > 0$, and set $T = 1/(2q)$. Traverse the velocities $R_{\sigma_k} = 2J_0 a_{\sigma_k}$ for times $T\beta_k$. The resulting curve z is closed because $\sum_k \beta_k a_{\sigma_k} = 0$. By Lemma 3.1,

$$2A(z) = 4T^2 Q_\sigma(\beta) = 2T.$$

Moreover, $h_K(a_i) = 1$ for each facet row, so $H_K^*(-J_0 R_i) = h_K(2a_i)^2/4 = 1$. Hence $I_K(z) = T = A(z)$: this curve is feasible for the dual problem (9). The dual action principle, Theorem 2.6, gives

$$c \leq I_K(z) = \frac{1}{2q}, \quad q \leq \frac{1}{2c}.$$

Nonpositive feasible values also satisfy this last inequality, since $c > 0$. Thus $Q_{\max} \leq 1/(2c)$.

For the reverse inequality, choose the simple minimum-action orbit supplied by Theorem 3.4. Its period is c . If its distinct facet word is σ and its dwell times are $\tau_k > 0$, put $\beta_k = \tau_k/c$. Closure and the period identity show that $\beta \in M_\sigma$. The same shoelace formula now gives

$$2c = 4c^2 Q_\sigma(\beta), \quad Q_\sigma(\beta) = \frac{1}{2c} > 0.$$

Consequently $Q_{\max} \geq 1/(2c)$, proving both the formula and positivity. The maximum exists: there are finitely many words, each M_σ is compact, and the simple minimizer supplies a nonempty feasible set. $\square$

This is Haim–Kislev’s formula [Hai19, Theorem 1.1], with the change of weights and word convention described above. The derivation uses the dual action principle and Haim–Kislev’s simple-minimizer theorem as its foundations. The next proposition identifies when the dual curve constructed in the proof can be placed on the boundary.

Proposition 4.2 (From finite weights to a dual curve). *Let $\beta \in M_\sigma$ and put $q = Q_\sigma(\beta)$. If $q > 0$, define*

$$T = \frac{1}{2q}, \quad \tau_k = T\beta_k.$$

Traversing the velocities R_{σ_k} for the durations τ_k , and deleting zero-duration pieces, gives a closed simple dual curve z with

$$A(z) = I_K(z) = T.$$

If (σ, β) attains $Q_{\max}$ across all words, then $T = c_{\text{EHZ}}(K)$, and a translation of z is a minimum-action simple generalized Reeb orbit on ∂K , with the active word obtained from σ by deleting its zero-weight entries.

Conversely, if a simple generalized Reeb orbit has period T , active word σ , and positive dwell times τ_k , then $\beta_k = \tau_k/T$ lies in M_σ and $Q_\sigma(\beta) = 1/(2T)$.

Proof. The closure constraint gives

$$\sum_k \tau_k R_{\sigma_k} = 2T J_0 \sum_k \beta_k a_{\sigma_k} = 0,$$

while $\sum_k \tau_k = T$. The shoelace formula gives

$$2A(z) = 4T^2 Q_\sigma(\beta) = 2T.$$

Every positive-duration piece has velocity R_i , for which

$$H_K^*(-J_0 R_i) = \frac{1}{4} h_K^2(2a_i) = 1.$$

Hence $I_K(z) = T$, and z is feasible for the free-period dual problem.

If (σ, β) attains $Q_{\max}$, the Haim–Kislev formula gives $T = 1/(2Q_{\max}) = c_{\text{EHZ}}(K)$. Thus z is a dual minimizer. Its multiplier in (11) is $\nu = I_K(z)/T = 1$. The reconstruction in Theorem 2.6 therefore has scale one: it only translates z onto ∂K . Translation preserves the ordered pure-velocity pieces. On a piece with velocity R_i , the reconstructed inclusion is

$$2a_i = -J_0 R_i \in \partial H_K(\gamma) = 2 \operatorname{conv}\{a_j : F_j \text{ is active at } \gamma\}.$$

The irredundant rows a_i are extreme points of K° , so this convex combination forces F_i to be active there. Thus, after zero durations are deleted, the resulting minimum-action orbit is simple and has the reduced active word.

For the converse, closure of the orbit gives $\sum_k \beta_k a_{\sigma_k} = 0$, and positivity and $\sum_k \beta_k = 1$ are immediate. Since $A(\gamma) = T$, the shoelace formula yields

$$2T = 4T^2 \sum_{1 \leq r < s \leq m} \frac{\tau_r}{T} \frac{\tau_s}{T} \omega_0(a_{\sigma_r}, a_{\sigma_s}) = 4T^2 Q_\sigma(\beta), \quad \text{so} \quad Q_\sigma(\beta) = \frac{1}{2T}.$$

□

4.2 Solving one word and comparing the results

For a fixed word, the constraints are linear and the objective is quadratic. Write

$$C_\sigma = \begin{pmatrix} a_{\sigma_1} & \cdots & a_{\sigma_m} \\ 1 & \cdots & 1 \end{pmatrix}, \quad e = (0, 0, 0, 0, 1)^T, \quad Q_\sigma(\beta) = \frac{1}{2} \beta^T H_\sigma \beta.$$

Here H_σ is symmetric with zero diagonal and, for $j < k$, $(H_\sigma)_{jk} = (H_\sigma)_{kj} = \omega_0(a_{\sigma_j}, a_{\sigma_k})$. Thus the problem is to maximize this quadratic form subject to $C_\sigma \beta = e$ and $\beta \geq 0$.

First suppose that the maximizing weights are all positive. Small perturbations in $\ker C_\sigma$ remain feasible, so the first derivative must vanish on that kernel. Equivalently, there is a vector of multipliers λ such that

$$H_\sigma \beta = C_\sigma^T \lambda, \quad C_\sigma \beta = e.$$

These are the equality-constrained Karush–Kuhn–Tucker (KKT) equations:

$$\begin{pmatrix} H_\sigma & -C_\sigma^T \\ C_\sigma & 0 \end{pmatrix} \begin{pmatrix} \beta \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ e \end{pmatrix}. \quad (20)$$

They are a linear system even though the objective is quadratic. After solving it, one checks whether the weights are positive and evaluates Q . A stationary point can be a minimum or a saddle, so this calculation does not yet identify the maximum.

Example 4.3 (A four-facet calculation). Take $K = [-1, 1]^4$ and the ordered facet rows $(e_1, e_3, -e_1, -e_3)$, where the coordinate order is (q_1, q_2, p_1, p_2) . Closure forces the weights to have the form (x, y, x, y) , and normalization gives $x + y = 1/2$. The ordered symplectic pairings give

$$Q = 2xy = 2x(1/2 - x), \quad 0 \leq x \leq 1/2.$$

Thus stationarity reduces to $1 - 4x = 0$. The second derivative is -4 , so $x = y = 1/4$ maximizes this word, with $Q = 1/8$; both boundary choices have value zero. The corresponding capacity upper bound is 4. Identifying the capacity still requires a global comparison or another argument giving the matching lower bound.

If a maximum has zero weights, delete those entries from the word. The remaining weights are positive, and deleting zeros changes neither closure nor the objective. It follows that solving all active words covers maxima on every face of the nonnegative feasible set. Conversely, a shorter word can be padded with zero weights to recover a full-facet permutation. Comparing the feasible stationary values over this complete family therefore gives the global maximum in Haim–Kislev’s formula.

A singular system requires care, but not a search over infinitely many objective values. Suppose (β, λ) and $(\beta + z, \lambda + \mu)$ solve (20). Then $C_\sigma z = 0$ and $H_\sigma z = C_\sigma^T \mu$, whence

$$Q_\sigma(\beta + z) - Q_\sigma(\beta) = z^T H_\sigma \beta + \frac{1}{2} z^T H_\sigma z = \lambda^T C_\sigma z + \frac{1}{2} \mu^T C_\sigma z = 0.$$

Thus all solutions of the same stationary system have the same quadratic value. One must still decide whether that affine solution set contains positive weights. A singular matrix by itself means neither infeasibility nor failure of the capacity formula. This observation will also be used in the pentagon classification.

A feasible positive value q gives a dual curve with value $1/(2q)$. Capacity is the infimum of the dual values, so $c_{\text{EHZ}}(K) \leq 1/(2q)$. Once q is proved to be the largest value across all words, that curve is a dual minimizer and can be translated onto the boundary. For a nonglobal candidate, boundary placement instead requires the conditions in Lemma 3.2.

The full enumeration can be reduced when a geometric argument proves that a smaller family still contains a capacity minimizer. For example, an actual transition from facet F_i to F_j requires an intersection of those facets and the directed sign $\omega_0(a_i, a_j) \geq 0$. These necessary conditions can exclude words before solving their systems. They need not hold for every feasible dual curve; what makes them valid for computing capacity is the existence of a boundary-realized global minimizer. A stronger reduction for Lagrangian products is proved next.

4.3 Finite enumeration for Lagrangian products

Let $K_q, K_p \subset \mathbb{R}^2$ be nondegenerate convex polygons containing the origin in their interiors, and let

$$K = K_q \times_L K_p \subset \mathbb{R}_q^2 \oplus \mathbb{R}_p^2$$

be their Lagrangian product. A facet of K is either a q -facet, coming from an edge of K_q , or a p -facet, coming from an edge of K_p . We call a single facet a one-facet block, and an ordered adjacent pair of facets of the same type a two-facet block.

Theorem 4.4 (Finite enumeration for Lagrangian products). *For a Lagrangian product $K = K_q \times_L K_p$ of nondegenerate planar convex polygons containing the origin in their interiors, $c_{\text{EHZ}}(K)$ is obtained by maximizing Q_σ over the Haim–Kislev quadratic programs associated to cyclic words formed by k alternating q -blocks and p -blocks, where $k \in \{2, 3\}$. Each block contains one or two facets from its factor, distinct blocks use disjoint facets, and a two-facet block must use adjacent polygon edges. Thus every word in the family uses at most twelve facets. This is an exhaustive candidate family for the capacity value; it may contain overcounting or infeasible words, and it is not a classification of all minimum-action simple Reeb orbits.*

Proof. Put $T = c_{\text{EHZ}}(K)$. Rudolf’s nonsmooth billiard theorem gives a minimizing Minkowski billiard in K_q , together with its polygonal dual in K_p , having at most three bounce points and length T [Rud24, Theorem 1]. Write its vertices as q_j and its dual vertices as p_j , with indices understood cyclically. In the notation needed here, write

$$N_C(x) = \{u : \langle u, y - x \rangle \leq 0 \text{ for every } y \in C\}$$

for the outward normal cone. Rudolf calls this a strong billiard because the paired polygons satisfy the following normal-cone relations:

$$q_{j+1} - q_j \in N_{K_p}(p_j), \quad p_{j+1} - p_j \in -N_{K_q}(q_{j+1}),$$

Its length is

$$\ell_{K_p}(q) = \sum_j h_{K_p}(q_{j+1} - q_j) = \sum_j \langle p_j, q_{j+1} - q_j \rangle.$$

The last equality follows from the first normal-cone relation and $p_j \in \partial K_p$. These relations say precisely that the alternating broken line through

$$(q_j, p_j), (q_{j+1}, p_j), (q_{j+1}, p_{j+1})$$

is a generalized characteristic for Rudolf's complex-structure convention. That convention is opposite to ours, so reverse the loop. Since λ_0 differs from $-p dq$ by an exact form, its action in our convention is

$$-\oint_{\text{rev}} p dq = \oint_{\text{orig}} p dq = \sum_j \langle p_j, q_{j+1} - q_j \rangle = \ell_{K_p}(q) = T.$$

Thus the reversed lift is a capacity-realizing generalized characteristic on $\partial(K_q \times_L K_p)$ whose legs alternately move only one factor. Parametrize it in the contact normalization. Its period is then T , and, as a dual curve z , it satisfies

$$A(z) = I_K(z) = T.$$

The following surgery is performed on z as a competitor in the dual variational problem, not as an orbit already constrained to ∂K . In particular, splitting a normal-cone direction into pure facet directions does not assert that the corresponding broken path, drawn directly in K_q or K_p , would remain inside that polygon. Boundary placement is recovered only after the surgery has been shown to preserve dual minimality.

Apply the splitting–merging–rescaling part of the proof of Theorem 3.4 to z . Approximation and compactness are unnecessary here because the billiard lift is already a finite broken line. The only extra point to retain from the billiard is its block bound. On each leg, the polygonal normal cone is generated by one facet normal or by the two adjacent normals at a vertex, so splitting replaces that leg by one or two pure directions inside the same block. When a repeated direction is merged, it is removed from one same-type block and absorbed into another block that already contains it. Thus merging cannot increase the number of blocks of either factor, although it may decrease them. If the source block is a singleton, deleting it coalesces its two neighboring opposite-type blocks and reduces both block counts by one. This possible loss is harmless: only the inherited upper bound of three blocks of each type is needed below.

The rescaling argument in the cited proof uses dual minimality to show that the resulting distinct-facet curve is again a dual minimizer. Indeed, write $A' \geq T$ for its action before rescaling; its dual value is still T . Multiplying every duration by T/A' gives a dual-feasible curve with

$$A = I_K = \frac{T^2}{A'}.$$

The dual minimum is T , so $T^2/A' \geq T$, hence $A' \leq T$. Therefore $A' = T$: no rescaling is needed, and the distinct-facet curve z' is a dual minimizer with $A(z') = I_K(z') = T$. By the preceding observations, its cyclic word still has at most three blocks of each type.

Only now do we return to the primal problem. Normalize the positive dwell times of z' by their sum. The resulting finite weights have QP value $1/(2T) = Q_{\max}$, so Proposition 4.2 translates z' to a simple generalized Reeb orbit with the same active word on $\partial(K_q \times_L K_p)$. Its factor coordinates therefore remain in K_q and K_p ; this containment comes from dual-minimizer reconstruction, not from the splitting operation. A simple orbit cannot contain three consecutive facets of the same type. Indeed, along three consecutive q -facet segments the q -coordinate is fixed; the two breakpoints would place it at both endpoints of the middle edge of K_q , a contradiction. The same holds for p -facets. Hence every maximal block in the final word has length one or two, and a two-facet block uses adjacent edges.

If only one factor type occurred, the whole word would be a single block of length at most two, whose positive closure equation would require one outward normal, or a positive combination of two adjacent outward normals, to vanish. This is impossible. Thus both types occur. Their maximal blocks alternate cyclically, so there is a common number $k \leq 3$ of q - and p -blocks. One q -block and one p -block cannot close, by the same argument in each factor, so $k \neq 1$. Hence $k \in \{2, 3\}$. Since each block contains at most two facets, the resulting simple

word uses at most $4k \leq 12$ distinct facets. This proves that the restricted family contains a capacity maximizer. It does not exclude other minimum-action orbits with more blocks. $\square$

Algorithm 4.5 (Lagrangian-product capacity enumeration). *For each factor, enumerate all one-facet blocks and all ordered adjacent two-facet blocks. For $k = 2, 3$, choose k non-overlapping q -blocks and k non-overlapping p -blocks, order them cyclically in the alternating pattern q -block, p -block, q -block, p -block, and so on. Read the facet labels in that order, discard infeasible words, and solve the corresponding Haim–Kislev quadratic program. The capacity is $1/(2Q_{\max})$, where $Q_{\max}$ is the largest objective value among the feasible words in this candidate family.*

4.4 A six-facet reduction for the product capacity

Algorithms for polygonal Lagrangian products have also been developed through Minkowski billiards. Krupp and Rudolf enumerate face choices for two- and three-bounce trajectories and solve the resulting geometric problems using linear programming [KR22, Section 7.3]. The reduction below works instead with the balanced facet weights in the Haim–Kislev formula: it replaces the continuous weight search by pairs of vertices of two closure polytopes.

The product structure gives a reduction that is unavailable for a general polytope. The q - and p -coordinate planes are Lagrangian, so pairings between two q -facets or two p -facets vanish. After separating the total weight in each factor from its normalized distribution, the objective is therefore linear in either distribution when the other is fixed. Maximizing these two linear functions at vertices is the mechanism behind both the six-facet bound and the following algorithm.

Let $u_1, \dots, u_r$ be the planar dual vertices of K_q , and let $v_1, \dots, v_s$ be those of K_p . Define the normalized closure polytopes

$$C_q = \left\{ \alpha \in \mathbb{R}_{\geq 0}^r : \sum_i \alpha_i = 1, \sum_i \alpha_i u_i = 0 \right\},$$

$$C_p = \left\{ \gamma \in \mathbb{R}_{\geq 0}^s : \sum_j \gamma_j = 1, \sum_j \gamma_j v_j = 0 \right\}.$$

Theorem 4.6 (Six-facet product maximizer). *For a Lagrangian product $K_q \times_L K_p$ of nondegenerate planar convex polygons containing the origin in their interiors, the complete Haim–Kislev family has a global maximizer using at most three q -facets and at most three p -facets. The total weight in each factor is $1/2$. Consequently the capacity is represented by a word of length at most six.*

Algorithm 4.7 (Product capacity from closure vertices). *Enumerate the vertices of C_q and C_p . For each pair, enumerate the cyclic orders of the union of their supports, give each factor total weight $1/2$, and evaluate Q_σ . The largest value is $Q_{\max}$. Each support union has at most six facets, so fixing one cyclic starting point leaves at most $5! = 120$ orders.*

Proof. Choose a global maximizer and let I_q and I_p be its q - and p -positions. Its objective is positive, hence both factor weights are nonzero. Put

$$t = \sum_{k \in I_q} \beta_k, \quad \alpha_k = \frac{\beta_k}{t} \ (k \in I_q), \quad \gamma_k = \frac{\beta_k}{1-t} \ (k \in I_p).$$

Closure splits between the factors, so the zero-extended vectors α and γ lie in the faces

$$C_q(\sigma) = \{ \alpha \in C_q : \alpha_i = 0 \text{ if } q\text{-facet } i \text{ is omitted from } \sigma \},$$

an arbitrary global maximizer (σ, β)
 $\downarrow$ fix γ , maximize the linear function in α
 α is a vertex of C_q , hence $|\text{supp } \alpha| \leq 3$
 $\downarrow$ fix α , maximize the linear function in γ
 γ is a vertex of C_p , hence $|\text{supp } \gamma| \leq 3$
 $\downarrow$ maximize $t(1-t)$ and delete zero weights
 a global maximizer with $t = \frac{1}{2}$ and at most six facets

Figure 3: The proof mechanism for the six-facet reduction. Linearity in each factor distribution is the product-specific step.

and likewise $C_p(\sigma) \subset C_p$. Since the symplectic form vanishes on each factor, there is a bilinear function

$$B_\sigma : C_q(\sigma) \times C_p(\sigma) \longrightarrow \mathbb{R}$$

such that

$$Q_\sigma(\beta) = t(1-t)B_\sigma(\alpha, \gamma). \quad (21)$$

Its coefficients are the signed pairings $\langle u_i, v_j \rangle$, with signs determined by the order in σ .

With γ fixed, the bilinear factor is linear in α , so it has a maximizer at a vertex of $C_q(\sigma)$. Replacing α by that vertex preserves global maximality. Fixing the new α and repeating the argument in $C_p(\sigma)$ gives a global maximizer for which both normalized factor weights are vertices of the support-restricted closure polytopes. Coordinate restrictions define faces of C_q and C_p , and a vertex of a face is a vertex of the ambient polytope.

A vertex of

$$\left\{ x \geq 0 : \sum_i x_i = 1, \sum_i x_i w_i = 0 \right\}, \quad w_i \in \mathbb{R}^2,$$

has at most three positive coordinates. Indeed, more than three positive columns in the three scalar equality constraints have a nontrivial linear dependence, which gives a feasible line through the point and contradicts extremality. Thus each factor uses at most three facets.

For the selected vertices the bilinear value is positive. Varying only the total factor weight gives

$$\tau(1-\tau)B_\sigma(\alpha, \gamma), \quad 0 \leq \tau \leq 1,$$

which is maximal at $\tau = 1/2$. Delete the remaining zero-weight positions. The relative cyclic order and objective are unchanged, and the complete Haim–Kislev family contains the resulting word of length at most six.

The same argument proves Algorithm 4.7: the selected vertices occur in the enumeration, while every enumerated candidate is feasible for the complete Haim–Kislev family and therefore cannot exceed $Q_{\max}$. $\square$

Remark 4.8 (Why the reduction is product-specific). *The argument does not extend directly to a general polytope. It uses that each factor is Lagrangian: same-factor pairings vanish, leaving a bilinear function which is linear in one normalized closure distribution after the other is fixed. Without such a splitting, the objective remains quadratic in the full weight vector, and moving to a vertex of the closure polytope need not preserve or improve it.*

The theorem asserts the existence of one sparse capacity maximizer. It neither classifies all minimizing orbits nor says that every maximizer has support at most six. The preceding twelve-facet block family remains useful when an argument must compare a specified family of geometric branches. No genericity assumption is needed: the closure polytopes are closed, and degeneracy can only lower the rank used in the support bound.

4.5 Use in later certificates

The later proofs use these results in two ways. The HKO local-maximum proof uses selected active words to build smooth upper bounds. The rotated-pentagon proof compares fixed feasible choices at different angles using the quadratic formula directly.

The numerical implementation follows the same distinction between general polytopes and products. In the general case it examines the candidate words and their stationary points, using proved exclusion criteria to reduce the search. It returns an interval containing the capacity. For a recognized Lagrangian product it instead enumerates the closure-polytope vertices from Algorithm 4.7 and compares the resulting objective values. This route returns the exact capacity of the supplied input, whose floating-point coordinates are treated as exact rational numbers. Both routes use interval bounds to control numerical error and exact arithmetic when a comparison cannot otherwise be decided. Section 13 describes those checks. The HKO proof uses a separate SageMath program over its number field.

The QP is therefore the appropriate language when the global objective value, its competing branches, or inequalities between them matter. The next section keeps the same simple facet words but asks a different geometric question: it propagates actual breakpoints through facet-pair sections and detects a closed orbit as a fixed point of an affine return map. This second route is conditional on explicit regularity assumptions, but it exposes the trajectory geometry and provides an independent implementation comparison.

5 Computing closed trajectories with a flow graph

The quadratic program computes capacity by comparing possible facet orders and weights. A second approach follows trajectories through the facets themselves. In the relative interior of a facet, the contact-normalized velocity is constant. Following a chosen facet direction carries a point on an incoming two-dimensional section to an outgoing section by an affine map. Composing these maps around a facet word produces a return map; a fixed point gives a closed trajectory.

We use the local passage geometry of Chaidez and Hutchings [CH21]. The exhaustive search below is justified by the simple-minimizer theorem: some capacity-minimizing trajectory uses each of its facet directions once. It is therefore enough to search finitely many cyclic words, retaining the conditions that make each passage possible. We state the regularity hypotheses before the correctness theorem and discuss the rational implementation after its proof.

5.1 Affine tubes

Let

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1, \ i = 1, \dots, F\}$$

be a bounded full-dimensional polytope with $0 \in \text{int } K$, and assume that all listed inequalities define genuine facets F_i . Put

$$R_i = 2J_0 a_i.$$

As in Section 3, a trajectory on F_i with velocity R_i has action equal to its elapsed time.

For each ordered pair (i, j) for which a_i, a_j are linearly independent, choose affine coordinates

$$\psi_{ij} : \mathbb{R}^2 \longrightarrow H_i \cap H_j, \quad H_i = \{x : \langle a_i, x \rangle = 1\},$$

and let

$$P_{ij} = \{u \in \mathbb{R}^2 : \psi_{ij}(u) \in K\}.$$

The set P_{ij} is allowed to be empty or lower-dimensional. Thus the construction also permits a breakpoint to lie on additional facets; it does not assume that every breakpoint lies in the relative interior of a two-face.

Consider a passage from facet F_p along F_i to F_n , with three distinct facet labels. We consider the case where the incoming direction approaches H_i and the direction along F_i approaches H_n :

$$\langle R_p, a_i \rangle > 0, \quad \langle R_i, a_n \rangle > 0.$$

For $x \in F_p \cap F_i$, the time at which the line $x + tR_i$ reaches H_n is

$$\tau_{pin}(x) = \frac{1 - \langle a_n, x \rangle}{\langle a_n, R_i \rangle}. \quad (22)$$

The denominator is positive. Restrict the starting section to those points for which $x \in K$, $x + \tau_{pin}(x)R_i \in K$, and $\tau_{pin}(x) \geq 0$. On this closed polygonal domain, the endpoint map

$$\varphi_{pin}(x) = x + \tau_{pin}(x)R_i$$

and the action contribution τ_{pin} are affine. Convexity of K ensures that the whole segment lies in $K \cap H_i$.

We describe a tube by a closed starting polygon P , an affine endpoint map Φ , and an affine total-action function A . If (P_1, Φ_1, A_1) and (P_2, Φ_2, A_2) are consecutive tubes in matching coordinates, their concatenation is

$$(P_1 \cap \Phi_1^{-1}(P_2), \Phi_2 \circ \Phi_1, A_1 + A_2 \circ \Phi_1). \quad (23)$$

Thus an empty intermediate intersection excludes every longer word containing that partial tube.

A *simple cyclic facet word* is a cyclic word $w = [s_1, \dots, s_k]$ with distinct entries. To close it, concatenate the primitive tubes for

$$(s_1, \dots, s_k, s_1, s_2).$$

Write the resulting data as (P_w, Φ_w, A_w) . A point $u \in P_w$ is a closed search candidate precisely when $\Phi_w(u) = u$. It represents the displayed word as a strict generalized Reeb orbit precisely when every reconstructed segment time is positive. In that case its action is $A_w(u)$.

Lemma 5.1 (Tube semantics). *For every simple cyclic word whose primitive denominators are positive, the fixed points of Φ_w in P_w are exactly its closed nonnegative-time tube realizations. The fixed points with all segment times positive are exactly the strict closed generalized Reeb orbits represented by w , and their action is A_w .*

Proof. Equation (22) gives the unique next breakpoint and elapsed time for one passage. The endpoint and the elapsed time are affine, and the domain inequalities state exactly that the endpoints and the passage belong to K . Formula (23) says that two passages can be concatenated exactly when their common breakpoint agrees; it also composes their endpoint maps and adds their elapsed times. Induction over the word proves the assertion for the closed tube. Its starting and ending sections have the same coordinates, so closure is equivalent to $\Phi_w(u) = u$. Finally, the closed tube was defined with nonnegative times, whereas the active word records exactly the positive-duration pieces. This gives the strict-time condition. Since $\lambda_0(R_i) = 1$ on F_i , the sum of the elapsed times is the Reeb action. $\square$

Example 5.2 (A return map on a two-dimensional section). *Figure 4 follows the word $w = [1, 2, 4, 5, 3]$ on a six-facet polytope. For this example we number the facets from 0 to 5, as in the figure. The normalized facet rows, rounded here to four decimal places, are*

$$\begin{pmatrix} -0.2091 & -0.8088 & 1.6079 & -0.0729 \\ 0.2405 & 0.5222 & 0.6397 & -0.2308 \\ -0.8794 & -0.3786 & 0.2356 & 0.0382 \\ 0.2108 & -0.2660 & 0.1605 & -0.3711 \\ -0.3527 & 0.1313 & -0.4414 & 0.0589 \\ -0.1226 & 0.1707 & -0.1480 & 0.4331 \end{pmatrix}.$$

The exact dyadic rows and all rational calculations for this example are provided with the flow-graph data described in Section 14. The rounded numbers below illustrate the calculation; the exact values decide membership and closure.

Start on $F_1 \cap F_2$, immediately before the passage along F_2 . A convenient affine chart uses the last two ambient coordinates:

$$\psi_{12}(u, v) \simeq \begin{pmatrix} -2.446722 \\ 3.041760 \\ 0 \\ 0 \end{pmatrix} + u \begin{pmatrix} 0.991996 \\ -1.681765 \\ 1 \\ 0 \end{pmatrix} + v \begin{pmatrix} -0.183156 \\ 0.526320 \\ 0 \\ 1 \end{pmatrix}.$$

The five passages take this section successively to $F_2 \cap F_4$, $F_4 \cap F_5$, $F_5 \cap F_3$, $F_3 \cap F_1$, and back to $F_1 \cap F_2$. Composing their affine maps gives

$$\Phi_w(u, v) \simeq \begin{pmatrix} -1.601002 & 1.364891 \\ 4.482887 & -4.446374 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} + \begin{pmatrix} 1.626615 \\ -1.821973 \end{pmatrix},$$

$$A_w(u, v) \simeq 36.063319 + 6.721729u - 6.397030v.$$

The fixed point is $(u, v) \simeq (0.791856, 0.317243)$. It belongs to the closed tube, and its five passage times are approximately

$$0.538394, \quad 5.134160, \quad 14.681452, \quad 16.367140, \quad 2.635397.$$

They are all positive and sum to the orbit's action, approximately 39.356542. Thus the fixed-point calculation produces a closed orbit. Finding the capacity still requires comparison with the other words.

Closed tube for $\sigma = (1, 2, 4, 5, 3)$

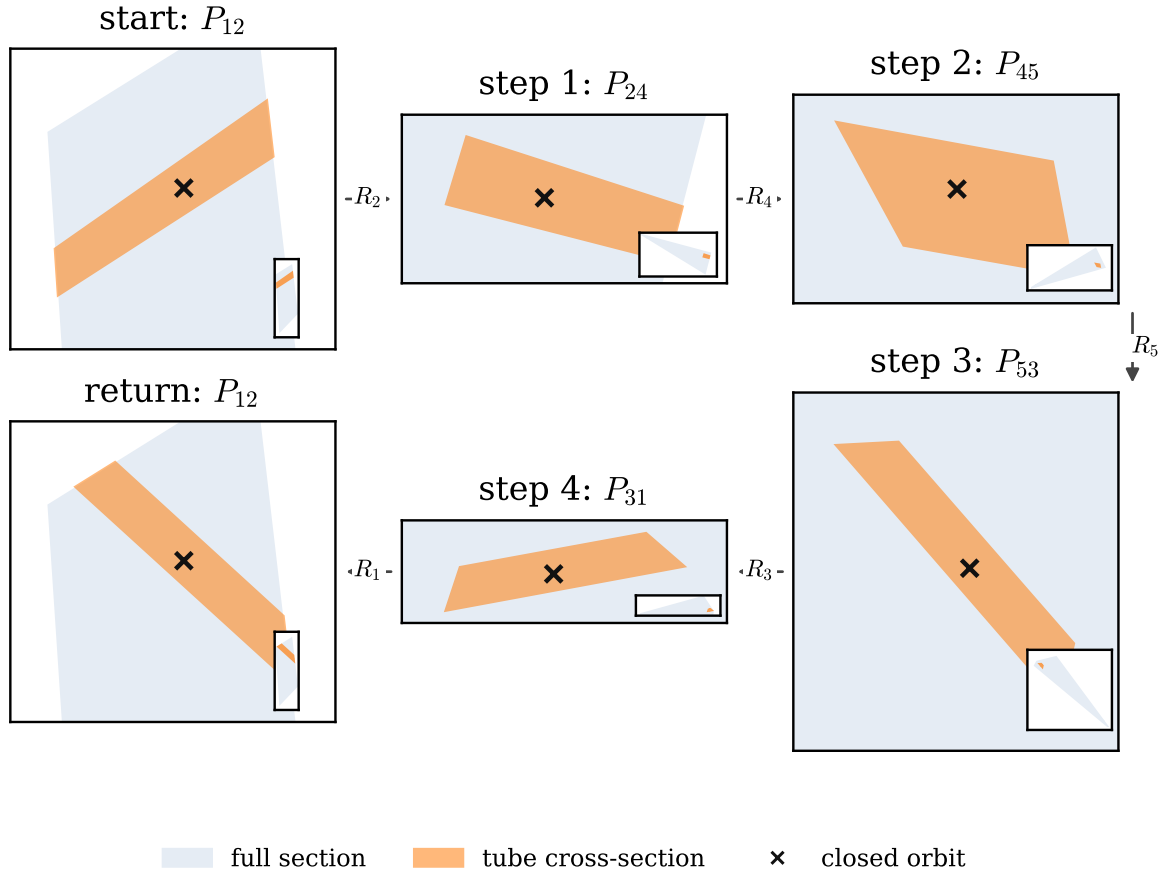


Figure 4: The five passages of Example 5.2. The colored polygons are the successive sections of the tube; each inset locates the displayed region in the full facet-pair section. The black cross follows the closed orbit. Each panel uses a Euclidean-orthonormal frame, so its coordinates differ from the affine chart used in the text. The start and return panels share a frame and scale: closure returns the cross to the same position.

5.2 Exhaustive search and correctness

The fixed-point calculation is simplest when every surviving return map has a unique fixed point. The following hypotheses make this precise and also justify the two pruning rules used in the search.

Definition 5.3 (Flow-graph regular presentation). *The facet presentation above is flow-graph regular if:*

1. *for distinct i, j , $F_i \cap F_j \neq \emptyset$ implies $\omega_0(a_i, a_j) \neq 0$;*
2. *every sublist of at most four of the vectors $a_1, \dots, a_F$ is linearly independent; and*
3. *for every simple cyclic word of length at least five which survives the facet-intersection and positive-transition tests and has nonempty closed tube domain, the linear part M_w of Φ_w satisfies $\det(I - M_w) \neq 0$.*

The last condition is called finite-orbit regularity.

Algorithm 5.4 (Idealized exhaustive flow-graph search). *Enumerate every simple cyclic facet word once. For each word:*

1. *discard it if an adjacent facet intersection is empty or an adjacent transition sign is not positive;*
2. *discard it if its length is at most four: under the regularity hypothesis, these facet directions are linearly independent and cannot form a closed positive-time path;*
3. *construct its closed tube (P_w, Φ_w, A_w) ; discard the word if $P_w = \emptyset$, and otherwise solve $\Phi_w(u) = u$;*
4. *reject a singular fixed-point equation as outside the regular case, and otherwise record the unique solution only if it lies in P_w and every reconstructed segment time is positive.*

Return the least positive recorded action.

We justify both pruning steps. If a strict trajectory changes from R_i to R_j at $x \in F_i \cap F_j$, then the inequalities for K immediately before the breakpoint give

$$\langle a_j, R_i \rangle \geq 0.$$

The first condition in Definition 5.3 excludes equality. Thus every trajectory transition has the positive sign used by the search, even when x lies on additional facets. Second, a strict closed orbit with active word $(s_1, \dots, s_k)$ satisfies

$$\sum_{r=1}^k \tau_r R_{s_r} = 0, \quad \tau_r > 0.$$

If $k \leq 4$, this contradicts the linear independence in condition 2.

Theorem 5.5 (Idealized real-arithmetic flow-graph correctness). *Let $K \subset \mathbb{R}^4$ be a bounded full-dimensional convex polytope with $0 \in \text{int } K$, equipped with a flow-graph regular ordered normalized irredundant facet presentation. Then the finite procedure in Algorithm 5.4, with its algebraic and order operations performed exactly over $\mathbb{R}$, returns*

$$c_{\text{EHZ}}(K).$$

Proof. There are finitely many simple cyclic words. By Lemma 5.1, every recorded point is a strict closed generalized Reeb orbit with the recorded action. The two arguments preceding the theorem show that the transition and short-word pruning cannot remove such an orbit. Empty tubes contain no orbit and are discarded before solving a fixed-point equation. Finite-orbit regularity makes every remaining fixed-point equation nonsingular, so no long word reaches the rejection branch and each word has at most one fixed point.

By Theorem 3.4, a capacity-minimizing orbit can be chosen with an active cyclic word in which every used facet direction occurs in one positive-duration interval. Its breakpoints lie in the corresponding facet-pair sections, its transitions have the required positive signs, and its passages satisfy the primitive tube equations. Lemma 5.1 therefore represents it by a strict fixed point with action $c_{\text{EHZ}}(K)$. The returned minimum is no larger than this action, while soundness of every recorded orbit makes it no smaller. $\square$

Corollary 5.6 (Exact rational-arithmetic flow-graph correctness). *Under the hypotheses of Theorem 5.5, assume in addition that $a_i \in \mathbb{Q}^4$ for every facet row. Then rational affine coordinates can be chosen for every facet-pair plane, and Algorithm 5.4 is a finite exact rational-arithmetic algorithm returning $c_{\text{EHZ}}(K)$.*

Proof. Rational Gaussian elimination gives a rational base point and basis for every facet-pair plane. In these coordinates, the section inequalities, primitive times and maps, tube intersections and compositions, and nonsingular fixed-point systems all have rational coefficients. Their feasibility, signs, solutions, and action comparisons can therefore be decided in exact rational arithmetic. The conclusion follows from Theorem 5.5. $\square$

5.3 Why the regular class is generic

The regularity assumptions are explicit algebraic nonvanishing conditions. For fixed facet count there are only finitely many of them. Pairwise symplectic degeneracy and short-word dependence are ordinary polynomial conditions. The only non-immediate point is that the long-word determinant is not the zero rational function.

When all cyclic primitive denominators of a word are nonzero, let M_w^{alg} denote the composition of the primitive linear maps, regardless of their signs or whether the geometric tube domain is nonempty. For a searched tube, this is the linear part M_w of its affine return map.

Proposition 5.7 (Presentation-chamber genericity). *Fix F , and let $\mathcal{C} \subset (\mathbb{R}^4)^F$ be a nonempty Euclidean-open set of ordered normalized dual-row lists defining bounded full-dimensional polytopes with 0 in the interior and all F inequalities irredundant. An open dense subset of $\mathcal{C}$ satisfies the stronger conditions*

1. $\omega_0(a_i, a_j) \neq 0$ for every $i \neq j$;
2. every sublist of at most four rows is linearly independent; and
3. $\det(I - M_w^{\text{alg}}) \neq 0$ for every simple cyclic word w of length at least five.

Consequently every presentation in this subset satisfies Theorem 5.5, and every rational presentation in it satisfies Corollary 5.6.

Proof. The first two families are finite collections of nonzero polynomial conditions. For a fixed word $w = (s_1, \dots, s_k)$, each primitive linear map is

$$\xi \mapsto \xi - \frac{\langle a_{s_{r+2}}, \xi \rangle}{\langle a_{s_{r+2}}, 2J_0 a_{s_{r+1}} \rangle} 2J_0 a_{s_{r+1}}.$$

On the domain where the denominators are nonzero, coordinate bases for the facet-pair planes and hence $\det(I - M_w)$ depend rationally on the rows.

To see that this rational function is nonzero for every $k \geq 5$, use

$$\begin{aligned} A &= (2, 3, -4, -3), & B &= (0, 1, 2, -1), & C &= (-4, -3, 0, -1), \\ D &= (4, -1, 1, 1), & E &= (-4, 2, -4, -4). \end{aligned}$$

All pairings needed below are nonzero. Direct substitution gives

$$\det(I - M_{ABCDE}) = -\frac{10643}{600}, \quad \det(I - M_{ACDE}) = \frac{1}{168}.$$

At the linear-map level, replacing a cyclic subword x, A, B, A, y by x, A, y leaves the surrounding return map unchanged: the middle transition is the identity on $\ker a_A$, and the two flows in the A -direction combine algebraically. Hence the specializations $(A, B)^m, C, D, E$ cover every odd length at least five, while $(A, B)^m, A, C, D, E$ reduce every even length at least six to the second displayed value. Thus no long-word determinant is identically zero.

The common denominator domain is a nonempty Zariski-open subset of an irreducible affine parameter space. Thus a nonzero rational function remains nonzero on every nonempty coordinate-minor chart. The complement of each nonvanishing condition is therefore a proper algebraic set. There are only finitely many sublists and simple words for fixed F , so their common nonzero locus is open dense. Intersecting it with the Euclidean-open chamber $\mathcal{C}$ preserves density. The all-pairs condition is stronger than the nonempty-pair condition in Definition 5.3, and the remaining conditions imply finite-orbit regularity. $\square$

This is chamber-relative genericity for ordered normalized irredundant presentations. It is not the conjectural CH2021 claim that generic polytopes have no Type 2 orbit below an action bound.

5.4 Relation to CH2021

Chaidez–Hutchings study a finite model that compares Reeb dynamics on a four-dimensional polytope with Reeb dynamics on its smoothings [CH21]. Their setting and proof route differ from the general Haim–Kislev formula. They assume that the polytope is *symplectic*; in particular, its two-faces are not Lagrangian, so the relevant combinatorial Reeb direction on each two-face is one-dimensional.

A Chaidez–Hutchings combinatorial Reeb orbit is a cyclic sequence of oriented line segments in the boundary. Their symplectic flow graph records the orbits which avoid the one-skeleton: its vertices are open two-faces, an edge is a Reeb passage through a three-face, and periodic graph orbits correspond to these Type 1 combinatorial Reeb orbits with the same action [CH21, Proposition 2.14]. Their smoothing theory relates this model to Reeb orbits on smooth convex approximations and yields a capacity formula involving Type 1 and Type 2 combinatorial orbits [CH21, Theorems 1.11 and 1.12, Corollary 1.15].

The rational-arithmetic algorithm developed here uses part of the same local geometry but a different global argument. Its nodes are closed facet-pair sections, and it searches simple facet words whose affine return maps have fixed points. The simple-minimizer theorem supplies completeness on the stated regular class. Consequently this algorithm neither implements the CH2021 rotation pruning nor imports the Type 1/Type 2/Type 3 classification as its capacity proof. In particular, an action cutoff alone would not control arbitrarily long helical combinatorial words in the CH2021 model; the finite search here instead comes from simple words with no repeated facet direction.

5.5 Scope and rational implementation

The regularity hypotheses exclude some important structured examples. In a Lagrangian polygon product, all facet normals from either factor lie in a two-dimensional subspace. Any

three of them are linearly dependent, violating condition 2 of Definition 5.3. The genericity proposition therefore does not make this algorithm applicable to the products studied later in the thesis. The product quadratic program of Section 4 provides a separate method for those bodies.

The rational implementation follows the construction above: it enumerates simple directed cycles with positive transitions, intersects and composes affine tubes, and reconstructs the segment times at a return-map fixed point. It assumes a bounded, irredundant rational facet presentation in dimension four, with correct facet-incidence and symplectic-sign matrices supplied by the caller. These are input requirements, not conclusions of the tube search. It checks short-word independence and rejects a nonempty facet pair with zero symplectic pairing. A singular fixed-point equation is harmless if it is inconsistent or if its fixed set misses the tube; otherwise the implementation reports an unsupported case.

On selected deterministic five-, six-, and seven-facet examples, this search agrees with the certified quadratic-program capacity computation. The tests also exercise tube composition, positive-time reconstruction, and unsupported inputs. These comparisons check the correspondence between the rational program and the theorem on particular inputs. They do not establish a performance advantage over the quadratic program; no common-input benchmark is claimed here. The input data and test entry points are listed with the other computational material in Section 14.

6 First-Order Variation in Facet Coordinates

We now let the supporting hyperplanes move and ask how the systolic ratio changes. The volume has a direct derivative in a neighborhood with fixed simple face structure. Capacity is more delicate: the Haim–Kislev formula expresses it through a maximum over words, and the maximizing word can change.

There are two useful calculations. A smooth feasible choice of weights gives a differentiable upper bound for the ratio. If the constrained optimizer is nondegenerate, its weights also vary smoothly, and we can differentiate its optimized value. The first construction needs less: it will supply the upper bounds in the HKO proof even at singular stationarity systems. For local search, the second construction gives branch gradients that can be compared before proposing a step.

Leipold studies a related optimization problem for volume-preserving linear images of a simplex [Lei26, Sections 9.2, 11.3, and 12.3]. There the balance weights are fixed, capacity evaluation reduces to a linear-ordering integer program, and active permutation branches provide first-order information for numerical search. Here the facets move independently, so both the feasibility constraints and the optimizing weights can change. We derive the corresponding derivatives in facet coordinates.

6.1 Row coordinates and the volume variation

Fix a labelled presentation

$$K(a) = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1, i = 1, \dots, F\}, \quad a = (a_1, \dots, a_F). \quad (24)$$

The vectors a_i are the dual vertices used in Section 4. A row perturbation is a tuple $h = (h_1, \dots, h_F) \in (\mathbb{R}^4)^F$, and we write $a(t) = a + th$. Throughout this section we work in a neighborhood in which $K(a)$ remains a bounded full-dimensional polytope, the listed rows remain irredundant, and the same simple face combinatorics persists. In particular, $V(a) > 0$, and the Haim–Kislev formula applies. These are open conditions at a simple labelled polytope.

For the facet F_i , let S_i be its three-dimensional Euclidean volume and let $\bar{x}_i$ be its centroid. The volume derivative has a particularly simple expression in row coordinates.

Proposition 6.1 (Volume first variation). *In the stable simple chart above, with $V(a) = \text{vol}_4(K(a))$,*

$$DV(a)[h] = - \sum_{i=1}^F \frac{S_i}{\|a_i\|} \langle \bar{x}_i, h_i \rangle. \quad (25)$$

Proof. At a point $x \in F_i$, differentiating the moving hyperplane equation $\langle a_i(t), x(t) \rangle = 1$ gives

$$\langle a_i, \dot{x} \rangle = -\langle h_i, x \rangle.$$

Thus the outward normal velocity of this facet is $-\langle h_i, x \rangle / \|a_i\|$. The first variation of volume is the integral of the outward normal velocity over the boundary. Integrating on each facet and using $\int_{F_i} x \, dS = S_i \bar{x}_i$ gives (25). Intersections of facets have three-dimensional measure zero and do not contribute to the boundary integral. $\square$

The formula displays both parts of a row perturbation at once: changing $\|a_i\|$ moves the supporting hyperplane, while changing its direction tilts that hyperplane. It is also convenient computationally, because it needs only facet volumes and centroids once the incidence data are known.

6.2 Feasible sections and smooth maxima

Fix an ordered word $\sigma = (\sigma_1, \dots, \sigma_m)$ with distinct facets. Write

$$C_\sigma(a)\beta = \left(\sum_{k=1}^m \beta_k a_{\sigma_k}, \sum_{k=1}^m \beta_k \right), \quad e = (0, 0, 0, 0, 1),$$

so that the Haim–Kislev feasibility equations are $C_\sigma(a)\beta = e$ and $\beta \geq 0$. We use the quadratic objective from (19):

$$Q_\sigma(a, \beta) = \sum_{1 \leq j < k \leq m} \beta_j \beta_k \omega_0(a_{\sigma_j}, a_{\sigma_k}). \quad (26)$$

For later differentiation, its explicit row variation at fixed β is

$$D_a Q_\sigma(a, \beta)[h] = \sum_{1 \leq j < k \leq m} \beta_j \beta_k (\omega_0(h_{\sigma_j}, a_{\sigma_k}) + \omega_0(a_{\sigma_j}, h_{\sigma_k})). \quad (27)$$

First consider any smooth feasible choice, without imposing stationarity.

Lemma 6.2 (Feasible-section upper function). *Suppose that $\beta(a)$ is smooth near a_0 ,*

$$C_\sigma(a)\beta(a) = e, \quad \beta(a) > 0, \quad Q_\sigma(a, \beta(a)) > 0.$$

Define

$$A_\sigma(a) = \frac{1}{2Q_\sigma(a, \beta(a))}, \quad U_\sigma(a) = \frac{A_\sigma(a)^2}{2V(a)}.$$

Then

$$\text{sys}(K(a)) \leq U_\sigma(a) \quad (28)$$

throughout this neighborhood. If the chosen weights attain the global Haim–Kislev maximum at a_0 , then equality holds in (28) at a_0 .

Proof. The chosen $\beta(a)$ is feasible for the global maximum defining $Q_{\max}(a)$. Hence $Q_{\max}(a) \geq Q_\sigma(a, \beta(a))$. The Haim–Kislev formula gives

$$c_{\text{EHZ}}(K(a)) = \frac{1}{2Q_{\max}(a)} \leq A_\sigma(a).$$

Squaring and dividing by the positive number $2V(a)$ proves the upper bound. A globally maximizing choice gives equality at the base point. $\square$

The lemma requires no KKT system. For example, if $C_\sigma(a_0)$ has rank five and $\beta_0 > 0$ is feasible, choose an invertible five-column minor and hold the other $m - 5$ coordinates of β fixed. Solving the five feasibility equations gives a smooth local section through β_0 ; positivity persists after shrinking the neighborhood. This elementary construction is the one used for the singular seven-facet data in the HKO certificate.

At a nondegenerate constrained maximum, the KKT equations instead choose a canonical optimizing branch and remove the derivative of β from the value derivative. Let $H_\sigma(a)$ denote the Hessian of $Q_\sigma(a, \cdot)$ with respect to β .

Definition 6.3 (Nondegenerate Haim–Kislev branch). *A feasible pair (σ, β_0) at a_0 is a nondegenerate branch if*

1. $C_\sigma(a_0)$ has rank five;
2. $\beta_0 > 0$ and $Q_\sigma(a_0, \beta_0) > 0$;
3. β_0 is a constrained critical point of $Q_\sigma(a_0, \cdot)$; and
4. $H_\sigma(a_0)$ is negative definite on $\ker C_\sigma(a_0)$.

For $m = 5$, the kernel is zero and the last condition is vacuous.

Proposition 6.4 (Derivative of a nondegenerate branch). *Let (σ, β_0) be a nondegenerate branch at a_0 . Then, after shrinking the row chart, there is a unique smooth constrained maximizer $\beta_\sigma(a) > 0$ near β_0 . Put*

$$q_\sigma(a) = Q_\sigma(a, \beta_\sigma(a)), \quad U_\sigma(a) = \frac{1}{2V(a)} \left(\frac{1}{2q_\sigma(a)} \right)^2.$$

If $\lambda_\sigma \in \mathbb{R}^5$ is chosen so that the KKT equation is

$$D_\beta Q_\sigma(a_0, \beta_0) = C_\sigma(a_0)^\top \lambda_\sigma,$$

then

$$Dq_\sigma(a_0)[h] = D_a Q_\sigma(a_0, \beta_0)[h] - \langle \lambda_\sigma, D_a C_\sigma(a_0)[h] \beta_0 \rangle. \quad (29)$$

Consequently,

$$\frac{DU_\sigma(a_0)[h]}{U_\sigma(a_0)} = -2 \frac{Dq_\sigma(a_0)[h]}{q_\sigma(a_0)} - \frac{DV(a_0)[h]}{V(a_0)}. \quad (30)$$

Proof. Full rank and negative definiteness make the saddle-point KKT matrix nonsingular. The implicit function theorem therefore gives a unique smooth critical point near β_0 . Negative definiteness and positivity are open conditions, so this point remains the positive constrained maximum.

Differentiate $q_\sigma(a) = Q_\sigma(a, \beta_\sigma(a))$. The term containing $D\beta_\sigma$ is

$$D_\beta Q_\sigma D\beta_\sigma = \lambda_\sigma^\top C_\sigma D\beta_\sigma.$$

Differentiating $C_\sigma(a)\beta_\sigma(a) = e$ gives $C_\sigma D\beta_\sigma = -(D_a C_\sigma)\beta_\sigma$, which proves (29). The logarithmic derivative of $U_\sigma = (2V)^{-1}(2q_\sigma)^{-2}$ is (30). $\square$

Equations (27), (25), and (29) are the practical first-order calculation. They use the current rows, the word, its positive KKT weights and multipliers, and facet volumes and centroids. They do not require finite differences of the full capacity computation.

6.3 From branches to the systolic ratio

A nondegenerate branch always gives the upper function U_σ , but it gives the actual systolic ratio only while it realizes the global Haim–Kislev maximum.

Suppose that finitely many nondegenerate branches have fixed-word maxima $q_1, \dots, q_r$ on a neighborhood of a_0 , and that they cover the global finite problem there in the explicit sense

$$Q_{\max}(a) = \max_{1 \leq j \leq r} q_j(a).$$

For example, this identity follows from a complete retained word list and a persistent value gap for every discarded word. Then

$$\text{sys}(K(a)) = \min_{1 \leq j \leq r} U_j(a). \quad (31)$$

If $I(a_0) = \{j : U_j(a_0) = \text{sys}(K(a_0))\}$, the one-sided directional derivative is

$$D^+ \text{sys}(a_0; h) = \min_{j \in I(a_0)} DU_j(a_0)[h]. \quad (32)$$

Indeed, this is the first-order expansion of a finite minimum of smooth functions. In particular, a unique active nondegenerate branch gives an ordinary gradient. At a tie of nondegenerate branches, the directional derivative is generally only the lower envelope of the tied branch slopes.

The hypothesis preceding (31) contains real mathematical work. A list returned by one numerical solve, or a tolerance window around its smallest actions, is not by itself a certificate of the displayed identity.

6.4 What the first-order model contributes to search

On a nondegenerate branch patch with active branch rows $r_j = DU_j(a_0)$, a common first-order ascent direction in a chosen linear slice S can be sought through the maximin problem

$$\max_{h \in S, \|h\| \leq 1} \min_{j \in I(a_0)} r_j(h). \quad (33)$$

The slice removes directions generated by the systolic-ratio symmetries from Definition 2.2. The true ratio is constant on the corresponding group orbits, so its derivative vanishes on their tangent space; S is chosen as a complementary linear slice. A positive value in (33) predicts improvement of every active nondegenerate branch. The prediction still has to be checked by recomputing the capacity after a finite step.

At finite distance, currently inactive branches can matter before they are active. A useful semi-local model therefore retains their base gaps:

$$\min_{j \in W} (U_j(a_0) - \text{sys}(K(a_0)) + DU_j(a_0)[h]), \quad (34)$$

where W is a chosen branch window. Formula (34) explains why a single active gradient can miss a nearby branch switch, but it is a prediction model rather than an infinitesimal theorem. Its reliability depends on the window containing the branches that become relevant, on the branches remaining feasible, and on the linearization error at the proposed step.

One retained finite experiment illustrates this distinction. On a fixed panel of twelve random ten-facet starts, the tested multi-direction method accepted a positive move at each of eight configured trace iterations, but all runs stopped at their iteration cap and every endpoint scan still found a further above-threshold move. Thus the evidence supports finite-step progress and a measured computation cost on that panel, not endpoint local maximality. A later comparison tests seven fixed policies on sixty-four matched starts, including policies based on

finite-gap and branch-history models. This is a separate experiment on a historical numerical objective, not an enlargement of the twelve-start panel. Its inputs and results are recorded in `experiments/dev-gradient-ascent/optimizer-comparison/`.

A retained three-case diagnostic also explains a failure of the affine branch-window prediction. At the tested finite step, two named KKT branches change too rapidly for their base-point derivatives to predict even the sign of the action change. Smaller-step differences approach the analytical slopes, while the volume calculation remains consistent. Thus a small Euclidean step alone need not make the branch model accurate; its useful scale also depends on the conditioning of the KKT system. The decomposition and its three cases are retained in `experiments/dev-gradient-ascent/endpoint-model-audit/`.

6.5 The non-generic boundary

The smooth branch calculation can break down in four ways.

- If a maximizing weight is zero, deleting that coordinate is valid for the capacity value at the base point. It need not preserve the nearby slope: the longer word may acquire a positive weight on one side of the perturbation and define a different branch there.
- If the constraint rank drops or the KKT matrix is singular, the implicit function theorem does not provide a unique smooth optimizing branch. A continuum of optimizers may nevertheless have a well-defined value.
- A tie is described by (32) only when all relevant tied branches extend smoothly and the stated coverage and gap hypotheses hold. Base-point branch enumeration alone does not prove this.
- When facet labels become redundant or the face combinatorics changes, both the branch family and the smooth volume formula require a new local chart. Hausdorff continuity of capacity and volume across that boundary does not imply differentiability there.

All constraints and objectives are semialgebraic. A possible route to a fully general one-sided theory would organize germs of feasible optimizing branches by a finite decomposition of direction space. Such a construction would have to record support appearances, singular optimizer sets, and dominance of every discarded germ. Obtaining a complete first-order description at an arbitrary non-generic polytope remains open here.

6.6 Application at HKO

The HKO argument uses the feasible-section calculation. It constructs twenty-six upper bounds that agree with the true ratio at the base body. Together their derivatives force at least one bound to decrease in every direction transverse to the symmetries. Section 7 gives the construction and proves that this directional statement yields a strict local maximum. No continuation of an optimizing branch is required.

6.7 Implementation and validation status

The implementation evaluates the branch and volume formulas in dual-vertex coordinates. Regression tests compare the volume derivative with central finite differences and compare a fixed-word capacity derivative with central finite differences on known fixtures. An exact rational implementation of the capacity-gradient formula is also compared with the floating-point formula on a rational simplex fixture. These checks test the correspondence between the formulas and their implementations; they do not certify branch coverage, finite-step prediction, or local maximality.

For the HKO theorem, the evidence standard is stronger and different. The SageMath verifier checks the selected feasible sections and their derivative rows exactly in the relevant algebraic number field. The mathematical use of those rows is the upper-bound and transverse-slice argument in Section 7.

7 Local maximality of the HKO pentagon product

Let P be the regular pentagon of circumradius one, centred at the origin and with vertex $(1, 0)$. In coordinates (q_1, q_2, p_1, p_2) put $K_0 = P \times R(-\pi/2)P$, with symplectic form $\omega_0 = \sum_j dq_j \wedge dp_j$. The Haim–Kislev–Ostrover capacity formula [HO26] for this body gives

$$c_{\text{EHZ}}(K_0) = 2 \cos(\pi/10)(1 + \cos(\pi/5)), \quad \text{sys}(K_0) = \frac{3 + \sqrt{5}}{5} > 1.$$

Here $\text{sys}(K) = c_{\text{EHZ}}(K)^2 / (2 \text{vol}_4 K)$. The counterexample itself is due to Haim–Kislev and Ostrover; the result proved here concerns its local maximality when the number of facets stays ten.

Theorem 7.1. *There is a Hausdorff neighborhood $\mathcal{U}$ of K_0 among convex four-dimensional polytopes with exactly ten facets such that $\text{sys}(K) \leq \text{sys}(K_0)$ for $K \in \mathcal{U}$. Equality holds precisely for those bodies in $\mathcal{U}$ obtained from K_0 by translation, positive dilation and a linear symplectic map.*

7.1 The proof strategy

The ten supporting hyperplanes may move in all four coordinate directions, so the nearby bodies include polytopes that are no longer products. Write

$$K(a) = \{x \in \mathbb{R}^4 : a_i \cdot x \leq 1, i = 1, \dots, 10\}, \quad a = (a_1, \dots, a_{10}),$$

and let a_0 be the row list of K_0 .

We construct twenty-six smooth upper bounds U_j for the systolic ratio, all equal to it at a_0 . If one of these bounds decreases, the ratio decreases with it. Different bounds handle different directions. The proof first identifies their derivatives and the symmetry directions. A linear-algebra lemma then gives a uniform negative slope transverse to symmetry. Only at the end do we pass from these derivatives to a finite neighborhood and recover the equality cases.

7.2 Local coordinates and volume

Lemma 7.2 (A simple row neighborhood). *There is a neighborhood $\mathcal{N}$ of a_0 in $(\mathbb{R}^4)^{10}$ on which $K(a)$ is bounded and simple, with the same vertex–facet incidences as K_0 . Every sufficiently Hausdorff-close ten-facet polytope has a labelled presentation with $a \in \mathcal{N}$, and its rows tend to a_0 as the body tends to K_0 .*

Proof. The origin stays in the interior under sufficiently small Hausdorff perturbations of $K_0 = K(a_0)$. One way to see this is through support functions: $h_K(u) = \max_{x \in K} u \cdot x$. Write B for the Euclidean unit ball. If $rB \subset K_0$ and the Hausdorff distance from K to K_0 is less than $\epsilon < r$, then $h_K(u) \geq h_{K_0}(u) - \epsilon \geq r - \epsilon$ for every unit vector u . Thus $(r - \epsilon)B \subset K$. On such a neighborhood polarity is continuous: the radial function, the distance from zero to the boundary along a unit direction, of K° is $1/h_K$ on the unit sphere, and the common positive lower bound on h_K makes these reciprocals converge uniformly. The polars are uniformly bounded convex bodies, so this also gives Hausdorff convergence.

The polar K_0° has the ten vertices $a_{0,i}$. Choose disjoint small balls about them. Each vertex is uniquely exposed by some linear functional ℓ_i . For polars sufficiently close to K_0° , every maximizer of ℓ_i lies in the corresponding ball. Otherwise a convergent sequence of maximizers outside that ball would limit to a maximizer on K_0° different from $a_{0,i}$. A maximizing face contains a vertex. A polytope K with exactly ten facets has a polar with exactly ten vertices; hence there is exactly one polar vertex in each ball and none elsewhere. Labelling these vertices a_i gives

$$K = \{x : a_i \cdot x \leq 1, i = 1, \dots, 10\}, \quad a_i \longrightarrow a_{0,i} \quad \text{as } K \longrightarrow K_0.$$

For completeness, the face structure is stable in this neighbourhood. The base body is a product of two polygons and is simple: precisely four facets meet at each vertex, with independent normals. Their four equations therefore determine a smoothly varying intersection under a small change of a . All the remaining inequalities have strict slack at that intersection, so it remains a vertex with exactly those incident facets.

No new vertex can appear. The convex hull of the base normals contains a ball about zero; the convex hull of sufficiently close normals contains a smaller common ball. Consequently the bodies $K(a)$ are uniformly bounded. If arbitrarily close parameters had a new vertex, choose four independent active constraints there. There are finitely many four-element subsets, so along a subsequence the same subset is active and the vertices converge to a point of K_0 where those four constraints are equalities. Simplicity forces them to be the incident facets of one of the original vertices. Their intersection is the continuation already constructed, contradicting newness. This argument also excludes a new vertex arising from a subset whose normals are dependent at the base point; no invertibility assumption for such a subset is needed.

The vertex–facet incidences therefore remain unchanged. In particular each inequality still defines a facet. Conversely, small changes of the ten normals give Hausdorff-small changes of $K(a)$: their convex hulls converge, contain a common ball about zero, and polarity is continuous there. □

Lemma 7.3 (Smooth volume and its derivative). *On a sufficiently small neighborhood $\mathcal{N}$ as above, $V(a) = \text{vol}_4 K(a)$ is smooth and positive. If $F_i(a)$ has centroid $\bar{x}_i(a)$, then*

$$DV(a)[h] = - \sum_{i=1}^{10} \frac{\text{vol}_3(F_i(a))}{\|a_i\|} \langle h_i, \bar{x}_i(a) \rangle.$$

Proof. The fixed incidences express each vertex smoothly in the rows. A fixed boundary triangulation then expresses volume smoothly in those vertices. On the moving facet, differentiating $\langle a_i + \epsilon h_i, x(\epsilon) \rangle = 1$ gives outward normal velocity $-\langle h_i, x \rangle / \|a_i\|$. Integrating this velocity over the facets and using their centroids gives the formula. □

The concrete base point and its volume

Put $t = \tan(\pi/5)$, $s = (5 - t^2)/2 = \sqrt{5}$, and

$$t^4 - 10t^2 + 5 = 0, \quad 0 < t < 1, \quad \alpha = (3 - s)/2, \quad b = t(1 + s)/2, \quad d = s - 1.$$

The selected root is unique in $(0, 1)$: the polynomial is strictly decreasing there and changes sign. For the circumradius-one pentagon with vertex $(1, 0)$, the edge normals have arguments $(2k + 1)\pi/5$ and lengths $\sec(\pi/5) = d$. Rotating the second factor by $-\pi/2$ gives the following ten rows $a_{0,i}$, in (q_1, q_2, p_1, p_2) order:

$$\begin{pmatrix} 1 & t & 0 & 0 \\ -\alpha & b & 0 & 0 \\ -d & 0 & 0 & 0 \\ -\alpha & -b & 0 & 0 \\ 1 & -t & 0 & 0 \\ 0 & 0 & t & -1 \\ 0 & 0 & b & \alpha \\ 0 & 0 & 0 & d \\ 0 & 0 & -b & \alpha \\ 0 & 0 & -t & -1 \end{pmatrix}.$$

These rows give the normalized facet presentation of K_0 . Writing $\theta = \pi/5$, the area of one pentagon is the sum of five triangles with central angle 2θ , hence $B = \frac{5}{2} \sin(2\theta) = 5 \sin \theta \cos \theta$. Fubini therefore gives

$$V_0 = B^2 = 25 \sin^2 \theta \cos^2 \theta = \frac{25(5+s)}{32} = \frac{25(15-t^2)}{64}.$$

To specialize Lemma 7.3, observe that F_i is an edge times the other centered pentagon. Its centroid is $\bar{x}_i = a_{0,i}/\|a_{0,i}\|^2$, its three-volume is $2 \sin \theta B$, and $\|a_{0,i}\| = \sec \theta$. Consequently

$$DV_0[h] = -\mu \sum_{i=1}^{10} a_{0,i} \cdot h_i, \quad \mu = 10 \sin^2 \theta \cos^4 \theta = \frac{25}{32} + \frac{5s}{16}.$$

This gives the volume derivative for every row perturbation at the base point.

7.3 The touching upper bounds

For an ordered facet word $\sigma = (\sigma_1, \dots, \sigma_m)$, write

$$C_\sigma(a) = \begin{pmatrix} a_{\sigma_1} & \cdots & a_{\sigma_m} \\ 1 & \cdots & 1 \end{pmatrix}, \quad e = (0, 0, 0, 0, 1)^T.$$

The equations $C_\sigma(a)\beta = e$ impose closure and normalization. For positive feasible weights let

$$Q_\sigma(a, \beta) = \sum_{j < i} \beta_i \beta_j \omega_0(a_{\sigma_j}, a_{\sigma_i}), \quad A_\sigma(a) = \frac{1}{2Q_\sigma(a, \beta(a))}, \quad U_\sigma(a) = \frac{A_\sigma(a)^2}{2V(a)}.$$

When $Q_\sigma > 0$, the variational capacity formula gives $\text{sys}(K(a)) \leq U_\sigma(a)$.

Let T be the tangent space at a_0 to the orbit generated by translations, positive dilations and linear symplectic maps. The following lemma records the finite calculation that will be used in the rest of the proof.

Lemma 7.4 (Exact upper bounds and symmetry directions). *The space T has dimension fifteen. On a neighborhood of a_0 there are twenty-six smooth functions U_j such that*

$$\text{sys}(K(a)) \leq U_j(a), \quad U_j(a_0) = \text{sys}(K_0).$$

Their derivative rows $r_j = DU_j(a_0)$ vanish on T , have rank twenty-five, and satisfy

$$\sum_{j=1}^{26} \lambda_j r_j = 0, \quad \lambda_j > 0, \quad \sum_{j=1}^{26} \lambda_j = 1.$$

The construction and computer-algebra verification of this lemma occupy the following calculations. Holding selected weights fixed and solving the five closure and normalization equations for the others produces the smooth sections: an invertible five-column minor gives smoothness, and strict positivity at the base point persists nearby.

One of the twenty-six upper bounds

We now work through one of the twenty-six upper bounds to illustrate the construction and differentiation. The other twenty-five are constructed in the same way. The exact computations for all twenty-six were verified using a computer algebra system. Appendix B.1 gives the complete program with explanations of its checks.

Consider the seven-facet word and position sets

$$\sigma = (2, 3, 8, 9, 5, 4, 6), \quad I = (1, 2, 3, 4, 5), \quad J = (6, 7).$$

We solve closure and normalization for the weights at positions I and hold the weights at positions J fixed. These are positions in the word, not facet labels. Set

$$r = \frac{6909830056250513}{10^{17}}, \quad c = \frac{t^2 + 5}{40}, \quad z = \frac{5 - t^2}{20} = \frac{s}{10}, \quad k = \frac{3 - t^2}{4}.$$

Fix $\beta_6 = r$ and $\beta_7 = z$; the decimal-looking rational is an exact rational choice, not a rounding assertion. The complete base vector is

$$\beta_0 = (c + kr, c - r, c, c, z - kr, r, z)^T.$$

At this word the closure matrix's selected minor is

$$M = C_\sigma(a_0)_I = \begin{pmatrix} -\alpha & -d & 0 & 0 & 1 \\ b & 0 & 0 & 0 & -t \\ 0 & 0 & 0 & -b & 0 \\ 0 & 0 & d & \alpha & 0 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}, \quad \det M = 10(1 - t^2) > 0.$$

The full matrix is obtained by adjoining columns $(a_4, 1)^T$ and $(a_6, 1)^T$. Writing $e = (0, 0, 0, 0, 1)^T$, define the nearby section by

$$\beta_J(a) = (r, z)^T, \quad \beta_I(a) = C_\sigma(a)_I^{-1}(e - C_\sigma(a)_J(r, z)^T).$$

Direct substitution gives $C_\sigma(a_0)\beta_0 = e$ and $\beta_0 > 0$. For example $0.726 < t < 0.727$ and $0.069 < r < 0.070$ suffice to check all seven signs from the displayed formulas. Invertibility and positivity persist in a neighborhood.

Use $\omega(x, y) = x_{q_1}y_{p_1} + x_{q_2}y_{p_2} - x_{p_1}y_{q_1} - x_{p_2}y_{q_2}$ and

$$Q_\sigma(a, \beta) = \sum_{j < i} \beta_i \beta_j \omega(a_{\sigma_j}, a_{\sigma_i}).$$

At the fixed base geometry, solving closure with arbitrary $(\beta_6, \beta_7) = (u, v)$ gives the particularly simple identity

$$Q_\sigma = -4tv^2 + (2t - 2t^3/5)v.$$

It is independent of u and has its maximum at $v = z$; its value is $Q_0 = t/5$. Thus its Hessian on the two-dimensional closure affine space is singular (in the (u, v) coordinates it is $\text{diag}(0, -8t)$). The five-by-five closure minor is nevertheless invertible, so the displayed feasible section continues smoothly and supplies an upper bound nearby.

$$A_0 = \frac{1}{2Q_0} = \frac{5}{2t} = 5t - \frac{t^3}{2}, \quad U_0 = \frac{A_0^2}{2V_0} = \frac{11 - t^2}{10} > 1.$$

The HKO capacity formula gives $c_{\text{EHZ}}(K(a_0)) = 2 \cos(\pi/10)(1 + \cos(\pi/5)) = A_0$. Thus the constructed upper bound agrees with the systolic ratio at the base point.

A worked coordinate outside product-preserving perturbations

We now choose a perturbation that varies the first momentum component of the second q -facet's normal:

$$h_2 = (0, 0, 1, 0), \quad h_i = 0 \quad (i \neq 2).$$

Put $w = c + kr = \beta_{0,1}$. Differentiated closure gives

$$D\beta_J[h] = 0, \quad D\beta_I[h] = -wM^{-1}(0, 0, 1, 0, 0)^T = w \begin{pmatrix} t^3/20 - t/4 \\ t^3/20 - t/4 \\ t^3/40 - 3t/8 \\ -3t^3/20 + 5t/4 \\ t^3/40 - 3t/8 \end{pmatrix}.$$

Inserting these values into the product rule for Q gives the exact scalar

$$DQ[h] = \frac{3-t^2}{80} + \left(\frac{3t^2}{20} - \frac{1}{4} \right) r - \frac{t^2+5}{4} r^2 =: q_h.$$

Since a_2 has zero momentum coordinates, $DV_0[h] = 0$, while

$$DA[h] = -\frac{25}{2t^2}q_h, \quad DU[h] = -\frac{125}{4t^3V_0}q_h.$$

This coefficient of the upper-bound derivative is approximately -0.1801707325 . Changing only the two polygon factors keeps every q -facet normal in the q -plane. This perturbation adds a p_1 -component to one of those normals, so the derivative describes a change of the upper bound outside that product family.

7.4 Computing the twenty-six derivatives

The accompanying table gives the exact fixed values for every section, including the worked entry. Except at entry 22, the minor positions are $I = (1, 2, 3, 4, 5)$ and $J = (6, \dots, m)$. Entry 22 instead has $I = (1, 2, 3, 4, 6)$ and $J = (5, 7)$. The table is ordered by the witness's zero-based certificate index. For each table row set $\beta_J = u$ and compute

$$\beta_I = C_I^{-1}(e - C_J u), \quad (D\beta)_J[h] = 0, \quad (D\beta)_I[h] = -C_I^{-1}(DC[h])\beta.$$

For each of the forty coordinate directions, compute

$$\begin{aligned} DQ[h] &= \sum_{j < i} ((D\beta_i[h])\beta_j + \beta_i(D\beta_j[h]))\omega(a_{\sigma_j}, a_{\sigma_i}) \\ &\quad + \sum_{j < i} \beta_i\beta_j(\omega(h_{\sigma_j}, a_{\sigma_i}) + \omega(a_{\sigma_j}, h_{\sigma_i})), \\ R[h] &= -\frac{A_0}{2V_0Q_0^2}DQ[h] - \frac{A_0^2}{2V_0^2}DV_0[h]. \end{aligned}$$

Together with the explicit table these formulas specify the entire 26×40 matrix R exactly.

The finite predicates are: every word has distinct valid labels; $\det C_I \neq 0$; $C\beta = e$; all $\beta_i > 0$; $Q = t/5$; the differentiated closure equations hold; the rows annihilate the fifteen symmetry columns; $\text{rank } R = 25$; and the one-dimensional kernel of R^T has a vector whose entries have one strict sign. Normalize that vector to sum to one, producing $\lambda > 0$, $\lambda^T R = 0$. The symmetry columns are explicitly $h_i = -a_{i,j}a_i$ for $j = 1, \dots, 4$, $h_i = -a_i$, and $h_i = -X^T a_i$ for the ten generators

$$X = \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix}, \quad B = B^T, \quad C = C^T,$$

using the four matrix units for A and the three symmetric matrix units for each of B, C . Their column rank is required to be fifteen. These exact arithmetic checks establish the asserted independence, rank and positive relation. The constructions above give smooth feasible sections on a common neighborhood, and the HKO capacity formula gives the touching identities. This proves Lemma 7.4.

7.5 The transverse slope estimate

The exact calculation is now summarized by Lemma 7.4. We use only its rank and positivity statements in the remaining argument.

Lemma 7.5 (Uniformly negative transverse slopes). *Choose a linear complement S to the symmetry tangent space T , so that $(\mathbb{R}^4)^{10} = T \oplus S$ and $\dim S = 25$. There is $\eta > 0$ such that*

$$\min_{1 \leq j \leq 26} r_j(v) \leq -\eta \|v\| \quad \text{for every } v \in S.$$

Proof. The rows have rank twenty-five and vanish on the fifteen-dimensional space T . Their common kernel is therefore exactly T , and their restrictions span S^* . For $v \in S \setminus \{0\}$, suppose that every $r_j(v)$ were nonnegative. Their positively weighted sum is zero, so each would have to vanish. Then $v \in T \cap S = \{0\}$, a contradiction. Thus $\min_j r_j(v) < 0$ for every nonzero transverse direction.

The function $v \mapsto \min_j r_j(v)$ is continuous on the unit sphere of S . Its maximum there is strictly negative by compactness. Write that maximum as $-\eta$ and use homogeneity to obtain the assertion. $\square$

Figure 5 illustrates the slope condition in one and two transverse dimensions.

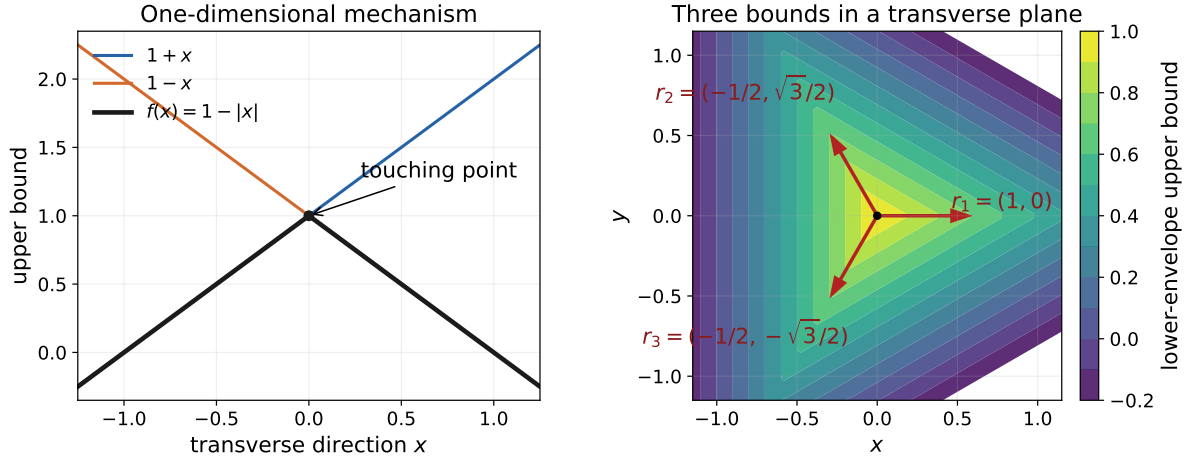


Figure 5: Toy models of touching upper bounds. Left: the lower envelope of $1+x$ and $1-x$ is $1-|x|$. Right: level sets of the lower envelope of three affine functions with gradients $(1,0)$, $(-1/2, \sqrt{3}/2)$ and $(-1/2, -\sqrt{3}/2)$. The gradients span the plane and sum to zero, and the envelope decreases in every nonzero direction. In the proof the systolic ratio lies below the envelope and agrees with it at the base point.

7.6 From slopes to a neighborhood

We now pass from the derivative estimate to nearby bodies. The uniform constant η lets us choose one neighborhood for all transverse directions.

Lemma 7.6 (Strict decrease on the slice). *For all sufficiently small nonzero $v \in S$,*

$$\text{sys}(K(a_0 + v)) \leq \text{sys}(K_0) - \frac{\eta}{2} \|v\| < \text{sys}(K_0).$$

Proof. There are only twenty-six smooth upper bounds. Their differentiability at a_0 therefore supplies a common neighborhood in which

$$|U_j(a_0 + v) - U_j(a_0) - r_j(v)| \leq \frac{\eta}{2} \|v\| \quad (j = 1, \dots, 26).$$

Shrink it so that all bounds are defined and dominate the ratio there. For each $v \in S$, Lemma 7.5 supplies an index with $r_j(v) \leq -\eta\|v\|$. Since $U_j(a_0) = \text{sys}(K_0)$,

$$\text{sys}(K(a_0 + v)) \leq U_j(a_0 + v) \leq \text{sys}(K_0) - \frac{\eta}{2}\|v\|. \quad \square$$

Lemma 7.7 (Local symmetry decomposition). *Every row list sufficiently close to a_0 is the image of some $a_0 + v$, with $v \in S$ small, under a symmetry close to the identity. The symmetry and v can be chosen smoothly in local group coordinates.*

Proof. Translation by y , dilation by $\rho > 0$ and a linear symplectic map M act on the rows, respectively, by

$$a_i \mapsto \frac{a_i}{1 + a_i \cdot y}, \quad a_i \mapsto \rho^{-1}a_i, \quad a_i \mapsto M^{-T}a_i.$$

These actions are smooth near the identity. Choose fifteen local coordinates θ for this symmetry group and write its action as $g(\theta)$. The map

$$\Phi(\theta, v) = g(\theta) \cdot (a_0 + v), \quad v \in S,$$

has derivative at $(0, 0)$ with image T in the group variables and image S in the slice variables. The fifteen symmetry derivatives are independent by Lemma 7.4. Thus $D\Phi(0, 0)$ is an isomorphism, and the inverse function theorem gives the decomposition. $\square$

The toy model in Figure 6 shows how a strict transverse maximum becomes a family of equal maxima along symmetry.

Symmetry extension of a strict transverse maximum

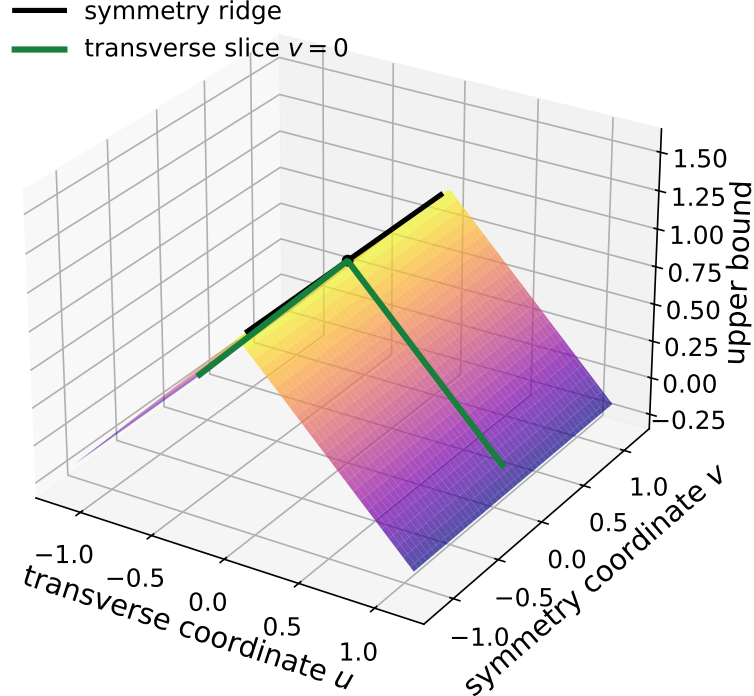


Figure 6: The toy function $f(u, v) = 1 - |u|$ has a strict maximum on the transverse slice $v = 0$ and is constant along the symmetry coordinate v . Its maxima form the ridge $u = 0$. In the HKO proof the symmetry tangent space T has dimension fifteen and the transverse space S has dimension twenty-five. The picture illustrates their roles in the local decomposition.

Proof of Theorem 7.1. By Lemma 7.2, a sufficiently Hausdorff-close ten-facet body has a row list in the neighborhood of Lemma 7.7. It is therefore a symmetry image of $K(a_0 + v)$ for some small $v \in S$. The ratio is unchanged by that symmetry. Lemma 7.6 gives a strict decrease if $v \neq 0$, and equality if $v = 0$. Equality consequently forces the body to be a symmetry image of K_0 . Conversely, every such image has the same ratio as K_0 . $\square$

The twenty-six exact choices

The constants $c = (t^2 + 5)/40$ and $z = (5 - t^2)/20$ are as above. Facet labels and positions start at one; entry numbers start at zero to identify the rows in the accompanying computational witness. Every finite decimal in the table denotes its exact rational value.

Entry	Word σ	Fixed positions J	Exact fixed values
0	(1, 2, 8, 4, 10, 6)	(6)	(c)
1	(1, 2, 8, 7, 4, 10)	(6)	(z)
2	(2, 3, 8, 9, 5, 4, 6)	(6, 7)	(0.06909830056250513, z)
3	(1, 8, 7, 3, 4, 10)	(6)	(z)
4	(2, 9, 8, 5, 4, 6)	(6)	(z)
5	(2, 8, 4, 5, 6, 10)	(6)	(c)
6	(1, 6, 2, 8, 5, 4, 10)	(6, 7)	(0.180901699437495, c)
7	(1, 7, 8, 4, 3, 9, 10)	(6, 7)	(0.0202254248593736825 t^2 + 0.0584220259843842225, 0.18090169943749473)
8	(1, 5, 7, 6, 3, 10, 9)	(6, 7)	(0.020225424859373655 t^2 + 0.058422025984384415, 0.18090169943749462)
9	(2, 8, 4, 3, 9, 5, 6)	(6, 7)	(0.1809016994374947, z)
10	(2, 3, 9, 8, 4, 5, 6)	(6, 7)	(0.18090169943749512, z)
11	(2, 8, 9, 5, 7, 6, 3)	(6, 7)	(0.18090169943749473, c)
12	(1, 7, 8, 4, 10, 2)	(6)	(c)
13	(1, 7, 4, 3, 9, 10)	(6)	(c)
14	(2, 9, 5, 6, 7, 3)	(6)	(c)
15	(2, 8, 3, 9, 5, 6)	(6)	(z)
16	(2, 8, 4, 5, 10, 6)	(6)	(c)
17	(1, 7, 3, 2, 8, 4, 10)	(6, 7)	(0.18090169943749507, z)
18	(1, 5, 6, 7, 3, 2, 9)	(6, 7)	(0.06909830056250547, z)
19	(1, 7, 3, 9, 5, 6, 2)	(6, 7)	(c , 0.06909830056250552)
20	(1, 8, 5, 4, 6, 10, 2)	(6, 7)	(c , 0.14926352961534042)
21	(1, 6, 2, 8, 7, 4, 10)	(6, 7)	(z , 0.1809016994374947)
22	(1, 5, 7, 4, 3, 9, 10)	(5, 7)	(0.18090169943749482, c)
23	(1, 7, 3, 4, 10, 9, 5)	(6, 7)	(c , 0.06909830056250518)
24	(1, 7, 6, 3, 10, 9, 5)	(6, 7)	(0.1809016994374948, c)
25	(2, 8, 9, 5, 4, 6, 3)	(6, 7)	(z , 0.12028993467659169)

8 Finding mathematical patterns with data science

We use data science to look for geometric relationships involving symplectic capacity. We sample random polytopes and compute a broad range of readily available geometric quantities, including symplectic invariants and measurements that can change under symplectic transformations. The sampling distributions are specified in Section 8.3.

We then look for simple statistical patterns without requiring a geometric explanation in advance. When a pattern emerges, we return to the definitions and try to understand why the quantities might be related. This can suggest a tentative conjecture, which further examples may help sharpen or refute. The aim is to formulate a precise statement and find a proof that explains the observed relationship.

The experiments reported below carried out parts of this program. They found a strong association between symplectic ridge area and the numerical systolic ratio. Examining the descriptor geometrically clarified what it measures, but did not establish a general relation to capacity. The pattern also served a complementary aim: finding bodies with large ratios. Selection rules based on it improved the results of fresh sampling before we had an explanation for their success. The chapter follows this signal from its statistical discovery through geometric interpretation and search tests, then considers search methods that change the bodies themselves.

8.1 Ridge area and systolic ratio

The initial sample contains generic four-dimensional polytopes and Lagrangian products of planar polygons. We compare their systolic ratios with measurements of face structure and geometry. For numerical evaluations we write $\widehat{\text{sys}} = c_{\text{num}}^2 / (2V_{\text{num}})$; the normalization is the same as for sys , but volume is computed in floating-point arithmetic.

A two-dimensional face of a four-dimensional polytope is called a ridge. For a ridge F with cyclically ordered vertices $v_0, \dots, v_{r-1}$, define its absolute symplectic area by

$$A_\omega(F) = \frac{1}{2} \left| \sum_{i=0}^{r-1} \omega_0(v_i, v_{i+1}) \right|, \quad v_r = v_0, \quad \omega_0 = dq_1 \wedge dp_1 + dq_2 \wedge dp_2.$$

This is the polygonal area formula with the symplectic pairing in place of the planar determinant. It measures the area seen by ω_0 ; a ridge can have positive Euclidean area and zero symplectic area. Two summaries proved useful:

$$R(K) = \frac{\sum_F A_\omega(F)}{\sqrt{\text{vol}_4(K)}}, \quad C(K) = \frac{\max_F A_\omega(F)}{\sum_F A_\omega(F)}.$$

The first records total symplectic ridge area relative to volume. The second records the fraction concentrated on the largest ridge, when the total is positive. Both are unchanged by translation, common dilation and linear symplectic maps. The area formula is translation invariant because the terms involving the translation vector cancel around the closed polygon.

8.2 Looking for patterns in the measurements

The pooled rank correlation between R and the numerical systolic ratio is about -0.94 . Rank correlation compares the orderings of the bodies by the two quantities; a value of -1 would mean exactly reversed orderings. The ordinary linear correlation is much weaker, about -0.21 . Thus the principal observation concerns which bodies have larger systolic ratio, rather than how much the systolic ratio changes per unit of ridge area. The inverse association also occurs within every sampled facet-count or polygon-size group: the rank correlations range from about -0.84 to -1.00 . Thus it is not solely a consequence of pooling these groups. Figures 7 and 8 show the separate populations.

8.3 The sampled bodies and the measured ratio

We sampled generic polytopes with prescribed facet counts, and Lagrangian products of planar polygons. In both constructions we drew independent unit facet normals and support heights in $[0.8, 1.2)$, retaining bounded bodies with every proposed facet present. The detailed generation

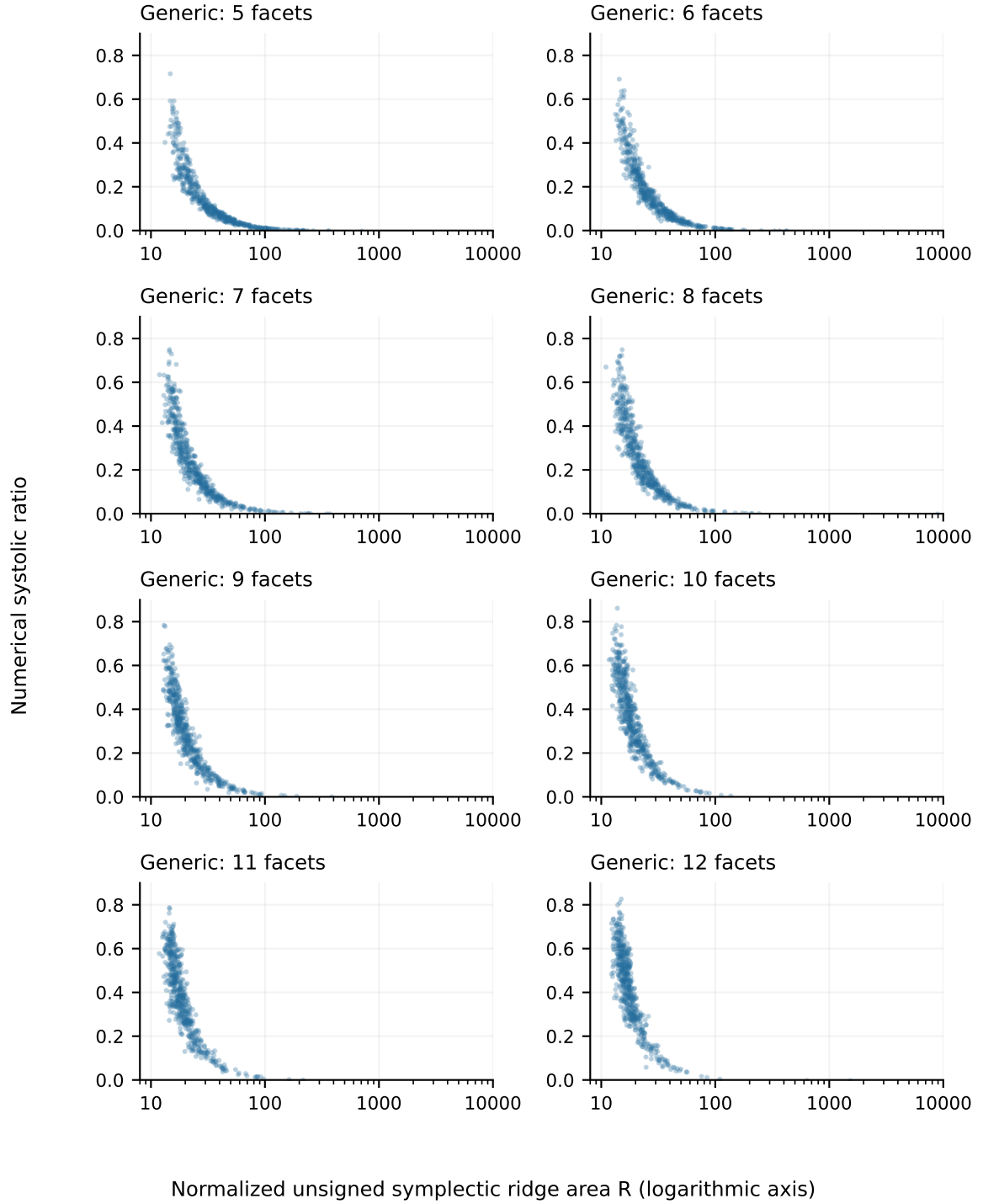


Figure 7: Ridge area and numerical systolic ratio for the 4,096 generic polytopes, separated by facet count. Each point is one body; the horizontal axis is logarithmic. All original numerical values are retained (Appendix A.2).

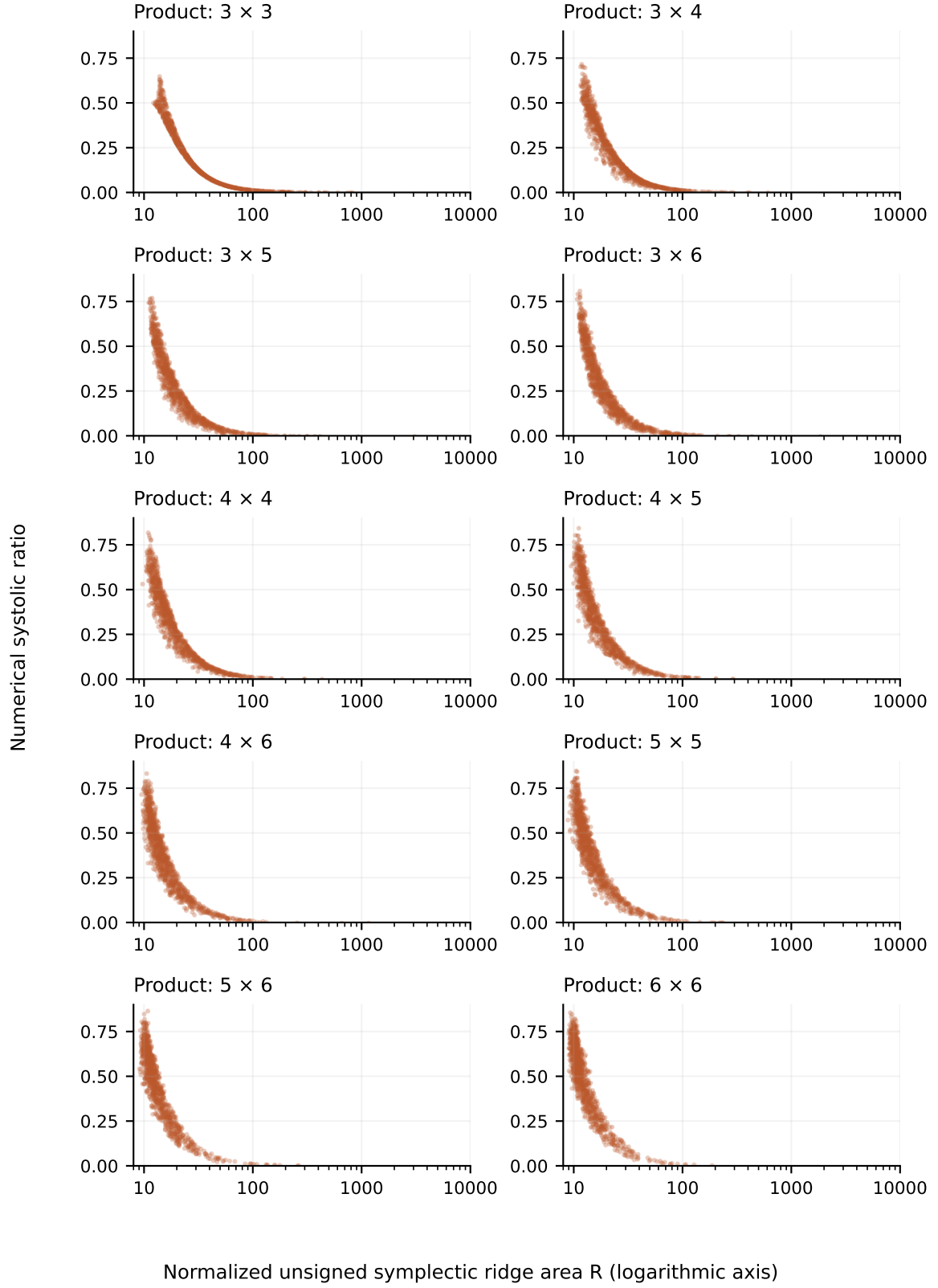


Figure 8: The same comparison for the 10,240 Lagrangian products, separated by factor side counts. Each group contains 1,024 bodies.

and rejection rules are given in Appendix A.1. Thus the experiment studies these particular distributions of bodies.

The second family consisted of Lagrangian products $P \times Q$. In coordinates (q_1, q_2, p_1, p_2) , each factor was a planar polygon constructed in the same way, using independent uniform angles for its unit normals. A $k \times m$ product has k sides in its first factor and m in its second. We call a fixed choice of these side counts, or of F in the generic family, a sample group. The retained exploratory data contained the following groups.

Family	Groups	Bodies per group	Total
Generic, $F = 5, \dots, 12$	8	512	4,096
Products, $3 \leq k \leq m \leq 6$	10	1,024	10,240

Seven of the ten product groups have a triangle or quadrilateral factor. For these products, the Viterbo inequality is already known [BMP26, Theorem 1.2 and Section 4]. They can reveal geometric patterns and near-equality behavior, but cannot supply counterexamples. The remaining groups are 5×5 , 5×6 and 6×6 ; their side counts alone do not imply the same bound. In particular, the theorem for an affinely regular hexagonal factor does not cover an arbitrary hexagon.

The largest numerical ratio in this sample was 0.863, and none exceeded one. Appendix A.2 records the recovered geometry, rebuilt descriptors and current capacity reevaluation. All but one of the 14,336 bodies now have current capacity intervals; the original numerical values are retained for the statistical analyses reported here.

8.4 Predictive information in ridge area

Prediction experiments gave another view of the signal. A random forest [Bre01], which averages predictions from many decision trees, used 39 descriptors: 27 describing face counts and incidences, and 12 summarizing ridge areas. The test split held out groups defined by capacity-source label and facet count, leaving 8,192 training rows and 6,144 test rows. Its test coefficient of determination was $R^2 = 0.885$. Here

$$R^2 = 1 - \frac{\sum_{i \in \text{test}} (s_i - \hat{s}_i)^2}{\sum_{i \in \text{test}} (s_i - \bar{s}_{\text{test}})^2}$$

compares squared prediction error with prediction by the test-set mean. Here s_i is the observed numerical systolic ratio and $\hat{s}_i$ is the model prediction. The mean absolute error was 0.0495. This was one group split, not repeated cross-validation across independent datasets.

A separate feature-family comparison used histogram-based gradient-boosted regression [Sci], which successively fits trees to improve the prediction. Under its common training/test split, ridge-area features alone gave $R^2 = 0.887$, whereas combinatorial counts alone gave 0.043. The comparison points to information in the ridge areas beyond those counts. Its split and feature list differ from the random-forest experiment, so the two scores do not rank the learning algorithms against each other.

8.5 Returning from a pattern to geometry

The ridge result suggests a first, deliberately imprecise idea: reducing symplectic area on the two-dimensional faces might favor a large systolic ratio. This is a starting point for interpretation. It leaves open what should be held fixed, whether a statement concerns individual bodies or a sampling distribution, and how face areas could constrain capacity.

The HKO body supplies a useful reference. Its normalized ridge sum is 8.944, close to the low end of the retained sample. However, the sample minimum is smaller, 8.926. Thus even this particularly successful body does not give a separating rule of the form “smaller ridge sum than

every ordinary example.” The observed association motivates selection; it does not establish a monotone geometric law relating R to capacity.

Products allow us to unpack the descriptor further. Let u_i and v_j be the oriented edge vectors of P and Q . A ridge lying entirely in a q - or p -plane has zero symplectic area. A mixed ridge is the rectangle formed from one edge in each factor, and its symplectic area is $|u_i \cdot v_j|$. Consequently

$$R(P \times Q) = \frac{\sum_{i,j} |u_i \cdot v_j|}{\sqrt{\text{area}(P) \text{area}(Q)}}.$$

For a fixed u_i , the sum over the edges of Q is the total variation of the projection of its boundary onto the direction of u_i . Convexity gives one increasing and one decreasing chain between the extreme projections, so

$$\sum_j |u_i \cdot v_j| = 2\|u_i\| \text{width}_Q(u_i/\|u_i\|).$$

The descriptor therefore combines the edge lengths of one factor with the widths of the other in the edge directions. This gives concrete geometry to investigate instead of treating the successful column as an unexplained score.

It also helps test the scope of the tentative idea. For two aligned unit squares, eight edge pairs have absolute scalar product one and the rest zero, so $R = 8$. Their product is the unit four-dimensional cube, whose capacity is one and whose systolic ratio is $1/2$.¹ HKO has the larger ridge score 8.944 and a ratio greater than one. Thus a universal claim that a smaller ridge score forces a larger systolic ratio is false. The strong negative association in the sampled distributions needs a more specific explanation.

Later analyses added ordinary Euclidean ridge area $E(F)$ and the ratio $\kappa(F) = A_\omega(F)/E(F)$ for a ridge of positive area. These separate its size from the alignment of its plane with the symplectic form, relative to the Euclidean metric. They need not survive a nonorthogonal symplectic transformation. Normalizing $E(F)$ by the square root of volume removes common scaling, but does not give symplectic invariance. Thus the broader collection contained both invariants and non-invariants. These later measurements were used to interpret the ridge signal; they were not inputs to the original 39-feature prediction experiment.

The factorization

$$A_\omega(F) = E(F)\kappa(F)$$

separates these two contributions. In the generic ten-facet selection described below, the low-ridge panel had smaller mean normalized Euclidean ridge area than controls (0.917 versus 1.365), and smaller Euclidean-area-weighted alignment factor (0.377 versus 0.503). Both components therefore moved with the coarse filter. Within the selected panel, however, the ten most extreme ridge scores had slightly larger ordinary area and smaller alignment factor than the other ninety, without a larger mean ratio. This decomposition does not isolate a cause. It explains why the next question cannot simply be which component to drive lower without considering the rest of the shape.

For rotated regular pentagons, the descriptor does have an exact relation to capacity. Let $K_\theta = P_5 \times \text{Rot}(\theta)P_5$, with circumradius one, and let $d(\theta) = \min_{k \in \mathbb{Z}} |\theta - k\pi/5|$. Each edge has length $2 \sin(\pi/5)$ and each factor has area $5 \sin(\pi/5) \cos(\pi/5)$. The mixed-edge formula therefore gives

$$R(K_\theta) = 4 \tan(\pi/5) \sum_{k=0}^4 |\cos(\theta + 2\pi k/5)| = 16 \sin(\pi/5) \cos d(\theta).$$

¹For the centered unit cube the dual rows are $\pm 2e_i$. In the variational formula, closure forces equal weights x_j on the two q_j rows and equal weights y_j on the two p_j rows, with $\sum_{j=1}^2 (x_j + y_j) = 1/2$. The order of the four rows in a coordinate pair gives a contribution at most $8x_j y_j$ to Q . Thus $Q \leq 8 \sum_j x_j y_j \leq 1/2$. A cyclic word on one coordinate pair with its four weights equal to $1/4$ attains $Q = 1/2$, giving $c = 1/(2Q) = 1$.

For the second equality, the absolute-cosine sum is even and has period $\pi/5$. On $[-\pi/10, \pi/10]$, the terms indexed by 0, 1, 4 are nonnegative and those indexed by 2, 3 are nonpositive. Pairing opposite indices reduces the sum to $[1 + 2\cos(2\pi/5) - 2\cos(4\pi/5)]\cos\theta = 4\cos(\pi/5)\cos\theta$. Combining this with the capacity profile proved in Theorem 9.1 yields

$$\text{sys}(K_\theta)R(K_\theta)^2 = 16(3 + \sqrt{5}). \quad (35)$$

Thus smaller normalized total unsigned ridge area means larger systolic ratio within this rotation family. This supplies a geometric explanation for one instance of the observed inverse association.

Other rotation families behave differently. In products with side counts 3×6 and 4×4 , both R and the numerical systolic ratio decrease across the four sampled positions in each family. Together with the square–HKO comparison, these examples show why reducing ridge area cannot serve as a universal ascent rule. We next test a weaker possibility: using ridge area to select promising new bodies from a population.

8.6 Testing whether a pattern can guide fresh selection

Alongside geometric interpretation, we tested the signal as a search rule. A predictor can describe an existing dataset without helping on new bodies. We therefore changed the order of computation: generate fresh geometry, compute its descriptors, fix the selected bodies and comparison bodies, and only then evaluate their ratios. Selection cannot use the systolic ratios at this stage.

An initial pool of 100,000 product candidates tested thirty scalar selection sets. These included low ridge sums and other ridge summaries, as well as high face counts, selecting globally or separately within side-count groups. For example, the per-group low-sum rule selected the ten smallest values of R in each group. The union of all selections and their baselines required 1,675 evaluations, not 100,000. Its maximum recorded ratio was 0.868. Several rules selected the same bodies; their outcomes cannot be added as independent repetitions. The next experiments isolated more specific rules.

The concentration experiment generated a new pool of 100,000 candidates. Within each side-count group, it first retained the lowest one percent by R . It then divided this tail into two equally sized parts according to C . The question was whether spreading the already small total symplectic area more evenly would select better bodies. The half with smaller C had a mean recorded ratio larger by 0.0340 than the other half. The difference was positive in eight of the ten groups. The run evaluated 2,490 bodies and reached a maximum numerical ratio of 0.902. Appendix A.3 records the selection and continuation rules.

A different descriptor came from the primal vertices. Let Γ be their population covariance matrix: subtract the mean vertex and average the outer products of the centered vectors. Its two Williamson eigenvalues $0 < \nu_1 \leq \nu_2$ describe its symplectic scales. Equivalently, the eigenvalues of $J\Gamma$ are $\pm i\nu_1, \pm i\nu_2$, where $J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$. We used

$$\rho(K) = \nu_2/\nu_1$$

to favor bodies whose vertex covariance has similar scales in the two symplectic directions. This uses geometry alone. It is unchanged by common dilation and linear symplectic transformations, as well as translation because the covariance is centered.

Two independently seeded pools of 50,000 products compared three selections in every side-count group: the lowest 0.5% by ρ , the ridge rule above (lowest 1% by R , then its lower half by C), and 25 disjoint controls. There were 500 memberships in each selection across the two pools. Because 64 bodies belonged to both geometric selections, 1,436 distinct bodies were evaluated.

The mean difference between the covariance selection and controls was 0.333, with an interval $[0.310, 0.356]$. Against the ridge selection, the difference was 0.0145, with interval $[-0.0149, 0.0439]$. These intervals use the twenty seed-by-group differences and a Student- t calculation with nineteen degrees of freedom. They do not represent twenty independent seeds. The covariance rule selected much larger recorded ratios than controls, while this comparison did not establish an advantage over ridge selection. No recorded value above one was found in these selection experiments.

8.7 Transfer and the limit of a stronger filter

The ridge observation first came from a mixture of generic polytopes and products. To test a concrete selection rule on generic ten-facet bodies, we generated 10,000 fresh candidates and selected the lowest one percent by *mean* normalized ridge area, $R(K)$ divided by the ridge count. Generic bodies with the same facet count can have different ridge counts, so this differs from selecting by the sum. The 100 selected bodies had mean recorded ratio 0.626, compared with 0.339 for 100 disjoint controls. The difference was 0.287, with a percentile-bootstrap interval $[0.250, 0.324]$ from 20,000 resamples of the realized panels.

The same experiment asked whether a more extreme filter would help further. The ten bodies in the lowest 0.1% were compared with the other ninety selected bodies. Their mean difference was -0.0318 , with interval $[-0.0946, 0.0331]$. This failed the criterion for expanding that particular search to a larger pool. It did not establish that stronger filtering is harmful: the interval permits effects of either sign. The useful contrast is that a coarse geometric filter worked well while the next restriction did not show an additional gain.

We also changed the product sampling law. Starting from the original random normals and heights, we replaced every support height in both factors by one. Each side of the resulting polygon lies on a tangent line to the unit circle. We required both the original factors and the changed factors to be admissible, and then separately normalized the two changed factors to area one. This produces a conditioned family of tangential polygons.

On one fresh pool with 3,200 eligible bodies in each of the 4×6 and 6×6 groups, ridge and covariance selection were compared with shared, disjoint controls. Each selection had sixteen memberships per group; five selected bodies overlapped, giving 91 actual evaluations. Averaging the two group effects equally, relative to controls, covariance selection gained 0.259 and ridge selection gained 0.225. Their respective resampling intervals were $[0.192, 0.329]$ and $[0.160, 0.296]$. These intervals condition on the realized panels; one seed and two side-count groups do not measure variation across arbitrary source laws. The shared controls also make the comparisons dependent. The experiment nevertheless gives a concrete instance in which both rules remained useful after changing how bodies were generated.

8.8 Changing support heights and symplectic alignment

The transfer experiment selected among tangential products. It did not isolate the effect of making a particular polygon tangential. For that question we kept the normals fixed and compared the original product with three changes: equalize the first factor's heights, equalize the second factor's heights, or equalize both. Sixteen independently generated pairs, eight of type 4×4 and eight of type 4×6 , supplied 64 bodies. We accepted a pair only if all four constructions were valid, before evaluating any ratio. The paired result therefore concerns this jointly admissible source.

Each factor was then scaled to area one. Separate factor dilations preserve the mathematical ratio, since

$$\text{diag}(aI_2, bI_2) = \sqrt{ab} \text{diag}(\sqrt{a/b} I_2, \sqrt{b/a} I_2)$$

is a common dilation followed by a symplectic map. Replacing unequal support heights is the

shape change being tested. Changing both factors raised the mean numerical ratio by 0.0153: eleven pairs improved and five declined. The mean changes were 0.0040 for 4×4 and 0.0266 for 4×6 . All 64 evaluations succeeded. An average improvement therefore did not give a rule that equalizing the heights helps every body.

Rotation changes a different aspect of the geometry. An orthogonal map preserves Euclidean shape but need not preserve ω_0 . A rotated Lagrangian product can consequently have a different capacity and need no longer be a Lagrangian product for the fixed symplectic form.

We compared two allocations of sixteen evaluations. One evaluated a fresh product, a symplectic rotation as a control, and fourteen prescribed orthogonal rotations. The other evaluated sixteen fresh products, sharing the first body with the rotation allocation. We repeated the comparison four times, twice for 3×3 and twice for 4×4 . The rotation schedule and all source bodies were fixed before target evaluation. Shared starts gave 124 distinct requests and 128 charged memberships. The symplectic controls preserved the numerical ratios within $4 \cdot 10^{-16}$.

Rotation improved three of the four starting bodies. Fresh sampling still produced the larger maximum in every comparison. In one 4×4 case, rotation raised the ratio from 0.328 to 0.529, while fresh sampling reached 0.774. The rotation evaluations also took longer. For these four starts and this prescribed schedule, the same number of evaluations was more effective when spent on fresh bodies.

8.9 Adapting the sampling distribution

Fresh sampling won the rotation comparison, but it repeatedly draws from the same law. A further experiment tested whether evaluated bodies could improve that law. The cross-entropy method [Boe+05] used here repeatedly fits a simple distribution to the best part of a population, then samples another population from it.

This experiment used 5×5 products. We described each pentagon by the cyclic gaps between its five normal angles and the relative lengths of its five dual normals. There are four independent angular-gap coordinates and four independent logarithmic lengths per factor. One relative angle between the factors gives seventeen coordinates altogether. Separate factor scales and a simultaneous planar rotation do not change the mathematical ratio, which allows their removal from this description. Appendix A.5 specifies the coordinates and update.

Both arms began with the same 64 independent products from the original angle-and-height source. The adaptive arm fitted a diagonal Gaussian in these coordinates, then moved its mean and coordinate variances halfway toward those of the sixteen bodies with largest numerical ratio. It drew another 64 bodies, evaluated them, and repeated this update. Each arm had four populations including initialization, for 256 charged evaluations. The comparison arm continued independent sampling throughout. A variance floor kept the adaptive sampler from eliminating a coordinate’s variation entirely.

Three seeds produced the following results. The top-eight statistic is the median of the eight highest ratios after removing repeated, exactly equal dual-vertex arrays in an arm; different bodies with equal scores remain. It tests whether improvement extends beyond a single best observation.

Seed	Maximum		Top-eight median	
	IID	Adaptive	IID	Adaptive
1	.760	.799	.729	.782
2	.811	.831	.754	.776
3	.760	.856	.722	.814

The adaptive arm was better on both statistics for all three seeds. The shared initial populations account for 1,344 actual evaluations versus 1,536 charged arm memberships. All actual requests

succeeded and none gave a numerical ratio above one. Three paired runs give a useful small comparison, without establishing superiority across other generators or choices of population size and update rule.

Both arms rejected bodies outside the evaluator’s supported coordinate range. Because the adaptive arm reconstructs a normalized representation, this condition acts differently on its proposals and the original IID coordinates; the reported comparison includes that conditioning. The complete run took 216 seconds, most of it in construction and admission checks. Appendix A.5 gives the admission rule, its revision after an initial failed run, and the measured costs.

8.10 Local search from matched starting bodies

The adaptive sampler changes a distribution over products. A separate, earlier experiment changed the dual normals of generic ten-facet bodies directly. Seven fixed implementations started from the same 64 bodies. Each run allowed at most 128 charged full evaluations and a one-second accumulated evaluator-plus-optimizer time ceiling, excluding the initial evaluation. Invalid proposals still consumed calls. Work begun below the time ceiling was allowed to finish, so terminal results can include a small overrun; they are not values available at exactly one second.

The capacity calculation suggests local branch functions, each associated with a particular candidate in the variational problem. When several smooth functions compete through their minimum, following the gradient of one selected function can fail to increase the minimum. For example, the minimum of x and $-x$ decreases in both directions away from zero, although either individual function has a direction of increase. This motivates methods that include more than the currently selected branch.

We compared a literal single-branch gradient, an improvement-only version of that gradient, and three methods that model several competing branches. The finite-gap method uses branches close to the current minimum; the two history methods also retain branches encountered at earlier points. One history method predicts additional branches from transitions. CMA-ES [Han16], which adapts a Gaussian search population, and a poll along coordinate directions provide two further comparisons. The precise steps, symmetry directions removed, and acceptance rules are recorded in Appendix A.

Implementation	Median best value	Across-start 10–90% range
Four-anchor history	.985	.957–.999
History with transition prediction	.982	.956–.994
Finite-gap model	.977	.948–.995
Literal gradient	.881	.790–.924
Safeguarded gradient	.855	.780–.916
CMA-ES	.818	.718–.869
Coordinate poll	.601	.358–.787

Each entry summarizes 64 terminal best-so-far recorded values. The ranges show variation across starts, not confidence intervals for the medians. All 448 values remained below one. Comparing methods within the same start, the median history-minus-finite-gap difference was 0.00883, with an existing percentile-bootstrap interval [0.00335, 0.01533]. The bootstrap resampled the 64 matched starts 4,000 times. This paired median is not the difference of the two medians in the table.

The table ranks these implementations under the recorded allocation and historical evaluator. Its resampling does not account for configuration selection, untested starting distributions or evaluator error. The original producer tree and the individual proposal/evaluation trace tables are not locally available; retained run records and analysis identities support the reported

aggregates, rather than a replay of the executable. The historical objective used floating-point geometry and volume. These limits prevent interpreting the table as a certified ordering of mathematical capacities or as a comparison of optimally tuned algorithm families.

8.11 What a failed local probe tells us

A large terminal value also raises a geometric question: has search arrived at a local maximum? Evaluating a finite set of nearby points can discover an improvement, but failing to find one does not prove local maximality. The endpoint diagnostics gave concrete examples of this distinction.

Sixteen endpoints of the history method were chosen across the observed performance range of the common starts. At each, we evaluated both signs of twenty-five transverse basis directions at relative radii 10^{-3} , 10^{-4} and 10^{-5} : 150 probes per endpoint. Seven endpoints had at least one raw positive change and nine had none. The sixteen endpoints were selected by outcome ranks, so these fractions do not estimate a population convergence probability.

After allowing a larger search budget on those starts, a further continuation experiment examined three selected optimizer endpoints and HKO. It tested branch-model and selected-gradient proposals at five decreasing step lengths, falling back to a signed basis poll, and allowed at most ten accepted improvements. All three optimizer endpoints took ten improving steps. One had previously decreased along every tested signed basis direction, yet improved by 0.00223 along a sequence of oblique moves with total normalized path length 0.01. HKO accepted no move. The theorem establishes HKO’s local status; the finite search supplies a control observation consistent with it.

Another fixed suite probed a regular triangle–hexagon product, a rotated square–square product and the CH2021 example, together with a regular pentagon product chosen on a known increasing rotation path. It combined signed transverse basis directions, projected random directions and prescribed rotations at three radii. All 1,482 probes were valid. None of the three target bodies had a nominal gain above 10^{-12} , while the pentagon control had twelve, all in the rotation probes. Its generic basis and random probes found none. Even this control therefore shows that a finite collection of generic directions can miss an available increase. The target misses remain numerical diagnostics; they do not exclude untested directions or changes in facet count. The historical run also has incomplete source provenance, and its broad capacity-derived bounds do not turn the nominal target misses into a proof.

8.12 What remains unexplained

Ridge area was useful in two different ways. It exposed a strong association in the initial sample, and selection based on it enriched fresh pools. For regular pentagon products, the edge formula and capacity profile explain an inverse-square relationship. The sampled population is much broader: its association persists within every group, but the square–HKO comparison and opposing rotation paths rule out a universal monotonicity explanation. An explanation for the population pattern must account for the distribution of shapes as well as the symplectic areas of their faces.

The search experiments also distinguish selection from deformation. A stronger filter did not always give a better selection. Equalizing support heights improved some bodies and harmed others. Rotations improved individual bodies but lost to fresh sampling under the tested allocation. Adapting the product distribution and retaining several local branches both helped in their respective comparisons. The endpoint tests show what the latter methods still missed: a basis poll could report no improvement even when nearby oblique directions allowed further ascent.

None of the random samples, prospective selections, adaptive runs or matched-start optimizer runs produced a numerical ratio above one. The deliberately supplied HKO and pentagon controls are separate. For future above-one search, products with a triangle or quadrilateral

factor are ruled out unless the deformation leaves the class covered by the theorem. The remaining product families and generic polytopes still present both questions pursued here: which geometric quantities explain capacity, and which changes in a body can increase its systolic ratio?

9 The rotated-pentagon profile

The Haim–Kislev–Ostrover counterexample is a Lagrangian product of two regular pentagons. We determine how its capacity changes when one pentagon rotates relative to the other. The volume stays constant, so this also determines which relative positions maximize the systolic ratio.

Let $P_5 \subset \mathbb{R}^2$ be a regular pentagon centred at the origin with circumradius one. Write $R(\theta)$ for counterclockwise rotation through angle θ , and put

$$K_\theta = P_5 \times_L R(\theta)P_5 \subset \mathbb{R}_q^2 \oplus \mathbb{R}_p^2.$$

The Lagrangian product here is the Cartesian product in the two coordinate planes, with symplectic form $\omega_0((q, p), (q', p')) = q \cdot p' - p \cdot q'$. Simultaneous rotation of both planes is symplectic, so the choice of initial orientation for P_5 does not affect the capacity. We use

$$\text{sys}(K) = \frac{c_{\text{EHZ}}(K)^2}{2 \text{vol}_4(K)}.$$

The sampled ratios in Figure 9 rise from the aligned position to a maximum at $\theta = \pi/10$, then fall symmetrically. They suggested a profile proportional to $\sec^2 \theta$ on $[0, \pi/10]$. We prove this formula by comparing the quadratic programs at different angles. Their feasible weights remain the same under rotation, and each fixed choice has a particularly simple angular dependence. The known capacity at the Haim–Kislev–Ostrover position will then bound every such choice throughout the interval.

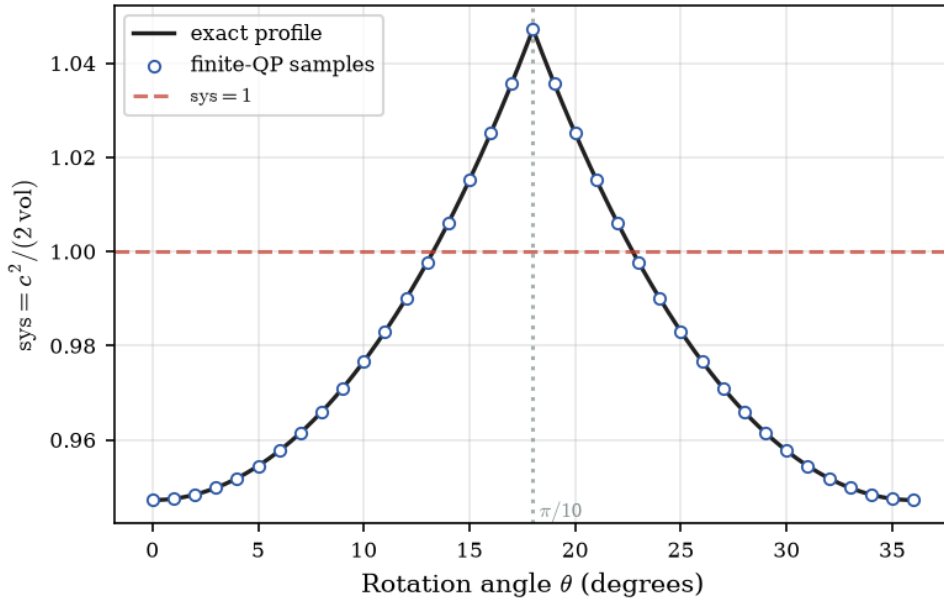


Figure 9: Systolic ratios for the rotating pentagon product. The blue markers are numerical evaluations of the finite quadratic programs; the black curve is the formula proved in this chapter. The horizontal line marks $\text{sys} = 1$.

To state the profile for all angles, define

$$d(\theta) = \min_{k \in \mathbb{Z}} \left| \theta - \frac{k\pi}{5} \right|.$$

This is the distance to the nearest multiple of $\pi/5$, and takes values in $[0, \pi/10]$.

Theorem 9.1 (Rotated-pentagon profile). *For every real θ ,*

$$c_{\text{EHZ}}(K_\theta) = (1 + \cos(\pi/5))^2 \sec d(\theta), \quad (36)$$

$$\text{sys}(K_\theta) = \frac{5 + 2\sqrt{5}}{10} \sec^2 d(\theta). \quad (37)$$

The maximum systolic ratio is $(3 + \sqrt{5})/5 > 1$, attained exactly at $\theta = \pi/10 + k\pi/5$, $k \in \mathbb{Z}$.

9.1 Symmetries and the Haim–Kislev–Ostrover position

We first reduce the angles to $[-\pi/10, \pi/10]$ and identify the capacity at its endpoints. The pentagon's rotational symmetry gives $K_{\theta+2\pi/5} = K_\theta$. If S is reflection in a symmetry axis of P_5 , the simultaneous reflection $(q, p) \mapsto (Sq, Sp)$ is symplectic and sends K_θ to $K_{-\theta}$. Thus capacity is even as a function of θ .

Exchanging the two factors gives a further symmetry. The symplectic map $(q, p) \mapsto (p, -q)$ sends K_θ to $R(\theta)P_5 \times_L (-P_5)$. Rotate both factors through $-\theta$ and use $-P_5 = R(\pi/5)P_5$ to obtain $K_{\pi/5-\theta}$. Together with evenness, this proves that capacity has period $\pi/5$. Every angle can therefore be reduced to $[-\pi/10, \pi/10]$ without changing capacity; evenness will give the final expression in terms of $d(\theta)$.

Haim–Kislev and Ostrover computed the capacity of the product with relative angle $-\pi/2$. Their convention for the symplectic form is $\sum_i dp_i \wedge dq_i = -\omega_0$. The map $(q, p) \mapsto (q, -p)$ changes this convention to ours and sends their body to $K_{\pi/2}$. Since $\pi/2 = \pi/10 + 2\pi/5$, their result [HO26, Proposition 1.4], together with the reflection symmetry, gives

$$c_{\text{EHZ}}(K_{-\pi/10}) = c_{\text{EHZ}}(K_{\pi/10}) = 2 \cos(\pi/10)(1 + \cos(\pi/5)). \quad (38)$$

9.2 Feasible weights under rotation

The quadratic capacity formula lets us compare different angles using the same facet order and weights. We spell out its variables for this family. Set

$$h = \cos(\pi/5), \quad n_i = \left(\cos\left(\frac{\pi}{2} + \frac{2\pi i}{5}\right), \sin\left(\frac{\pi}{2} + \frac{2\pi i}{5}\right) \right), \quad 0 \leq i < 5.$$

Choose the orientation $P_5 = \{q : n_i \cdot q \leq h, 0 \leq i < 5\}$. The ten facets of K_θ then have normalized normals

$$a_i(\theta) = h^{-1}(n_i, 0), \quad a_{i+5}(\theta) = h^{-1}(0, R(\theta)n_i), \quad 0 \leq i < 5,$$

so that their inequalities are $a_i(\theta) \cdot x \leq 1$.

A facet word $\sigma = (\sigma_1, \dots, \sigma_m)$ is an ordered list of distinct indices from $\{0, \dots, 9\}$. Its feasible weights satisfy

$$\beta_k \geq 0, \quad \sum_{k=1}^m \beta_k = 1, \quad \sum_{k=1}^m \beta_k a_{\sigma_k}(\theta) = 0. \quad (39)$$

The last equation is closure. For these weights, define

$$Q_{\sigma, \beta}(\theta) = \sum_{1 \leq j < k \leq m} \beta_j \beta_k \omega_0(a_{\sigma_j}(\theta), a_{\sigma_k}(\theta)). \quad (40)$$

This uses the traversal order of the quadratic-program chapter. The Haim–Kislev formula [Hai19] states that

$$c_{\text{EHZ}}(K_\theta) = \frac{1}{2Q_{\max}(\theta)}, \quad Q_{\max}(\theta) = \max_{\sigma, \beta} Q_{\sigma, \beta}(\theta) > 0, \quad (41)$$

where the maximum is over all words and their feasible weights. There are finitely many words, and each feasible set is a closed subset of a simplex, so the maximum exists. Zero weights allow facets to be omitted.

For this product, closure separates into equations in the two planes:

$$\sum_{\sigma_k < 5} \beta_k n_{\sigma_k} = 0, \quad R(\theta) \sum_{\sigma_k \geq 5} \beta_k n_{\sigma_k - 5} = 0.$$

Removing the invertible rotation from the second equation shows that feasibility is independent of θ . Thus a word and weights chosen at one angle remain feasible at every other angle.

We first exhibit a choice that gives the proposed capacity as an upper bound. Take

$$\sigma = (3, 8, 1, 0, 5, 6), \quad \beta = \frac{1}{4(1+h)}(2h, 2h, 1, 1, 1, 1).$$

These weights are positive and sum to one. The normal n_3 points opposite the bisector of n_0 and n_1 , and hence

$$n_0 + n_1 = -2hn_3.$$

This identity gives closure in both factors. To evaluate Q , group the consecutive entries $(1, 0)$ and $(5, 6)$ of the word. Normals from the same factor have zero symplectic pairing, so bilinearity lets us replace each group by its weighted sum. Writing

$$u = \frac{n_3}{2(1+h)}, \quad v = R(\theta)u,$$

the four weighted sums, in their order, are

$$(u, 0), \quad (0, v), \quad (-u, 0), \quad (0, -v).$$

Their ordered symplectic pairings sum to $2u \cdot v$. Therefore

$$Q_{\sigma, \beta}(\theta) = \frac{\cos \theta}{2(1+h)^2}. \quad (42)$$

For $|\theta| \leq \pi/10$ this is positive, and the capacity formula yields

$$c_{\text{EHZ}}(K_\theta) \leq (1+h)^2 \sec \theta. \quad (43)$$

To prove equality, we must show that no other feasible choice gives a larger quadratic value.

9.3 Bounding all choices by their endpoint values

Fix any facet word σ and feasible weights β . Same-factor pairings in (40) vanish, and each mixed pairing is, up to sign, $h^{-2}n_i \cdot R(\theta)n_j$. There are consequently constants $A_{\sigma, \beta}$ and $B_{\sigma, \beta}$ such that

$$Q_{\sigma, \beta}(\theta) = A_{\sigma, \beta} \cos \theta + B_{\sigma, \beta} \sin \theta.$$

This angular dependence determines the value inside our interval from the two endpoint values. The sine addition formulas give

$$\begin{aligned} Q_{\sigma, \beta}(\theta) &= \frac{\sin(\pi/10 - \theta)}{\sin(\pi/5)} Q_{\sigma, \beta}(-\pi/10) \\ &\quad + \frac{\sin(\pi/10 + \theta)}{\sin(\pi/5)} Q_{\sigma, \beta}(\pi/10). \end{aligned} \quad (44)$$

For $-\pi/10 \leq \theta \leq \pi/10$, both coefficients are nonnegative and their sum is $\cos \theta / \cos(\pi/10)$.

The fixed choice remains feasible at both endpoints, so its quadratic value there is at most the corresponding maximum. By (38) and (41),

$$Q_{\sigma,\beta}(\pm\pi/10) \leq Q_{\max}(\pm\pi/10) = \frac{1}{4\cos(\pi/10)(1+h)}.$$

We may substitute these upper bounds into (44) because its coefficients are nonnegative. This gives

$$Q_{\sigma,\beta}(\theta) \leq \frac{\cos \theta}{4\cos^2(\pi/10)(1+h)} = \frac{\cos \theta}{2(1+h)^2},$$

using $2\cos^2(\pi/10) = 1+h$. The bound holds for every feasible choice, and the choice in (42) attains it. We have proved

$$Q_{\max}(\theta) = \frac{\cos \theta}{2(1+h)^2}, \quad c_{\text{EHZ}}(K_\theta) = (1+h)^2 \sec \theta, \quad |\theta| \leq \pi/10.$$

Evenness and $\pi/5$ -periodicity give (36) for all real angles.

Finally, the five triangles joining the centre of P_5 to its edges give

$$\text{area}(P_5) = \frac{5}{2} \sin(2\pi/5), \quad \text{vol}_4(K_\theta) = \left(\frac{5}{2} \sin(2\pi/5) \right)^2.$$

Substitution into the definition of the systolic ratio yields

$$\text{sys}(K_\theta) = \frac{(1 + \cos(\pi/5))^4}{2 \left(\frac{5}{2} \sin(2\pi/5) \right)^2} \sec^2 d(\theta) = \frac{5 + 2\sqrt{5}}{10} \sec^2 d(\theta).$$

Since $\sec^2 t$ is strictly increasing on $[0, \pi/10]$, the maximum occurs precisely when $d(\theta) = \pi/10$. At those angles its value is $(3 + \sqrt{5})/5$. This completes the proof of Theorem 9.1.

10 Independent linear deformations of the pentagon factors

The rotation formula leaves open what happens when the two pentagons are stretched or sheared independently. Such changes generally alter both capacity and volume. We show that they cannot increase the systolic ratio beyond the Haim–Kislev–Ostrover value. Translations do not affect either quantity, so linear maps suffice to describe the extension to affinely regular pentagons.

We use a regular pentagon with circumradius one and outward facet normals

$$n_i = (\cos(2\pi i/5), \sin(2\pi i/5)), \quad P = \{q : n_i \cdot q \leq h, \ 0 \leq i < 5\}, \quad h = \cos(\pi/5).$$

This normal convention differs by a common rotation from the preceding rotation proof; applying that rotation to both coordinate planes is symplectic. Its area is $S = \frac{5}{2} \sin(2\pi/5)$. Throughout this section the product is Lagrangian, with form $\omega_0((q, p), (q', p')) = q \cdot p' - p \cdot q'$.

Theorem 10.1. *Let $G, H \in GL(2, \mathbb{R})$, and set*

$$M = G^{-1}H^{-T}, \quad m(M) = \max_{0 \leq i, j < 5} |n_i^T M n_j|.$$

Then

$$c_{\text{EHZ}}(GP \times HP) = \frac{(1+h)^2}{m(M)}.$$

Moreover,

$$\text{sys}(GP \times HP) := \frac{c_{\text{EHZ}}(GP \times HP)^2}{2 \text{area}(GP) \text{area}(HP)} \leq \frac{3 + \sqrt{5}}{5}.$$

Write $D = \text{conv}\{\pm n_i : 0 \leq i < 5\}$ and $\alpha = \pi/10$. Equality holds precisely when $M = tR_\alpha U$, where $t > 0$ and U is an orthogonal symmetry of the regular decagon D .

The formula requires only twenty-five absolute pairings, even though the capacity formula initially ranges over all facet orders and feasible weights. The reason is a containment between convex sets of matrices. Unlike the endpoint comparison for rotations, this containment is preserved by independent linear changes in the two factors.

10.1 Reducing the possible facet orders

Write a polygon as $\{q : u_i \cdot q \leq 1\}$. Its normalized closure polytope is

$$C = \{\gamma \geq 0 : \sum_i \gamma_i = 1, \sum_i \gamma_i u_i = 0\}.$$

For the pentagon, $u_i = n_i/h$. The Haim–Kislev formula [Hai19, Theorem 1.1], with earlier word entries as the first argument of ω_0 , reads

$$c_{\text{EHZ}} = \frac{1}{2Q_{\max}}, \quad Q = \sum_{j < k} \beta_j \beta_k \omega_0(a_{\sigma_j}, a_{\sigma_k}),$$

where the product covectors a_i are support-normalized, the weights have sum one, and their weighted covector sum vanishes.

For a product, same-factor pairings vanish. If the total weight in the first factor is t , the objective therefore has the form $t(1-t)B(\gamma, \eta)$, where B is bilinear in the normalized closure weights of the factors. At a positive global maximum, maximizing successively in γ and η allows both to be chosen at vertices of their closure polytopes. The global value cannot increase, so these replacements preserve it. A vertex has at most three positive entries: more than three columns of the three closure equations admit a two-sided feasible perturbation. The remaining factor $t(1-t)$ is maximal at $t = 1/2$.

Thus some maximizer uses at most three facets of each factor. This is an existence statement; it does not assert that all maximizing words are sparse. Moving the first weighted covector cyclically to the end preserves Q , since the change is its pairing with minus twice the sum of all the other weighted covectors, and closure makes that pairing zero. We may consequently group adjacent entries from the same factor cyclically. There are two, four, or six alternating groups; the two-group value is zero.

Put $x_i = \gamma_i u_i$ and $y_j = \eta_j v_j$. These are factor steps with sum zero; their actual product weights carry an additional factor $1/2$. A four-group order $x, y, -x, -y$ has value $x \cdot y/2$. A six-group order $x_1, y_1, x_2, y_2, x_3, y_3$ has value

$$Q = \frac{1}{2}(x_1 \cdot y_1 - x_3 \cdot y_2).$$

The latter identity follows by pairing each p -step with its preceding q -prefix: those prefixes are $x_1, -x_3, 0$. Represent these expressions by matrices

$$\frac{1}{2}xy^T, \quad \frac{1}{2}(x_1 y_1^T - x_3 y_2^T),$$

whose Frobenius pairing with the identity matrix is the objective. Under independent changes G, H , both types transform by the same linear map $T \mapsto G^{-T}TH^{-1}$. It remains to show that every six-group matrix for regular pentagons lies in the convex hull of the four-group matrices.

10.2 Two identities cover all six-group competitors

Set

$$\lambda = \frac{\sqrt{5}-1}{2} = \frac{1}{2h}, \quad s = \frac{1}{1+h}, \quad \rho = \frac{s^2}{2}, \quad \mathcal{D} = \text{conv}\{\pm n_i n_j^T : 0 \leq i, j < 5\}.$$

A positive closure triple has all cyclic angular gaps less than π . Its three positive integer index gaps therefore sum to five and are at most two, so they are cyclic permutations of $(1, 2, 2)$. There are exactly five triples, all rotations of $\{n_0, n_2, n_3\}$, and no opposite pairs. The normalized steps for this representative triple are

$$se, sa, sb, \quad e = n_0, \quad a = \lambda n_2, \quad b = \lambda n_3, \quad e + a + b = 0.$$

Indeed the closure weights are $h/(1+h), 1/[2(1+h)], 1/[2(1+h)]$. Every proper grouped sum of a triple is one step or its negative. The convex hull of all such sums, over the five triples, is $s\mathcal{D}$: all $\pm sn_i$ occur, and the remaining steps are their contractions by $\lambda < 1$. It follows, by expanding independent convex combinations in each factor, that the convex hull of the four-group matrices is exactly $\rho\mathcal{D}$. Every generator is feasible: choose a triple with the specified singled-out step, and group its other two steps together.

Lemma 10.2. *The two matrices $T_1 = ee^T - aa^T$ and $T_2 = eb^T - ae^T$ belong to $\mathcal{D}$.*

Proof. They have the identities

$$\begin{aligned} T_1 &= \lambda^2 ee^T - \lambda^2 en_3^T - \lambda^3 n_1 n_2^T, \\ T_2 &= \lambda^2 n_1 n_3^T - \lambda^2 n_2 n_4^T. \end{aligned}$$

To verify them, substitute $n_2 + n_3 = -\lambda^{-1}e$, $\lambda n_1 = e + n_2$, and $\lambda n_4 = e + n_3$. The first right-hand side becomes $(\lambda^2 + \lambda)ee^T - \lambda^2 n_2 n_2^T = T_1$; the second becomes $\lambda(en_3^T - n_2 e^T) = T_2$. The unsigned coefficients in the first identity sum to $2\lambda^2 + \lambda^3 = 1$. Those in the second sum to $2\lambda^2 < 1$, with the remaining mass assigned to $0 \in \mathcal{D}$. The signs are part of the defining generators, so these are convex combinations, not signed combinations with negative weights. $\square$

Lemma 10.3. *Every six-group matrix belongs to the four-group convex hull $\rho\mathcal{D}$.*

Proof. Rotate each triple to the representative above and cut the word at the q -step se . After division by ρ , the matrix is $ev^T - wz^T$, where $w \in \{a, b\}$ and v, z are distinct members of $\{e, a, b\}$. Reflection $F = \text{diag}(1, -1)$ on the left exchanges a, b and fixes e , so it suffices to take $w = a$. Its six possibilities are

(v, z)	(e, a)	(e, b)	(a, e)	(b, e)	(a, b)	(b, a)
$ev^T - az^T$	T_1	$T_1 F$	$T_2 F$	T_2	$-F T_1 F$	$-F T_1$

The last two equalities also follow from $e + a + b = 0$. The set $\mathcal{D}$ is invariant under sign, independent left and right reflections, and independent pentagon rotations. Lemma 10.2 therefore covers all twelve cyclic orders for every pair of closure triples. $\square$

10.3 Capacity and the sharp ratio

Proof of Theorem 10.1. The matrix containment survives the map $T \mapsto G^{-T} T H^{-1}$. Thus the maximum is attained among transformed four-group matrices, giving

$$Q_{\max} = \rho \max_{i,j} |n_i^T G^{-1} H^{-T} n_j|.$$

Taking $1/(2Q_{\max})$ proves the capacity formula. Six-group words may tie this value; the argument only excludes a strictly larger value.

Since areas multiply by absolute determinants, the systolic ratio becomes

$$\frac{(1+h)^4}{2S^2} \frac{|\det M|}{m(M)^2}.$$

Put $L = M/m(M)$. The inequalities defining $m(M)$ say exactly that $LD \subseteq D^\circ$. The polar regular decagon is $D^\circ = \sec \alpha R_\alpha D$, hence comparison of areas yields

$$|\det L| \leq \sec^2 \alpha.$$

Now

$$\frac{(1+h)^4}{2S^2} = \frac{5+2\sqrt{5}}{10}, \quad \frac{5+2\sqrt{5}}{10} \sec^2(\pi/10) = \frac{3+\sqrt{5}}{5},$$

which proves the bound. Equality of areas under convex-body containment holds precisely when $LD = D^\circ$. Then $\cos \alpha R_{-\alpha} L$ is a linear automorphism of D . Such an automorphism permutes vertices and preserves or reverses cyclic adjacency. Its action on two adjacent vertices determines it and agrees with an orthogonal dihedral symmetry. This proves the stated equality classification; conversely each listed matrix attains equality. $\square$

For $G = I$ and $H = R_\theta$, the formula reduces to the rotation profile: $m(R_\theta) = \cos d_5(\theta)$, where $d_5(\theta) = \min_{k \in \mathbb{Z}} |\theta - k\pi/5|$. The shorter endpoint proof remains useful because it establishes that profile without the matrix classification. The original exhaustive SageMath argument is retained as computational provenance in the repository's `experiments/regular-products/` directory, under `pentagon-rotation-formula-proof/`; it is not an input to either analytic argument presented here.

The affine theorem concerns all independent linear images of regular pentagons, including arbitrarily large deformations. The separate local maximality theorem concerns nearby ten-facet polytopes and includes changes that destroy the product structure. The two results control different sets of perturbations. Neither is a bound for all ten-facet polytopes or for products of arbitrary pentagons.

11 Rotating a product out of symplectic position

The capacity of a product of two planar convex bodies depends on how its factor planes sit in symplectic space. If the planes are symplectically orthogonal, the capacity is the smaller factor area. We show that this arrangement never maximizes capacity among rigid rotations of the product: a small rotation increases capacity while preserving volume.

Let $A, B \subset \mathbb{R}^2$ be convex bodies with nonempty interiors, with areas $a \leq b$. We keep the Euclidean body $A \times B$ fixed and vary the symplectic structure. This is equivalent to rotating the body while keeping the symplectic structure fixed. With

$$J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J_\theta = \begin{pmatrix} \cos \theta J_2 & \sin \theta I_2 \\ -\sin \theta I_2 & -\cos \theta J_2 \end{pmatrix}, \quad 0 \leq \theta \leq \frac{\pi}{2},$$

write c_θ for the Ekeland–Hofer–Zehnder capacity relative to J_θ . At $\theta = 0$ the factor planes are symplectic, with opposite orientations, and $c_0 = a$. At $\theta = \pi/2$ both factor planes are Lagrangian; write $\ell = c_{\pi/2}$. The relative coordinates of A and B are fixed throughout this path. The parameter θ describes the symplectic form: the corresponding rigid rotation has angle $\theta/2$, as verified below.

Theorem 11.1. *For these factors,*

$$c_\theta = \frac{a}{\cos \theta} \quad \text{whenever} \quad 0 \leq \tan \theta \leq \frac{\ell}{2a}.$$

Consequently, every sufficiently small positive rotation along this path increases both capacity and systolic ratio.

Proof. First suppose that A and B are polygons containing the origin in their interiors. For each facet write $u_i = n_i/h_i$ for its support-normalized normal. Haim-Kislev's formula [Hai19, Theorem 1.1] expresses reciprocal capacity as the maximum ordered symplectic pairing sum, with factor 2, over weights satisfying

$$\lambda_i \geq 0, \quad \sum_i \lambda_i = 1, \quad \sum_i \lambda_i u_i = 0.$$

For a product, the last equation holds separately in each factor. Let t be the total weight on facets from A . Dividing those weights by t gives a feasible planar configuration, so their contribution to reciprocal capacity is at most t^2/a . The corresponding bound for B is $(1-t)^2/b$. Reversing a planar order accommodates the opposite sign in the second diagonal block of J_0 .

For the mixed contribution, divide the A weights by $2t$ and the B weights by $2(1-t)$. The resulting weights have total mass one, satisfy closure, and are feasible for the Lagrangian endpoint. Its reciprocal capacity is $1/\ell$, so the original mixed contribution is at most $4t(1-t)/\ell$. The cases $t = 0, 1$ have no mixed contribution. Linearity in the symplectic form therefore gives

$$\frac{1}{c_\theta} \leq \max_{0 \leq t \leq 1} \left[\cos \theta \left(\frac{t^2}{a} + \frac{(1-t)^2}{b} \right) + \frac{4 \sin \theta}{\ell} t(1-t) \right]. \quad (45)$$

Putting all weight in A also gives $1/c_\theta \geq \cos \theta/a$.

Set $U = \cos \theta/a$, $V = \cos \theta/b$ and $W = 4 \sin \theta/\ell$. Then $V \leq U$, and the expression in brackets minus U is

$$(1-t)[V - U + t(W - U - V)].$$

The inner bracket is affine in t ; its endpoint values are $V - U$ and $W - 2U$. Both are nonpositive when $\tan \theta \leq \ell/(2a)$. The two reciprocal-capacity bounds then agree. Polygonal approximation and continuity of capacity extend the bounds, and hence the formula, to arbitrary planar convex bodies.

Finally, $\ell > 0$, so the interval contains positive angles. For each such angle, $a/\cos \theta > a$. The underlying rigid rotation preserves the four-dimensional volume ab , hence also increases $c_\theta^2/(2ab)$. $\square$

To see the rotation explicitly, put

$$K = \begin{pmatrix} 0 & J_2 \\ J_2 & 0 \end{pmatrix}, \quad R_\theta = \cos(\theta/2)I_4 + \sin(\theta/2)K.$$

Since $K^T = -K$ and $K^2 = -I_4$, the matrix R_θ is orthogonal with determinant one. Direct multiplication gives $J_\theta = R_\theta J_0 R_\theta^T$. Thus c_θ is the capacity of $R_\theta^T(A \times B)$ with respect to J_0 . This rotation is not symplectic with respect to J_0 for positive θ in the displayed interval; symplectic rotations would preserve capacity.

Example 11.2. *The Lagrangian arrangement can have smaller capacity than the symplectic arrangement. Take*

$$A = \text{conv}\{(-1, -1), (2, -1), (-1, 2)\}, \quad B = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix} A.$$

Both areas are $9/2$. To show that the Lagrangian capacity is smaller, it suffices to exhibit one feasible configuration in Haim-Kislev's formula. The support-normalized normals are

$$u_0 = (0, -1), \quad u_1 = (1, 1), \quad u_2 = (-1, 0), \quad v_0 = (0, -2), \quad v_1 = (1/2, 2), \quad v_2 = (-1/2, 0).$$

Give all six normals weight $1/6$ and order them as $(u_0, v_0, u_1, v_1, u_2, v_2)$. Both triples sum to zero, so the configuration satisfies closure. Its signed cross-pairing sum is 5, giving

$$\frac{1}{\ell} \geq 2 \left(\frac{1}{6} \right)^2 5 = \frac{5}{18}, \quad \text{and hence} \quad \ell \leq \frac{18}{5} < \frac{9}{2} = c_0.$$

Remark 11.3. *If $\ell \leq a$, Theorem 11.1 gives capacities larger than both endpoint values at small positive angles. By continuity, the maximum along this path is attained in its interior. The triangle example satisfies this condition.*

12 Three-Dimensional Visualization

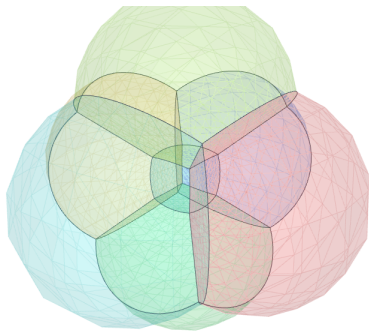
A four-dimensional polytope cannot be drawn directly, but its boundary can be viewed in three dimensions after choosing a projection. For a polytope $K \subset \mathbb{R}^4$ with the origin in its interior, we first apply the radial map

$$\partial K \longrightarrow S^3, \quad x \longmapsto \frac{x}{\|x\|},$$

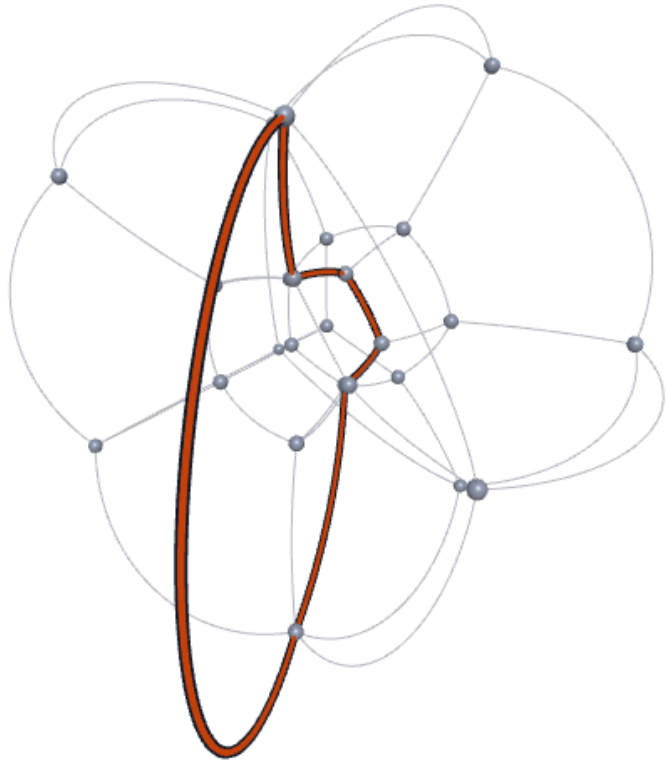
and then stereographically project S^3 from the north pole $e_4 = (0, 0, 0, 1)$ to $\mathbb{R}^3$. The viewer samples projected edges and Reeb segments along great-circle arcs. Its coloured two-face interiors are interpolated spherical meshes, included to make the boundary structure easier to see rather than to preserve metric geometry. The stereographic projection also distorts scale and clips arcs close to its pole.

Figure 10 shows projected two-faces of the four-dimensional cube and the projected one-skeleton of the HKO polytope K_{HKO} from Section 7 with one numerically recovered minimum-action generalized Reeb orbit. The producer exports only recovered orbits passing numerical closure and containment checks within fixed tolerances. The orbit shown is therefore a computed example, not a proof of a capacity statement.

The interactive views were also explored as a possible source of geometric patterns for our exploratory search. In practice, passing from four dimensions to a three-dimensional projection discarded too much visible Reeb-geometric information, and the impressions changed with the projection pole. We found no visual pattern that gave a reliable hypothesis or candidate-selection rule. For this exploration, the useful outcome is therefore limited to the explanatory role illustrated above.



(a) Projected two-face meshes of the four-dimensional cube.



(b) The HKO one-skeleton and a recovered minimum-action orbit.

Figure 10: Radial and stereographic views of four-dimensional polytopes. In (a), colour separates projected two-face meshes of the cube. In (b), the grey one-skeleton belongs to the Lagrangian product of two regular pentagons and the dark-outlined orange curve is a recovered six-segment closed Reeb orbit. Both panels use the pole e_4 and omit parts mapped beyond the viewer's clipping radius. They are qualitative explanatory views, not symplectic invariants or proof evidence.

13 Numerics

The computations in this thesis support two different kinds of claims. The HKO proof uses a separate SageMath packet over rational or ordered algebraic fields; the rotated-pentagon chapter uses an analytic proof. The Rust `capacity_4d` implementation computes the EHZ capacity of a four-dimensional polytope supplied in binary64 coordinates. It uses floating-point arithmetic for speed, but it does not accept an unverified floating-point sign or ordering decision: every such decision is enclosed by an outward error bound or resolved in exact rational arithmetic.

13.1 Input and output contracts

A finite binary64 number is a dyadic rational. The numerical capacity contract regards the supplied numbers as these exact rationals; it does not claim to recover unavailable source coordinates from which they may have been rounded. Before the QP search, exact geometry checks establish that

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1\}$$

is bounded, nonempty, full-dimensional, and described without redundant facets. The exact primal vertices and vertex–facet incidence are then constructed from the dyadic input. Floating-point filters may discard a subset only when a one-sided bound proves the exact conclusion; an indeterminate filter result is resolved exactly.

The production route accepts at most sixteen facets and requires every primal and dual vertex to have infinity norm in

$$[10^{-3}, 10^3].$$

These bounds are a deliberately narrow applicability policy, not a theorem about all polytopes. They exclude scales for which the current binary64 enclosures have not been justified and tested. The general candidate stream is additionally limited to 100 000 cyclic words. Violating any of these conditions returns an error rather than an uncertified capacity.

For a general polytope the result is an outward interval $[c_L, c_U]$ containing the exact capacity of the dyadic input. A scalar wrapper returns the midpoint only if this interval certifies the relative error requested by the caller. For a recognized Lagrangian product the specialized route returns the exact rational capacity of the dyadic input, together with outward binary64 endpoints for convenient downstream use.

13.2 Certified fixed-word decisions

For a word σ , write the equality-constrained stationary system in the symmetric form

$$\mathcal{K} \begin{pmatrix} \beta \\ \lambda \end{pmatrix} = \begin{pmatrix} H & C^\top \\ C & 0 \end{pmatrix} \begin{pmatrix} \beta \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ d \end{pmatrix}, \quad d = (0, 0, 0, 0, 1)^\top.$$

Here $C\beta = d$ is closure plus normalization and H is twice the quadratic form in (19). A numerical factorization supplies an approximate solution $\hat{x}$ and approximate inverse R . The following standard inverse-defect estimate is the central certificate.

Lemma 13.1 (Verified inverse defect). *Suppose outward arithmetic establishes*

$$\bar{r} \geq \|\mathcal{K}\hat{x} - b\|_\infty, \quad \bar{e} \geq \|I - \mathcal{K}R\|_\infty, \quad \bar{e} < 1.$$

Then $\mathcal{K}$ is invertible, and its exact solution x^ satisfies*

$$\|\hat{x} - x^*\|_\infty \leq \rho := \frac{\|R\|_\infty \bar{r}}{1 - \bar{e}}.$$

Proof. The Banach lemma applied to $\mathcal{K}R = I - (I - \mathcal{K}R)$ gives $\|\mathcal{K}^{-1}\|_\infty \leq \|R\|_\infty / (1 - \bar{\epsilon})$. Apply this bound to the residual $\mathcal{K}\hat{x} - b$. $\square$

Thus $\hat{\beta}_i - \rho > 0$ for every i certifies positive weights, while $\hat{\beta}_i + \rho \leq 0$ for some i certifies failure. If ξ is the multiplier for normalization, stationarity gives

$$2Q(\beta^*) + \xi^* = 0.$$

Consequently the same radius yields

$$Q(\beta^*) \in \left[-\frac{\hat{\xi} + \rho}{2}, -\frac{\hat{\xi} - \rho}{2} \right].$$

For $0 < q_L \leq Q \leq q_U$, monotonicity then gives the action enclosure

$$\frac{1}{2q_U} \leq \mathcal{A} \leq \frac{1}{2q_L}.$$

The implementation computes the residual and inverse-defect bounds with explicit roundoff and subnormal-error terms. It first uses a cheap induced norm bound and retries an indeterminate system with a tighter entrywise bound. The proof assumes round-to-nearest binary64 arithmetic with gradual underflow and the inspected matrix-multiplication evaluation order. The runtime checks gradual underflow; if the contract is unavailable, or if either bound remains indeterminate, the decision is made by exact rational arithmetic. A compiler, dependency, target, or arithmetic-contract change therefore requires the floating-point certificate tests to be repeated.

13.3 Complete general and product routes

The general search constructs the exact transition graph, pruned by facet incidence and symplectic signs, then enumerates its simple cyclic words. Candidate correctness is not obtained merely by solving every displayed KKT system. A fixed-word maximum must have nonpositive curvature in every tangent direction. Conversely, a certified vector $z \in \ker C_\tau$ with $z^\top H_\tau z > 0$ rules out τ . If τ injects into a longer word σ while preserving cyclic order, zero-extending z preserves both its tangent constraints and positive curvature. Hence σ is ruled out as well. Processing words by length allows these proved obstructions to skip many later factorizations.

For every word not excluded in this way, the route uses the verified fixed-word decision above, with exact fallback for every indeterminate case. The resulting objective intervals are aggregated outwards over the complete candidate family, and the reciprocal reverses the endpoints to enclose the capacity. Exact arithmetic is therefore a selective correctness fallback, not the default arithmetic for every word.

For a Lagrangian product, the inverse-defect route is correct but frequently indeterminate and unnecessarily expensive. The specialized implementation uses Algorithm 4.7. It encloses factor-closure weights and bilinear objectives in floating-point arithmetic, rejects a support or order only when the intervals prove it cannot win, and resolves ambiguous supports and every possible winner exactly. Because the final comparison is rational, this route returns the exact capacity and every tied sparse winner in the closure-vertex candidate family. It intentionally does not enumerate all minimizing geometric branches.

13.4 Empirical verification and limitations

The selected development route’s numerical audit compared the floating-point predicates with exact dyadic-rational arithmetic on 249 KKT systems: ordinary generated systems, scales from 10^{-2} through 10^3 , and systems approaching singularity down to binary64 coincidence. It found no incorrect determinate decision and no violation of a β - or Q -radius. Fifteen cases were

indeterminate and were resolved exactly. The largest observed ratios of actual error to certified radius were 0.00149 for β and 0.00081 for Q .

Complete selected-development-route comparisons used 88 words from three generated general polytopes and 1271 words from three product controls. Exact aggregation found no missed capacity and no missed minimizer class; every exact capacity lay in the returned interval. On the general cases, the largest difference between the printed central value and exact capacity was $2.31 \cdot 10^{-14}$. These finite comparisons test the implementation and exercise fallback, but the soundness claim rests on the preceding error bounds and complete-candidate arguments.

The selected product-development-route audit used 88 ordinary and adversarial products. Across 1835 closure weights and 173 496 objective intervals it found no interval violation, no incorrect determinate sign, and no disagreement between the hybrid and complete exact closure-vertex computations. Raw comparison with zero disagreed with the exact objective sign 23 559 times, mostly at exact zero or tiny values; the certified predicate returned indeterminate instead. A separate retained population of 10 240 random products had a literal capacity winner of length six in every case. This supports the implementation and stress-tests Theorem 4.6; it is not its proof and does not exclude longer tied minimizers.

A four-test correspondence check compares the selected development routes with the production implementation on fixed fixtures: general capacity bounds, an exact action window, derivatives of the retained words, and product capacities with their sparse winning words. All four tests pass; this is finite regression evidence, not all-input equivalence. The test source is `experiments/dev-quadratic-program/tests/selected_route_correspondence.rs`; its execution and fixture scope are recorded in `docs/numerics-correspondence-check/README.md`.

The guarantees above concern the exact dyadic polytope represented by the input. They do not propagate measurement error or prior rounding uncertainty in the coordinates, and they apply only within the stated facet, norm, and candidate-count limits. The HKO certificate instead represents its algebraic input in an exact ordered field. The analytic rotated-pentagon proof does not depend on these binary64 computations.

14 Availability of Code and Data

At the time of writing, the public development repository for this thesis is

<https://github.com/JoernStoehler/msc-math>.

The public repository was checked on 18 September 2026 at commit `1bec30637070`. The two certificate packets described below are present there. This review candidate and subsequent research additions are held in the working repository; they are not all included in that public commit. The repository URL follows ongoing development.

The HKO local-maximality proof and the original rotation computation have proof-facing packets in the public tree. The HKO packet in `experiments/hko-local-maximum/theorem/` contains the finite witness, the exact SageMath verifier, its recorded verification summary, and the Rust programs used to select the witness. Rust supplies finite choices; it is not trusted for theorem correctness. SageMath reconstructs the algebraic objects and checks the exact predicates stated in Section 7.4. Those predicates imply the theorem only together with the mathematical upper-function and slice argument given there.

The rotated-pentagon packet is stored under

`experiments/regular-products/
pentagon-rotation-formula-proof/`.

It contains the exact SageMath classifier and the recorded full-run transcript for all 3340 finite words left after the mathematical reduction. This is the original computational proof route; the analytic rotation and independent-linear-deformation proofs presented here do not use it. In

the computational route, the enumeration theorem, the translation from checked signs to action comparisons, and the continuity argument remain mathematical inputs. The repository exposes the finite computation and its output for inspection, with SageMath’s exact arithmetic and root isolation inside the software trust boundary. The maintained affine-pentagon argument and its source-convention check are in `formal/pentagon-affine-products/`.

Selected non-proof evidence is also tracked as ordinary repository files. In particular, the product-capacity implementation audit retains its 88-row numerical packet, and the historical first-order experiment retains its 12-run finite-step panel. The bounded generated-candidate experiment retains 1675 rows selected before the systolic ratio was computed, the corresponding 1675 evaluated rows, its frozen configuration, and compact reports. These records support only the finite experiments described in the corresponding chapters. They neither prove a population statement nor supply evidence for local maximality.

The repository is not a self-contained copy of every dataset used during the project. Its artifact registry identifies external snapshots and their intended local paths; a plain checkout does not materialize those files. For example, the 14336-row random/product provenance table is tracked, while the source random/product datasets and the method-facing invariant table are not tracked at those paths. The original source geometries have been recovered locally from the registered snapshot and matched against all 14336 table rows. They can be materialized through the `polytope-datasets` artifact; this recovery does not supply missing historical execution receipts. Separate current computations restored the feature schema for all bodies and returned capacity intervals for 14,335 of the 14,336 original geometries; one body remains outside the input-size policy. These new intervals do not certify the historical numerical ratios, whose volumes are floating-point values. Recovery and reevaluation receipts are in `docs/ds-retrospective-revalidation/`. Consequently, cloning the repository alone does not provide every input needed to repeat all empirical analyses, regenerate every figure, or reproduce every reported run. Packet-local instructions describe available producers and checks, but compatible software, dependencies, compute resources, and untracked inputs may still be required; timing and other environment-dependent fields need not reproduce byte for byte.

No frozen release, public Zenodo record, or DOI for this thesis is claimed here. Any archival deposit created later would be a separate, immutable object whose own public record, inventory, and hashes would have to establish its contents. The public commit identified above fixes which material was accessible at the time of this check; later local additions require a separate handoff.

15 AI in the research process

The use of AI in this project changed as the available models became more capable. My experience was that they supplied increasingly useful research ideas and could carry out longer tasks with fewer interruptions. This changed what I could delegate. A request could describe a mathematical or experimental outcome and leave the agent to choose intermediate steps, write code, inspect the results and revise its approach. The separate disclosure records AI use; here I discuss what this experience suggests about organizing mathematical work with agents.

15.1 Delegating an incomplete proof strategy

The local maximality proof is the clearest example of how much of a research task could be delegated. I asked a GPT-5-series model for a proof that the HKO body is a local maximum. I suggested looking for several upper bounds and somehow showing that together they control a neighborhood after quotienting out the symmetries. This was only the overall shape of a strategy. I had not specified a finite certificate or worked out how the upper bounds should be constructed.

The model developed essentially the entire argument from that starting point. It identified a sufficient configuration of twenty-six upper bounds in the twenty-five transverse dimensions. An initial derivative-based approach had provided only twenty-five nonsingular cases. The model then used the weights β in the quadratic objective to construct feasible sections: an invertible 5×5 closure minor lets five weights depend smoothly on the facet parameters while the remaining weights stay fixed. This supplied upper bounds even where the initial approach failed. It also completed proof steps involving the volume and wrote the SageMath code for the exact computations.

The assignment was therefore a conjecture and an incomplete proof idea; the result was a developed proof together with computational verification code. The derivative conditions and the form of the finite certificate were part of the model’s contribution, not conditions I supplied in the request. The useful specification was the mathematical outcome—a proof of local maximality—while the model had substantial freedom to discover what would establish it.

15.2 Matching the check to the claim

The different parts of this thesis require different evidence. In the HKO argument, arithmetic checks establish a finite statement about explicitly defined matrices and weights. The written proof still has to identify those matrices with derivatives of upper bounds for the geometric quantity. It also has to pass from labelled normals to a neighborhood of convex bodies. Omitting either connection would leave a mathematical gap even if every computed identity were correct.

For the search experiments, a high observed value answers a different question from a proof of a local maximum. The comparison of prescribed rotations with fresh sampling makes the distinction concrete: rotations improved several starting bodies, but fresh sampling reached the larger maximum in each of the four comparisons. An instruction merely to find an improvement could have returned a successful example while leaving the allocation question unanswered. Here the control determines what we learn from the computation.

The same issue occurs in more routine work. A retained code-repair example included a regression test that passed even when the original sign defect was restored. Testing the test against that defect revealed what an ordinary successful run did not establish. Such checks are useful because they target a particular possible mistake, rather than adding another undifferentiated approval of the result.

15.3 What still requires a reader

Mathematical exposition introduces a further difficulty. An agent can recover the meaning of a dense paragraph and judge it correct while a human reader has to stop repeatedly to decipher it. The intended reader of this thesis knows smooth symplectic geometry, but still needs an introduction to the nonsmooth variational problem and its finite reduction. A later definition does not remove the effort spent reading an earlier sentence that needed it.

This makes reading feedback valuable even when individual calculations have been checked. The useful feedback identifies where an explanation ceases to work: a missing implication, an unfamiliar object used before its definition, or a paragraph whose contribution remains unclear. Whether an agent can predict that feedback is itself an empirical question. Its ability to summarize a chapter does not establish that the chapter is comfortable to read. Nor does an automated check decide whether a mathematically correct side result deserves the space it occupies. These judgments concern the reader and the purpose of the thesis.

15.4 A connection with training on verifiable outcomes

There is a suggestive connection to reinforcement learning with verifiable rewards. Guo and collaborators trained DeepSeek-R1-Zero using rule-based accuracy and format rewards;

mathematical answers and code test outcomes supplied checkable feedback. Their later DeepSeek-R1 combined reinforcement learning with supervised fine-tuning and also used model-based rewards for general tasks [GYZ+25]. This is evidence that outcome feedback can shape learned problem-solving behavior, not that every useful judgment has a reliable automatic reward.

An outcome supplied in an agent’s task context is not a training reward: the task does not thereby update the model’s parameters. The practical connection is that both require a criterion aligned with the intended achievement. In the HKO example, the finite conditions that could be checked were themselves discovered during the research task. They were not a test suite supplied in advance. My experience motivates using agents for work with clear outcomes and checking whether the returned evidence actually supports them. It does not isolate the contribution of a particular training method or measure the time this project would have taken without AI.

16 Conclusion

Is the HKO body a local maximum of the systolic ratio among convex bodies? Our main theorem gives a partial answer: it is a local maximum among ten-facet polytopes. Every sufficiently nearby body has ratio at most $(3 + \sqrt{5})/5$, and equality occurs precisely under translations, positive dilations and linear symplectic maps. The restriction to ten facets is essential to the coordinate neighborhood used in the proof.

The argument also gives a useful way to work at a singular optimization problem. One can continue feasible choices and use their values as upper bounds, without continuing stationary optimizers. At HKO, finitely many such bounds suffice: after the symmetry directions are removed, their derivatives force a strict decrease in every direction. The algebraic calculation establishes the finite conditions, while the geometric proof identifies the upper bounds and relates their coordinates to nearby bodies.

The empirical work approached the problem from another direction. It used geometric measurements and standard data-science methods to look for patterns that might lead back to mathematical explanations. We found patterns below systolic ratio one and confirmed some of their usefulness on fresh samples, but did not establish an extension to ratios above one. A geometric calculation expresses the ridge-area measurement on products through edge lengths and widths of the factors. For rotated regular pentagons, equation 35 proves an exact inverse-square relation between this descriptor and the systolic ratio. This explains the association on that restricted family, not on the sampled population as a whole. The exploratory samples produced no new ratio above one.

We implemented, optimized and extended two methods for computing capacity: the Haim–Kislev quadratic program and the Chaidez–Hutchings flow graph. The expanded variational proof explains why the first problem is finite; the second method gives an alternative for cross-checking computations. For products of planar polygons, we proved reductions that make the finite search smaller. A short analytic proof gives the capacity throughout the relative-rotation family of two regular pentagons. The matrix argument in Section 10 extends the formula to independently linearly deformed regular pentagons and proves that HKO attains the sharp systolic bound throughout this larger family. The original exhaustive SageMath rotation proof remains available as computational provenance. A separate product-position argument shows that every symplectic product of two full-dimensional planar convex bodies admits a capacity-increasing rigid rotation, although the Lagrangian endpoint can have smaller capacity. The relative placement of the factor planes thus supplies another geometric direction to investigate.

Two directions remain particularly close to the questions studied here. First, the local theorem could be tested against deformations that change the facet count. Adding or splitting facets leaves the forty-dimensional chart used in the proof, so it requires a new argument. Second, the statistical associations invite geometric interpretation. A useful next conjecture would explain an association on a clearly specified family and survive attempts to break it by changing the sampling distribution.

The results therefore give a local theorem around the known counterexample, finite methods for investigating nearby and sampled polytopes, and measured successes and failures of several search ideas. The larger question of how capacity and volume constrain one another remains open beyond these families and neighborhoods.

A Data-science methods and additional results

This appendix gives the optimizer steps and additional failure diagnostics for the local-search comparison in Section 8.10. Sampling definitions and statistical results are presented with their interpretation in Section 8.

A.1 Sampling and retained data

A polytope was specified by inequalities

$$K = \{x : \langle n_i, x \rangle \leq h_i, \quad i = 1, \dots, F\}.$$

For the generic family, each n_i was independent and uniform on the unit sphere in $\mathbb{R}^4$: we normalized a vector of four independent standard Gaussian coordinates. The heights were independent and uniform on $[0.8, 1.2)$. A proposed body was retained only if it was bounded and every proposed inequality defined a facet. The construction also rejected zero or near-duplicate normals. These conditions are part of the sampling law; “random polytope” here does not mean a uniform choice from all convex bodies.

The original population contains 512 bodies for each generic facet count and 1,024 for each product side-count pair. The retained generator records identify seed 42; product seed/attempt metadata cannot be independently reconstructed from the recovered cache. The source geometries, row identifiers and stored numerical values are nevertheless matched for every body. The reproduction records are listed in Table 3.

A.2 Numerical evaluation and retrospective checks

The complete historical source geometry was recovered and matched to all 14,336 table rows. A geometry-only rebuild restored six later descriptors; all 645,120 overlapping table cells agreed. This retained the historical volume normalization and did not evaluate capacity.

A separate current-evaluator run attempted every body. It returned 14,245 successful evaluations, 54 input-policy failures and 37 scalar-tolerance failures. Among successful rows, the largest relative differences were $1.06 \cdot 10^{-14}$ in capacity, $3.14 \cdot 10^{-14}$ in volume and $3.23 \cdot 10^{-14}$ in the ratio. No row is dropped from the original statistical analysis. The original failures are retained alongside the separate follow-up below.

A subsequent interval-preserving evaluation recovered results for 90 of those refusals. The 37 scalar-tolerance refusals yielded capacity intervals without changing their input bodies. Exact power-of-two rescaling brought 53 of the 54 norm-policy failures into the admitted range; fresh geometry checks preceded evaluation, and homogeneity transferred their interval endpoints back to the original bodies. Across both runs, 14,289 bodies have accepted capacity scalars at tolerance 10^{-10} and another 46 have wider capacity intervals. One body, `random_F8_s3_45`, remains unresolved. No policy or tolerance was relaxed. The complete sample, including this body, remains in the historical statistical analysis.

The current evaluator reconstructs the geometry of the stored binary64 input and returns rational product capacities or outward capacity bounds. Its volume calculation remains floating-point. Agreement of central values does not mean that a narrow new interval contains an independently rounded old value: 5,583 old capacity scalars lie outside the new intervals despite the small differences above. The initial reevaluation corroborates the numerical values on its successful bodies; it does not certify the original execution or the complete ratio. Prospective selection experiments and optimizer traces have their own evidence records and are not validated by this baseline comparison.

A.3 Selection and comparison rules

The 100,000-product concentration experiment specified a positive aggregate mean difference and positive differences in at least seven of ten groups as its continuation criterion. It found eight positive groups and a mean difference of about 0.034. This is the descriptive design criterion; no significance claim is attached to passing it.

For the tangential-generator comparison, controls were selected by a fixed hash order, before capacity evaluation, from the eligible pool after removing the union of the ridge and covariance

selections. They are unselected same-generator comparisons, not an asserted random sample of the full pool.

The opposing 3×6 and 4×4 rotation paths contain four recorded positions each. Their simultaneous decrease in ridge score and numerical ratio is a finite observation; monotonicity at every intermediate angle was not established.

A.4 Finite-budget local optimizer comparison

The run configuration labels the 64 starting bodies as held out. The saved hashes identify the seven configurations used, but do not establish that these bodies were absent from earlier development or tuning. A later development configuration reuses 16 starts. Their earlier selection history is therefore unresolved; this does not show that the reported outcomes were used to select the configurations.

Every run used the same allocation rule: a nominal 1000 ms start-new-work cutoff for charged full-evaluator plus optimizer time and a cap of 128 full historical-evaluator calls. In-progress work could complete atomically past the cutoff, and methods could terminate early, so realized compute differs across runs. The 448 traces contain 28824 full calls. Using the historical numerical objective described in Section 8.10, four-anchor branch history has the largest comparison-block median best value, 0.9848, followed by transition prediction at 0.9816 and the finite-gap affine model at 0.9772. No run reaches a recorded numerical ratio of one.

The implemented optimizer steps. Except for the literal gradient, the methods remove the fifteen infinitesimal symmetry directions from the forty dual-normal coordinates. The local models and coordinate poll recompute this transverse space at their anchors; CMA-ES keeps its initial space. The literal gradient uses the ambient coordinates. Every outer proposal is checked by a fresh geometry reconstruction and the configured full orbit-word search. The implementations below are frozen comparison points; their parameters are not claimed to optimize the corresponding algorithm families.

Table 2: One outer iteration of each optimizer in the recorded comparison.

method	implemented step
literal and safeguarded branch gradients	Differentiate one currently minimizing branch. The literal method takes $a^+ = a + 0.01D_{as_\sigma}(a)$ even after a decrease. The safeguarded method projects and normalizes this gradient, accepts only a fully evaluated improvement, and expands or halves its distance after acceptance or rejection.
finite-gap affine model	Keep the current branch gaps in a 10% action window, linearize their values, and maximize the minimum affine value on a local ball. Fully evaluate the proposal and use the same expand-or-halve distance rule.
four-anchor branch history	Union candidate words from the four most recent full searches, using a 30% action window and normalized beta allowance 0.3. At a nearby point, re-solve the named KKT branches, discard transition-infeasible and beta-inadmissible solutions, linearize the remaining values, and maximize their affine minimum. Take at most two internal moves before the next full evaluation.
history with transition prediction	Use the preceding method, but for a normalized internal move of length at least 0.08, also predict newly feasible transitions in the move direction and evaluate the words thereby generated.
CMA-ES	Fully evaluate a population of eight Gaussian perturbations, then update the mean, covariance, and normalized scale from their ranking.

method	implemented step
coordinate direct search	Poll both signs of the 25 transverse basis axes. Move to the best improvement and double the radius, or halve the radius if no poll point improves.

The two history methods accept any usable outer evaluation, including a decrease; their reported best-so-far value therefore differs from their current iterate. The safeguarded gradient and finite-gap method accept only improvements, multiplying the relative step length by 1.25 on success and by 0.5 on failure. The history methods use up to two internal model steps between full evaluations.

The companion endpoint packet does not test local maximality. Signed polls along the transverse basis find an improving direction at seven of 16 one-second branch-history endpoints and two of the corresponding five-second endpoints. Separate validated continuation paths show that even an endpoint with no improving basis direction can retain appreciable ascent between the tested axes. These observations diagnose insufficient finite budgets; they do not estimate the distance to a local maximum.

Detailed HKO calibration. On 16 known perturbations along four development directions from HKO, the multi-branch continuation accepted no move at HKO and improved every perturbation. One step recovered 93.4%–100% of the known evaluator gap and removed 77.9%–100% of the distance to HKO. In contrast, the gradient of one current minimizing branch recovered –22.1%–41.4% of the gap.

Selected finite-step failure audit. At the eight highest four-anchor tuning endpoints, five admitted one validated improvement, by $1.57 \cdot 10^{-6}$ to $4.61 \cdot 10^{-5}$. Across the 80 multi-branch proposals, 52 decreased the recomputed evaluator value despite a represented target winner and positive affine prediction; 40 of these losses retained determinate, unchanged geometry. All 40 current-branch gradient proposals decreased.

The three-proposal audit compared two such failures with one positive control. At normalized radius 10^{-5} , the failures perturbed their named KKT matrices by 218 and 67 times the smallest base eigenvalue magnitude and crossed zero; the control had ratio 1.53, did not cross zero, and kept the correct sign. At the smallest tested radius, 10^{-8} , central differences approximated the implemented analytic derivatives with relative errors below 1% for the named action and 1.9% for the named branch ratio. Binary64 and exact-arithmetic geometry reconstruction agreed at all 39 audited points, and relative volume discrepancies were below 10^{-15} . For these selected cases the observed failure is finite-distance nonlinearity of an ill-conditioned KKT branch.

These diagnostics use selected controls rather than population samples. They do not provide a convergence theorem, a population error rate, or a universal conditioning cutoff.

A.5 Adaptive product sampling

For reproducibility, let $g_1, \dots, g_5 > 0$ be the cyclic normal-angle gaps of a factor and let r_i be its dual-normal lengths. Set

$$\ell_i = \log(g_i/g_5) \quad (i = 1, \dots, 4), \quad z_i = \log r_i - \frac{1}{5} \sum_{j=1}^5 \log r_j.$$

We store the first four z_i and recover $z_5 = -\sum_{i=1}^4 z_i$. Reconstruction uses

$$g_i = \frac{2\pi e^{\ell_i}}{1 + \sum_{j=1}^4 e^{\ell_j}}, \quad g_5 = \frac{2\pi}{1 + \sum_{j=1}^4 e^{\ell_j}}, \quad h_i = e^{-z_i}.$$

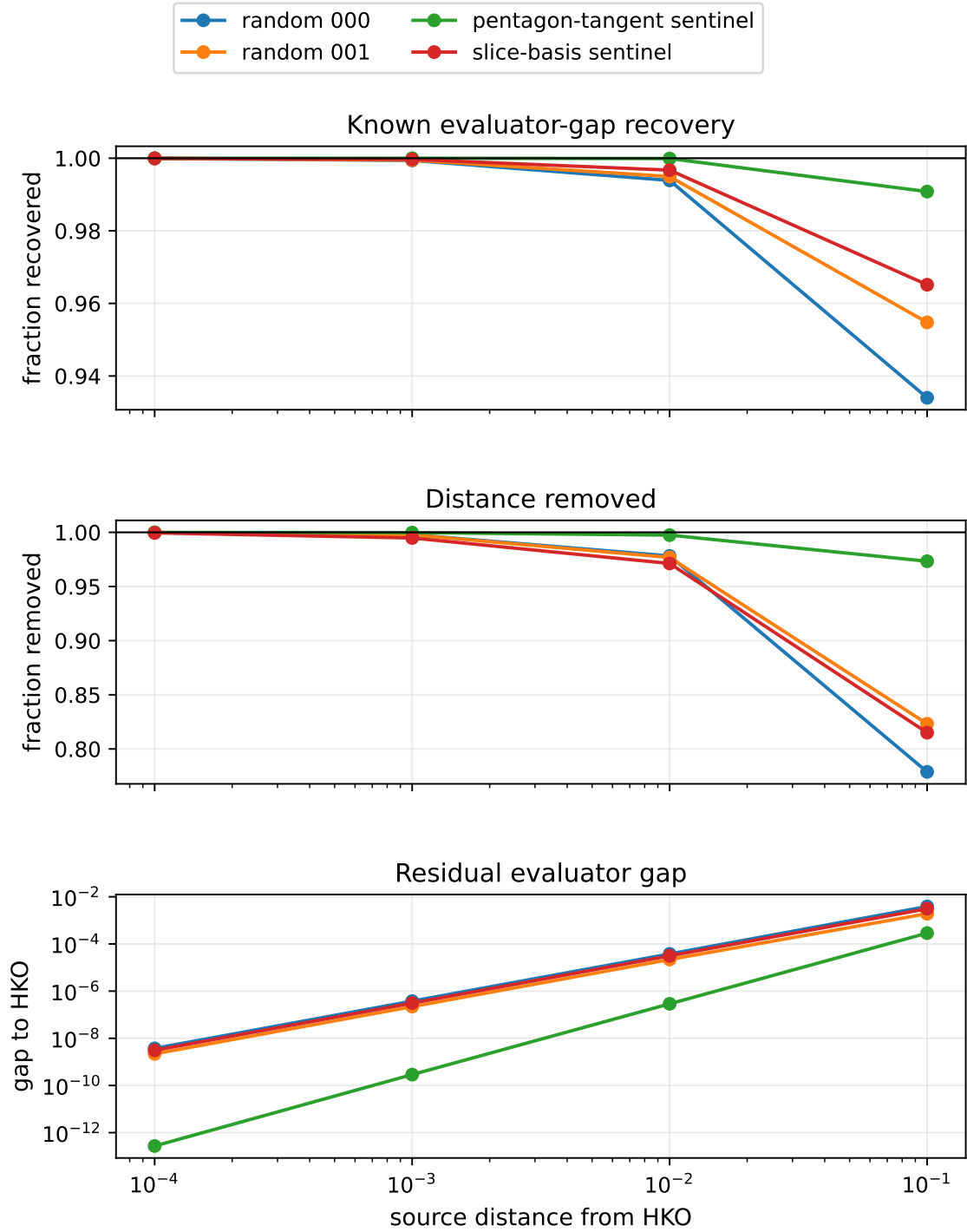


Figure 11: One-step calibration on known perturbations of HKO. From top to bottom, the panels show recovered evaluator gap, removed distance, and residual evaluator gap.

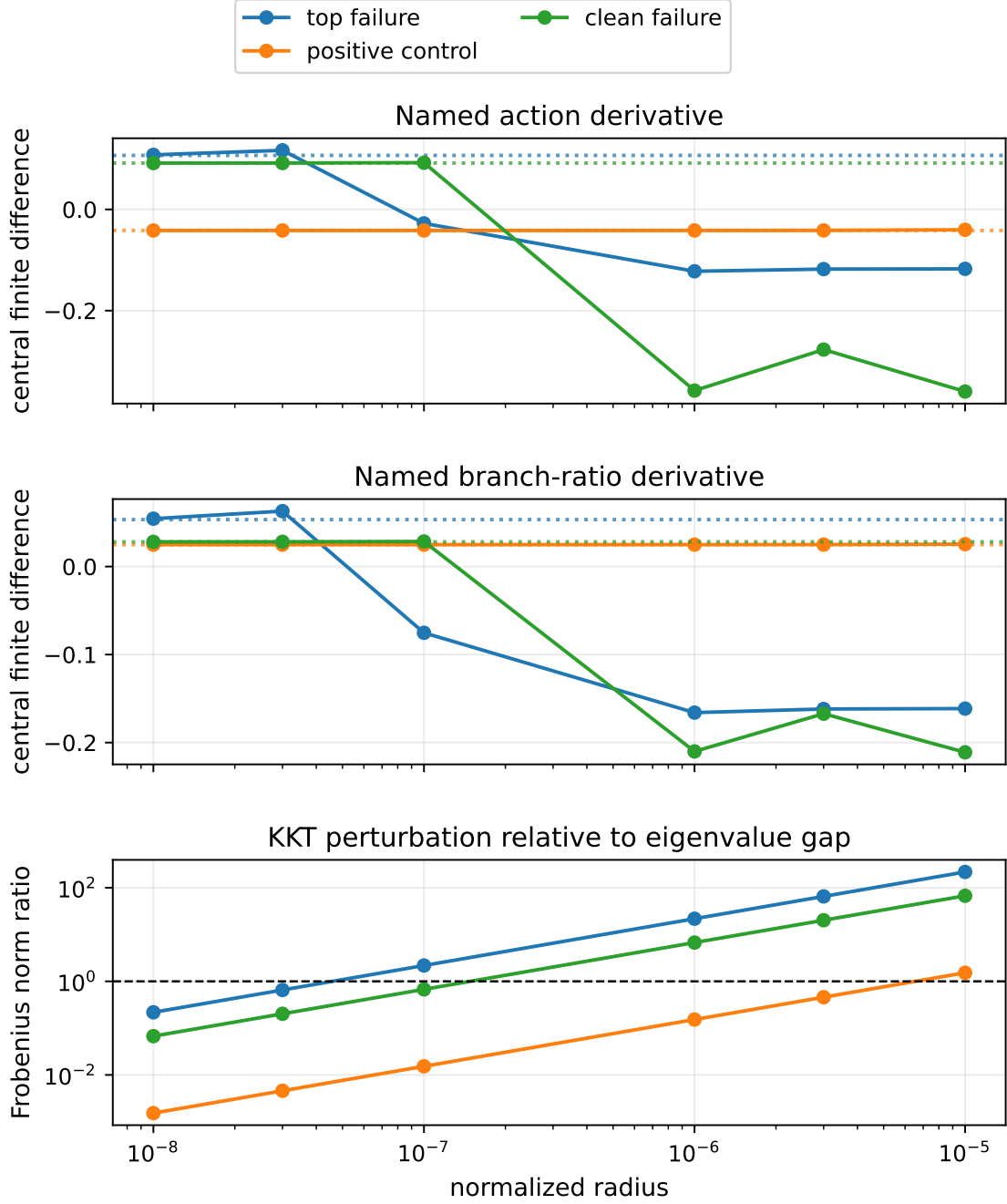


Figure 12: Finite differences for two failed affine proposals and one positive control. Prediction changes regime when the KKT perturbation becomes large relative to the smallest source eigenvalue magnitude (bottom). Dotted lines in the top two panels are the analytic derivatives.

The first normal angle is zero in the first factor and ϕ in the second. The initial IID encoding chooses a cyclic origin by lexicographically minimizing the sequence of gap/length pairs. Sampled adaptive coordinates are retained for the updates without another cyclic relabelling.

For an ordinary coordinate j , write $\hat{\mu}_{E,j}$ and $\hat{v}_{E,j}$ for the mean and population variance of the sixteen elites. The update is

$$\mu_j^+ = \frac{1}{2}\mu_j + \frac{1}{2}\hat{\mu}_{E,j}, \quad v_j^+ = \max\{\frac{1}{2}v_j + \frac{1}{2}\hat{v}_{E,j}, 0.05v_{0,j}\}.$$

Initialization uses the population moments of the shared 64 bodies. Coordinates are drawn independently with these Gaussian moments. The phase uses circular means, a unit-vector blend of old and elite means, and variance of angular deviations wrapped into $[-\pi, \pi)$. Samples are wrapped modulo 2π . A vanishing resultant retains the previous mean; initial fallback uses the smallest-ID candidate’s phase. Ties in elite selection are broken by candidate ID.

Each population permits at most 640 construction attempts for 64 accepted bodies. Failed geometry and unsupported evaluator inputs are rejected before target evaluation; an incomplete population does not trigger an update. The evaluator admission includes finite, nonredundant bounded geometry and primal and dual infinity-norm bounds between 10^{-3} and 10^3 . The three seeds are 202609140301, 202609140302 and 202609140303. The retained report records the stopped first version and the subsequent supported version separately.

The source restriction needs particular care. An initial version stopped when one body exceeded the evaluator’s admitted primal-vertex infinity-norm range. The completed version applied the evaluator’s existing admission conditions to both proposal arms before accepting a body for evaluation. This was a procedural revision after seeing the failure. Both arms reject unsupported inputs, but the admission test acts on the actual coordinate representation: IID bodies retain their original representation and adaptive bodies use the normalized reconstruction. It is not a symmetry-invariant condition on all pentagon products. The result is therefore for this implemented, admission-conditioned comparison.

The complete run took 215.8 seconds: 182.2 seconds for construction and admission checks, 32.9 seconds in evaluator processes, and 0.053 seconds for distribution updates. Including construction and charging initialization to each arm, the adaptive runs used about 35–36 seconds per seed and IID about 46–51 seconds. Equal evaluation counts did not imply equal cost. These measured costs favor the adaptive arm in this run; they are not the outcome of a separate equal-wall-time experiment.

A.6 Evidence and reproduction index

Paths in Table 3 are relative to the accompanying repository. Each packet identifies its inputs and analysis; the original historical optimizer executable and full proposal traces were not recovered.

Table 3: Source packets for the chapter.

Result	Repository source
Historical population and replay	docs/ds-evidence-closure/README.md;
Restored features and reevaluation	experiments/polytope-datasets/retained-lineage.md docs/ds-retrospective-revalidation/geometry-full/README.md; docs/ds-retrospective-revalidation/target-full/README.md;
Feature models	docs/ds-retrospective-revalidation/repair/README.md experiments/sys-datascience/methods/prediction-ranking/; experiments/sys-datascience/methods/standard-baseline-p2/

Result	Repository source
Fresh selection	experiments/sys-datascience/methods/ extreme-scalar-rejection-proposer/;
Transfer	experiments/sys-datascience/methods/ alternative-source-transfer/;
Paired support heights	experiments/sys-datascience/methods/ paired-tangentialization/
Orientation allocation	experiments/sys-datascience/methods/ orientation-allocation/
Adaptive product sampling	experiments/sys-datascience/methods/ diagonal-cem-pilot/
Local optimization and diagnostics	experiments/dev-gradient-ascent/optimizer-comparison/;
Product ridge geometry	docs/ds-evidence-closure/search-account.md docs/ds-evidence-closure/ridge-mathematics.md;
	experiments/sys-datascience/methods/ ridge-endpoint-path/

B HKO computation: an annotated exact program

B.1 Executable exact verification of the HKO bounds

This appendix isolates the finite certificate used in Lemma 7.4. The listings below form a complete SageMath program. Facet labels and positions in each word are displayed one-based; the executable converts them to zero-based indices at the section boundary.

The geometric lemmas in the main text establish that these ten labelled dual vertices form the relevant chart, that volume is smooth there, and that each positive feasible section supplies a touching upper bound. The finite CAS task is to instantiate 26 such sections, differentiate their upper bounds, and check the quotient certificate.

Exact field, geometry, and quadratic form. The first source block defines the ordered number field, the ten dual vertices, the symplectic form, the closure matrix, and the quadratic form.

```
#!/usr/bin/env sage -python
from sage.all import *
R = PolynomialRing(QQ, "x")
x = R.gen()
f = x**4 - 10*x**2 + 5
F = NumberField(f, 't', embedding=AA.polynomial_root(f, RIF(0, 1)))
t = F.gen(); s = (5 - t**2)/2; al = (3 - s)/2; be = t*(1 + s)/2; d = s - 1
a = [vector(F, v) for v in [(1, t, 0, 0), (-al, be, 0, 0), (-d, 0, 0, 0), (-al, -be, 0, 0), (1, -t, 0, 0), (0, 0, t, -1),
    (0, 0, be, al), (0, 0, 0, d), (0, 0, -be, al), (0, 0, -t, -1)]]
def om(u, v): return u[0]*v[2] + u[1]*v[3] - u[2]*v[0] - u[3]*v[1]
def C(w): return matrix(F, [[a[i][j] for i in w] for j in range(4)] + [[F(1)]*len(w)])
def q(w, b): return sum(b[i]*b[j]*om(a[w[j]], a[w[i]]) for i in range(1, len(w)) for j in range(i)
    )
```

All 26 exact assignments. The following is the complete `rows` object, including every word, solved coordinate set, fixed coordinate set, and exact fixed value. Values written with the caret character are strings parsed by Sage's exact evaluator; they are data, not Python exponentiation.

```
# (one-based word, solved positions I, fixed positions J, exact fixed values)
rows = [
    ([1, 2, 8, 4, 10, 6], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([1, 2, 8, 7, 4, 10], [1, 2, 3, 4, 5], [6], ['-t^2/20+1/4']),
    ([2, 3, 8, 9, 5, 4, 6], [1, 2, 3, 4, 5], [6, 7], ['6909830056250513/10^17', '-t^2/20+1/4']),
    ([1, 8, 7, 3, 4, 10], [1, 2, 3, 4, 5], [6], ['-t^2/20+1/4']),
    ([2, 9, 8, 5, 4, 6], [1, 2, 3, 4, 5], [6], ['-t^2/20+1/4']),
    ([2, 8, 4, 5, 6, 10], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([1, 6, 2, 8, 5, 4, 10], [1, 2, 3, 4, 5], [6, 7], ['36180339887499/(2*10^14)', 't^2/40+1/8']),
    ([1, 7, 8, 4, 3, 9, 10], [1, 2, 3, 4, 5], [6, 7], ['(8090169943749473*t^2+23368810393753689)/(4*10^17)', '18090169943749473/10^17']),
    ([1, 5, 7, 6, 3, 10, 9], [1, 2, 3, 4, 5], [6, 7], ['(4045084971874731*t^2+11684405196876883)/(2*10^17)', '9045084971874731/(5*10^16)']),
    ([2, 8, 4, 3, 9, 5, 6], [1, 2, 3, 4, 5], [6, 7], ['1809016994374947/10^16', '-t^2/20+1/4']),
    ([2, 3, 9, 8, 4, 5, 6], [1, 2, 3, 4, 5], [6, 7], ['2261271242968689/(125*10^14)', '-t^2/20+1/4']),
    ([2, 8, 9, 5, 7, 6, 3], [1, 2, 3, 4, 5], [6, 7], ['18090169943749473/10^17', 't^2/40+1/8']),
    ([1, 7, 8, 4, 10, 2], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([1, 7, 4, 3, 9, 10], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([2, 9, 5, 6, 7, 3], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([2, 8, 3, 9, 5, 6], [1, 2, 3, 4, 5], [6], ['-t^2/20+1/4']),
    ([2, 8, 4, 5, 10, 6], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([1, 7, 3, 2, 8, 4, 10], [1, 2, 3, 4, 5], [6, 7], ['18090169943749507/10^17', '-t^2/20+1/4']),
    ([1, 5, 6, 7, 3, 2, 9], [1, 2, 3, 4, 5], [6, 7], ['6909830056250547/10^17', '-t^2/20+1/4']),
    ([1, 7, 3, 9, 5, 6, 2], [1, 2, 3, 4, 5], [6, 7], ['t^2/40+1/8', '863728757031319/(125*10^14)']),
    ([1, 8, 5, 4, 6, 10, 2], [1, 2, 3, 4, 5], [6, 7], ['t^2/40+1/8', '7463176480767021/(5*10^16)']),
]
```

```
([1,6,2,8,7,4,10],[1,2,3,4,5],[6,7],[-t^2/20+1/4','1809016994374947/10^16']),
([1,5,7,4,3,9,10],[1,2,3,4,6],[5,7],[9045084971874741/(5*10^16)',t^2/40+1/8']),
([1,7,3,4,10,9,5],[1,2,3,4,5],[6,7],[t^2/40+1/8','3454915028125259/(5*10^16)']),
([1,7,6,3,10,9,5],[1,2,3,4,5],[6,7],[452254248593737/(25*10^14)',t^2/40+1/8']),
([2,8,9,5,4,6,3],[1,2,3,4,5],[6,7],[-t^2/20+1/4','12028993467659169/10^17'])]
```

Feasible sections and derivative rows. The next block is the actual section solver. It explicitly converts the one-based assignment data, solves $C_I \beta_I = e - C_J u$, restores $\beta_J = u$, and checks the minor's invertibility, closure, normalization, and positivity. The constants V , A , and vr are respectively the base volume, the known HKO capacity, and the volume derivative from Lemma 7.3. The derivative block computes $D\beta$, then differentiates Q and forms $r_j = DU_j(a_0)$ by the product rule for $U_j = A_j^2/(2V)$.

```
def val(z): return F(sage_eval(z, locals={'t': t}))
def section(W):
    w,I,J,u=W; w=[i-1 for i in w]; I=[i-1 for i in I]; J=[i-1 for i in J]
    assert len(set(w))==len(w) and all(0<=i<10 for i in w); assert sorted(I+J)==list(range(len(w)))
    cm=C(w); assert cm[:,I].det()!=0; U=vector(F,[val(z) for z in u]); b=[None]*len(w)
    x=cm[:,I].solve_right(vector(F,[0,0,0,0,1])-cm[:,J]*U)
    for k,i in enumerate(I): b[i]=x[k]
    for k,j in enumerate(J): b[j]=U[k]
    b=vector(F,b); assert cm*b==vector(F,[0,0,0,0,1]) and all(z>0 for z in b)
    return w,I,J,b
V=QQ(25)*(5+s)/QQ(32); A=5*t-t**3/qq(2); qm=1/(2*A)
vr=vector(F,sum((list(-(QQ(25)/QQ(32)+qq(5)*s/qq(16))*v) for v in a),[]))
def drow(w,I,J,b):
    cm=C(w); db=[[F(0)]]*40 for _ in w]
    for z in range(40):
        facet,coord=divmod(z,4); rhs=vector(F,[0]*5)
        if facet in w: rhs[coord]=-b[w.index(facet)]
        xi=cm[:,I].solve_right(rhs); assert cm[:,I]*xi==rhs
        for k,i in enumerate(I): db[i][z]=xi[k]
    for j in J: assert all(x==0 for x in db[j])
    dq=[]
    for z in range(40):
        facet,coord=divmod(z,4); u=vector(F,[1 if k==coord else 0 for k in range(4)]); x=F(0)
        for i in range(1,len(w)):
            for j in range(i):
                x+=(db[i][z]*b[j]+b[i]*db[j][z])*om(a[w[j]],a[w[i]])
                if w[j]==facet: x+=b[i]*b[j]*om(u,a[w[i]])
                if w[i]==facet: x+=b[i]*b[j]*om(a[w[j]],u)
            dq.append(x)
    return vector(F,[A/V*(-x/(2*qm**2))-A**2/(2*V**2)*vr[i] for i,x in enumerate(dq)])
```

Symmetry and finite conclusion. The final source block constructs the four translations, dilation, and ten $\mathfrak{sp}(4)$ columns, then checks symmetry annihilation, row rank 25, and a strictly positive normalized left-kernel relation. The check $q(w,b)==qm$ establishes that each section gives the known HKO action at the base point, so its upper bound touches the systolic ratio there. Thus the explicitly assigned rows, together with the computed rr and lam , witness the existence assertions of Lemma 7.4. The rank and positive relation supply the hypotheses of Lemma 7.5.

```
# Build the fifteen symmetry columns, then verify all rows, rank, and kernel.
Jm=matrix(F,[[0,0,-1,0],[0,0,0,-1],[1,0,0,0],[0,1,0,0]])
T=[vector(F,sum((-v[k]*v) for v in a),[])) for k in range(4)]
T += [vector(F,sum((-v) for v in a),[])]
G=[matrix(F,g) for g in [
    [[1,0,0,0],[0,0,0,0],[0,0,-1,0],[0,0,0,0]],[[0,1,0,0],[0,0,0,0],[0,0,0,0],[0,0,-1,0]],
    [[0,0,0,0],[1,0,0,0],[0,0,0,-1],[0,0,0,0]],[[0,0,0,0],[0,1,0,0],[0,0,0,0],[0,0,0,-1]],
```

```

[[0,0,1,0],[0,0,0,0],[0,0,0,0],[0,0,0,0]],[[0,0,0,1],[0,0,1,0],[0,0,0,0],[0,0,0,0]],
[[0,0,0,0],[0,0,0,1],[0,0,0,0],[0,0,0,0]],[[0,0,0,0],[0,0,0,0],[1,0,0,0],[0,0,0,0]],
[[0,0,0,0],[0,0,0,0],[0,1,0,0],[1,0,0,0]],[[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,1,0,0]]]
for X in G:
    assert X.transpose()*Jm+Jm*X==0
    T.append(vector(F,sum((list(-X.transpose()*v) for v in a),[])))
rr=[]
assert len(rows)==26
for W in rows:
    w,I,J,b=section(W); assert q(w,b)==qm; rr.append(drow(w,I,J,b))
M=matrix(F,[list(r) for r in rr]); assert M.rank()==25 and matrix(F,[list(c) for c in T]).rank
()==15
for r in rr:
    for c in T: assert r.dot_product(c)==0
kernel=M.transpose().right_kernel().basis(); assert len(kernel)==1
lam=kernel[0]
if all(x<0 for x in lam): lam=-lam
assert all(x>0 for x in lam); lam=lam/sum(lam); assert lam*M==0
print('HK0 appendix core passed: rows=26 rank=25 symmetry-rank=15')

```

The listed file is `thesis/candidate/recovered/hko-cas-appendix/hko_core.py`. Run it with `sage -python`, with Python optimization disabled so that its assertions are executed. The recorded run returned `HK0 appendix core passed: rows=26 rank=25 symmetry-rank=15`. These checks establish the finite hypotheses used in the main proof. The geometric chart and volume formula are supplied by Lemmas 7.2 and 7.3; Lemmas 7.5–7.7 then pass from the computed derivatives to local maximality.

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